OPEN LAB ACADEMY

Mathematics Research Companion

719 manuscripts, organized for understanding

719 / 372 / 17

active manuscripts / result families / disciplines

English report

Source revision: 7 October 2026 • Report prepared: 18 October 2026 (Taiwan)

Source commit / 1581d4cbdd69

A reading companion to manuscript claims - not a full proof translation, peer review or independent verification.

Revision, reading method and evidence

This report follows a pinned revision of the official openai/math catalogue: 722 manuscripts at the initial release; 719 active manuscripts in 372 result families and 17 disciplines at this revision; three withdrawal notices retained separately. A family is not a single paper, and constituent papers can differ in scope and hypotheses.

The catalogue section reproduces the official English abstracts. Statements such as 'We prove' belong to the original authors and do not signify independent confirmation by this report. The Chinese edition contains source-grounded paraphrases, explicit limitations and labeled research extensions, rather than full sentence-by-sentence translations of 719 proofs.

This material contains 17 discipline tracks and 15 main-paper deep-read starting points. The introductions, main statements and proof roadmaps were read from official source files. It does not claim independent full verification of all 719 manuscripts, nor reruns of all Lean or computation certificates. “Claim” denotes the manuscript’s assertion at this revision; examples teach concepts, and application paths are labeled inferences.

A plausible research application is not a delivered engineering tool, and a small numerical example is not a proof of a universal theorem. Major new assertions should be evaluated through their complete statements, dependencies, proofs and revision history.

The repository reports formalization material for part of the collection. This build did not rerun Lean, audit all axioms or execute every computation certificate. Formalization coverage and independent verification are separate statuses.

openai/math • 1581d4cbdd69a0faf34b2089614fe2a814316a6a

Catalogue and abstract level

Main-paper introduction/theorem deep read; not independently verified

Not reached here: independently reproduced proof

Contents

Revision, reading method and evidence 1
From foundations: 17 discipline tracks 3
15 main-paper deep-read starting points 9
Manuscript abstracts and original sources 24
Number theory 24
Algebraic and complex geometry 37
Real and complex analysis 56
Convex and metric geometry 61
Theoretical computer science 68
Dynamical systems and ergodic theory 84
Combinatorics 88
Algebra 98
Probability and statistical mechanics 104
Mathematical logic 128
Group theory 130
Mathematical physics 134
Operator algebras 149
Topology 156
Functional analysis 162
Differential geometry 167
Partial differential equations 178
Withdrawals and revision history 185
Sources and licensing 185

From foundations: 17 discipline tracks

Physics PhD route 268 → 271 → 265 → 266 → 221 → 362. Begin with finite spins and measurements, then thermodynamic limits, entanglement and field PDE.

Intuitive entry route 179 → 113 → 158 → 159 → 017. Compute small examples fully before tackling existence and asymptotic claims.

Algorithms and number theory route 107 → 102 → 003 → 002. First separate arithmetic complexity, bit time, hardness and arithmetic hypotheses.

Number theory

Prerequisites and sequence Integer division and congruences → series, complex numbers and calculus → groups, fields and complex analysis. Start with family 017, then 003 and 002.

A first worked example Compare the errors of 22/7 and 355/113 as approximations to π. A single exceptional approximation says nothing about infinitely many denominators.

Reading the formal statement Read the quantifiers: for every $\varepsilon>0$ there exists $Q(\varepsilon)$ such that for all $q \geq Q(\varepsilon)$… Distinguish existence, effective constants and algorithms.

What to audit Check analytic continuation, Euler factors, exceptional local primes, conditional dependencies and uniform estimates. Testing finitely many zeros cannot prove a zero-free infinite region.

Application context Established methods support arithmetic computation and coding. Proposed extensions here concern finite-field square roots and rational-point algorithms; they do not imply a break of elliptic-curve cryptography.

Algebraic and complex geometry

Prerequisites and sequence Polynomial equations → linear algebra, complex analysis and topology → manifolds, sheaves and cohomology.

A first worked example The circle $x^2+y^2=1$ has rational parametrization $x=(1-t^2)/(1+t^2)$, $y=2t/(1+t^2)$. This illustrates algebraic geometry, but is not a parametrization of a general elliptic curve.

Reading the formal statement Unpack a statement that a cohomology class comes from a cycle into its coefficient field, codimension, equivalence relation and realization functor.

What to audit Check smoothness, projectivity and characteristic assumptions, and whether specialization preserves every claimed realization. A rational-coefficient theorem is not an integral theorem.

Application context Established connections include algebraic statistics, error-correcting codes and mirror symmetry. Turning new cycle theorems into engineering tools requires additional effective algorithms and modeling.

Real and complex analysis

Prerequisites and sequence Limits, differentiation and integration → uniform convergence and measure → Fourier analysis and holomorphic functions.

A first worked example The functions $f_n(x)=x^n$ converge pointwise on [0,1], but their limit is discontinuous at 1. Pointwise convergence does not justify arbitrary interchange of operations.

Reading the formal statement For each estimate record the function space, norm, measure and constant dependence. Distinguish almost-everywhere statements from pointwise statements.

What to audit Check domination for each exchange of limits and integrals, endpoint exponents and the compactness actually available.

Application context Signal reconstruction, optics and PDE use analytic estimates. New bounds need finite-resolution error budgets before numerical deployment.

Convex and metric geometry

Prerequisites and sequence Vectors and distance → convex sets, inner products and integration → convex duality and high-dimensional geometry.

A first worked example A line segment joining two points of a disk stays in the disk; the same fails for an annulus. Convexity lets supporting hyperplanes convey global information.

Reading the formal statement Distinguish volume, Gaussian measure, covering density and lattice restrictions. An asymptotic Θ(n log n) order does not give the best constant for every finite n.

What to audit Check translation, affine and scale normalization, and whether optimization ranges over convex bodies, symmetric bodies or lattices.

Application context Connections include compressed sensing and coverage design. Existence of an extremizer still requires a constructive algorithm for implementation.

Theoretical computer science

Prerequisites and sequence Programming and discrete mathematics → complexity and probability → NP reductions, randomized algorithms and semidefinite programming.

A first worked example A triangle has maximum cut size 2; enumerate its two-way partitions. Finding a good cut, certifying optimality and counting all optima are different tasks.

Reading the formal statement Record input encoding, bit time versus arithmetic operations, error ε, failure probability δ and which parameters are fixed rather than part of the input.

What to audit An integrality gap is not an NP-hardness reduction. Small benchmarks do not prove an asymptotic exponent, and polynomial time need not be practically fast.

Application context Families 102, 107 and 113 connect to approximation limits, matrix multiplication and randomized counting; each requires auditing the full reduction or error analysis.

Dynamical systems and ergodic theory

Prerequisites and sequence Differential equations or iterated maps → measure and probability → invariant measures, ergodicity and entropy.

A first worked example For rotation $x \mapsto x+\alpha \bmod 1$, rational α gives periodic orbits and irrational α gives dense orbits. Density, equidistribution and mixing are distinct.

Reading the formal statement Separate time and space averages, topological and measure entropy, and C^1 from higher smoothness; the conclusions often depend on these choices.

What to audit Track exceptional null sets, order of limits and rates. A chaotic-looking finite trajectory does not prove positive entropy.

Application context This supplies foundations for statistical mechanics and nonlinear control. Qualitative recurrence is not a finite-time predictive return bound.

Combinatorics

Prerequisites and sequence Sets, counting and graphs → probabilistic method → extremal, additive and structural combinatorics.

A first worked example The set {1,2,4,5} has no nonconstant three-term arithmetic progression; {1,3,5} does, with common difference 2. Enumerate small examples before studying $r_k(N)$.

Reading the formal statement Specify simple versus multigraphs, nonzero progression differences and whether constants may depend on k.

What to audit A counterexample must satisfy every hypothesis of the conjecture. Finite search up to N cannot establish nonexistence for all sizes.

Application context Scheduling, communication codes and discrete design provide applications; families 158, 159 and 179 teach geometric coloring, arithmetic structure and autocorrelation.

Algebra

Prerequisites and sequence Matrices and polynomials → groups, rings and fields → modules, representations and homology.

A first worked example For finite square matrices, AB=I implies BA=I. On an infinite-dimensional space, left and right shifts show that a one-sided inverse need not be two-sided.

Reading the formal statement Retain characteristic, commutativity, finite-generation and identity assumptions. Similar terminology across rings, group rings and operator algebras needs care.

What to audit Check whether constructions remain in the specified class and whether manipulations silently assume commutativity or invertibility.

Application context Algebra supports coding, symbolic computation and symmetry analysis; a nonconstructive counterexample rarely becomes an application directly.

Probability and statistical mechanics

Prerequisites and sequence Distributions, expectation and conditioning → laws of large numbers and Gibbs measures → stochastic processes, limit theorems and phase transitions.

A first worked example For two Ising spins with $H=-Js_1s_2$, $Z=4\cosh(\beta J)$ and $\langle s_1s_2\rangle=\tanh(\beta J)$. This finite system can be checked exactly.

Reading the formal statement Distinguish quenched E log Z from annealed log E Z and track disorder, volume and temperature limits.

What to audit Check moment assumptions, independence, uniform integrability and exchange of limits. A variational upper bound is not automatically the true free energy.

Application context Family 221 links diluted spin glasses to random constraints. Message passing is a possible direction, but the variational formula does not guarantee efficient global optimization.

Mathematical logic

Prerequisites and sequence Propositions, quantifiers and induction → computability and set theory → model theory or proof theory.

A first worked example Being able to solve particular instances is not the existence of one algorithm for all instances. The halting-problem diagonal argument concerns this uniform requirement.

Reading the formal statement Specify the axiom system, definability, language size and relative consistency. Undecidability, independence and computational hardness are different notions.

What to audit Retain assumptions such as CH. A Lean theorem may use explicit axioms; successful compilation does not mean absence of mathematical premises.

Application context Applications concern specification languages and limits of automated proof. Undecidability does not make every finite instance uncomputable.

Group theory

Prerequisites and sequence Permutations and symmetry → actions and quotient groups → representations and geometric group theory.

A first worked example The six symmetries of a triangle form a noncommutative group: rotating then reflecting need not equal reflecting then rotating.

Reading the formal statement Separate abstract isomorphism, topological isomorphism, quasi-isometry and measure equivalence; the representation field is also part of the assumptions.

What to audit Check preservation of finite presentation, countability and residual finiteness under constructions and limits.

Application context Symmetry reduces physical and computational degrees of freedom; deep existence results need effective representations to become computational tools.

Mathematical physics

Prerequisites and sequence Classical, quantum and statistical mechanics → self-adjoint operators and tensor products → thermodynamic limits and many-body estimates.

A first worked example For two spin-1/2 particles, $S_1 \cdot S_2=(S_total^2-S_1{}^2-S_2{}^2)/2$ gives 1/4 on the triplet and −3/4 on the singlet, reducing a matrix problem by symmetry.

Reading the formal statement Fix $\hbar$, kB, coupling and boundary conditions. Finite-volume gaps, infinite-volume gaps, spontaneous magnetization and finite-temperature magnetization are distinct claims.

What to audit Audit the order volume→∞, field→0 and temperature→0, and distinguish physical approximation, numerical evidence and rigorous bounds.

Application context Prioritize 268, 271, 265 and 266 for spin chains, magnetism, tensor networks and quantum measurements, keeping applications within the stated hypotheses.

Operator algebras

Prerequisites and sequence Linear algebra and quantum operators → Hilbert spaces and bounded operators → C* and von Neumann algebras.

A first worked example The Pauli relation σxσz=−σzσx shows noncommutativity of observables; finite matrices are only the starting point for infinite-dimensional theory.

Reading the formal statement Distinguish norm, strong and weak operator topologies, traces and factor types; isomorphism and approximate embedding have different quantifiers.

What to audit Check completion, separability, normality and trace preservation. Finite-matrix approximations do not directly establish infinite-algebra isomorphism.

Application context There are established links to quantum information and many-body statistics; classification results need additional constructions for viable algorithms.

Topology

Prerequisites and sequence Continuous maps and geometry → fundamental groups and homology → manifolds and homotopy.

A first worked example A loop winding around a circle cannot contract within the circle, but can within a disk. Topology studies what allowed deformations preserve.

Reading the formal statement Distinguish homeomorphism, diffeomorphism and homotopy equivalence; four-dimensional and higher-dimensional results are not interchangeable.

What to audit Check compactness, orientation, local flatness and fundamental-group assumptions; identical homology is much weaker than equivalence of spaces.

Application context Topological data analysis and topological phases are established applications; new manifold results are usually foundations rather than ready-made data algorithms.

Functional analysis

Prerequisites and sequence Linear algebra, series and integration → normed spaces and completeness → operators, duality and weak convergence.

A first worked example In ℓ^2, every basis vector e_n has norm 1, yet e_n converges weakly to 0. Weak convergence need not imply norm convergence.

Reading the formal statement Identify Banach versus Hilbert structure, operator norms and compactness notions before applying a theorem.

What to audit Check boundedness versus uniform boundedness, dense domains and closed-graph hypotheses; finite-dimensional intuition often fails here.

Application context Applications include weak PDE solutions, inverse problems, kernel methods and optimization; numerical stability still requires discretization analysis.

Differential geometry

Prerequisites and sequence Multivariable calculus → curves, surfaces and tensors → curvature, geodesics and geometric analysis.

A first worked example A sphere of radius R has Gaussian curvature $1/R^2$; area scales like R^2 while curvature scales inversely, illustrating dimensional consistency.

Reading the formal statement Distinguish sectional, Ricci and scalar curvature, and completeness, compactness and simple connectivity.

What to audit Check singularities, boundaries and equality rigidity; local curvature control does not imply arbitrary global conclusions.

Application context Connections include general relativity, surface reconstruction and geometric optimization, with the same metric and boundary assumptions required.

Partial differential equations

Prerequisites and sequence Calculus, ODE and Fourier series → Sobolev spaces and energy methods → nonlinearity, regularity and compactness.

A first worked example Free transport $\partial_t f + v \cdot \nabla_x f = 0$ has solution $f(t,x,v) = f_0(x - tv, v)$. The density is constant along characteristics; a coupled field makes those characteristics unknown.

Reading the formal statement Separate existence, uniqueness, regularity, stability and scattering. Global weak existence is not global smooth existence.

What to audit Check initial compatibility, energy spaces, a priori bounds and continuation criteria. Bounds on each finite T need not be uniform in T.

Application context Family 362 introduces plasma PDE. A theorem is not a stable particle-in-cell solver and does not directly prove controllability of a physical device.

15 main-paper deep-read starting points

017-01 • The irrationality exponent of π: how good is exceptionally good?

Manuscript claim The manuscript claims μ(π)=2: for every ν>2 there is Q(ν) such that |π−p/q|≥ q^(−ν) for every integer p and q≥Q(ν). It derives convergence of the Flint-Hills series Σ1/(n^3 $\sin^2 n$), with radians. This does not limit high-precision computation of π.

Essential hypotheses Denominators are positive integers and the threshold depends on ν; the manuscript explicitly does not make it effective. Exponent two is weaker than a uniform c/q^2 lower bound and does not imply bounded continued-fraction partial quotients.

Why it matters The pigeonhole principle already gives μ≥2 for any irrational number. The difficulty is excluding infinitely many approximations better than q^(−2−ε). If correct, this places π at the minimum possible exponent and controls how sparsely extremely small sin n values occur.

Proposed proof roadmap Assume infinitely many exceptional approximations and choose widely separated denominators. Multivariable interpolation near logarithmic curves produces a nonzero determinant; clearing denominators gives an arithmetic lower bound. Taylor-degree collisions and approximation errors force an incompatible analytic upper bound. Dyadic grouping and spacing estimates then give series convergence.

A first worked example The error of 22/7 is about 1.264×10^{-3}; that of 355/113 is about 2.668×10^{-7}, much smaller than $113^{-2}\approx7.83\times10^{-5}$. A single example does not contradict an eventual bound with a large Q(ν). Exercise: plot log(error) against log(q) for 20 continued-fraction convergents; treat it only as finite evidence.

Learning and research extensions Start with continued fractions and Dirichlet approximation, then transcendence theory and interpolation determinants. A possible application is convergence criteria for oscillatory small-denominator series; an ineffective threshold does not yield practical best constants or a faster π calculator.

What to audit The key audit points are full rank in the interpolation theorem and thresholds independent of center locations: dimension counting alone is insufficient. Check for circular parameter choices. This guide reads the introduction and main statement; it does not independently verify the proof or Lean.

Source locators

main.tex: Introduction; Theorem thm:main; Corollary thm:flinthills; The proof in outline

Original PDF

003-01 • The quasi-Riemann hypothesis and a zero-free half-plane

Manuscript claim The manuscript claims that all Dirichlet L-functions and finite-order Hecke L-functions over $Q(\sqrt{-3})$ have no zeros for $\mathrm{Re}(s)>7/8$, allowing the principal pole at $s=1$. It does not place all nontrivial zeros on $\mathrm{Re}(s)=1/2$.

Essential hypotheses The boundary $\mathrm{Re}(s)=7/8$ is excluded. The boundary must be uniform in character, conductor and height, although constants and starting thresholds for a fixed character may depend on it. Outside $\mathrm{Re}(s)>1$, L denotes its analytic continuation.

Why it matters Zeros of $\zeta(s)$ govern prime-distribution errors. Classical regions that approach $\mathrm{Re}(s)=1$ differ from a fixed-width half-plane, and a density estimate can still permit exceptional zeros. The claimed advance is a uniform exclusion rather than a count of rare zeros.

Proposed proof roadmap The proposed proof first compares reflected and Poisson representations of a completed cubic-theta sum to obtain 11/12, then introduces prime compensation, asymmetric scales and two moment estimates to reach 7/8. The principal Poisson row contains a Mellin signal involving $1/L$; continuation rules out its poles. Norm characters and quadratic factorization transfer the result to Dirichlet L-functions.

A first worked example Modulo 5, the nonzero quadratic residues are 1 and 4, so 2 is a nonresidue. For a large prime p, sequential testing eventually finds a nonresidue; a worst-case polynomial bound in $\log p$ needs extra number theory. The paper derives $n(p)\leq C(\log p)^A$ and combines it with Tonelli-Shanks. The small example illustrates the application, not the zero-free proof.

Learning and research extensions Learn Euler products, complex analysis and character orthogonality, then Hecke L-functions, Mellin transforms and large sieves. A stated consequence concerns deterministic finite-field square roots; it is not RH, GRH or a general cryptographic breakthrough.

What to audit Audit the distinct sums in the two stages, omitted Euler factors, principal residues and uniform exponent margins. The repository's overall formalization percentage is not independent confirmation of this theorem.

Source locators

paper.tex: Introduction; Theorem thm:main; Proof overview; Corollary cor:nonresidues-square-roots

Original PDF

002-01 • The full BSD formula under low Selmer corank

Manuscript claim The manuscript claims that if the full q-power Selmer group of $E/\mathbb{Q}$ has $\mathbb{Z}q$-corank 0 or 1 for some prime q, then the Mordell-Weil rank, analytic rank and that corank agree, Sha is finite, and the complete BSD leading-term formula holds: $L^{(r)}(E,1)/r! = \Omega E \, \mathrm{Reg}E \, \#Sha \prod c_\ell / \#E(\mathbb{Q})tors^2$.

Essential hypotheses The hypothesis is low full Selmer corank, not merely low rational-point rank. L includes all finite Euler factors, ΩE integrates over every real component, and RegE uses the paper's stated height normalization.

Why it matters Rank predicts how many independent infinite-order rational points exist. The leading coefficient also records torsion, local Tamagawa numbers, the volume of the point lattice and Sha's local-to-global obstruction. These finite factors are the focus here.

Proposed proof roadmap The proof first uses companion low-corank converse and two-primary formulas. For each odd p it defines a valuation discrepancy $Xp(E)$, proves $Xp(E) \geq 0$, and chooses a quadratic twist with $Xp(E)+Xp(E^D)=0$. Nonnegativity forces equality. The difficult work is integral determinant lattices, theta comparisons and removal of quotient divisors, not cancellation in a fraction field.

A first worked example On $E:y^2=x^3-x$, (0,0), (1,0) and (−1,0) are points of order two, alongside the identity at infinity. They illustrate the torsion factor; these points alone do not determine analytic rank, Sha or BSD. Use SageMath to compute known invariants and compare them under a consistent normalization.

Learning and research extensions Start with the elliptic-curve group law and rational points, then Selmer groups, Galois cohomology and Iwasawa theory. Potential directions include certified curve databases and twists; the scope excludes unrestricted high rank and gives no discrete-logarithm attack.

What to audit Audit the companion papers as dependencies, especially unit arguments for integral quotients and period/height normalization factors. This guide explains the main statement and route without verifying every arithmetic dependency.

Source locators

sections/01-introduction.tex: Statement and normalization; Theorem thm:main; The arithmetic discrepancy; Integral comparisons

Original PDF

221-01 • Diluted spin glasses: what does the hierarchical cavity formula compute?

Manuscript claim For Poisson-diluted even-arity Ising models in the stated Panchenko-Talagrand class, the manuscript claims that the limiting quenched pressure lim $N^{-1}E$ log ZN equals $\inf_r$ Φr over all finite-depth hierarchical trial laws.

Essential hypotheses The arity p is even, M~Poisson(αN), and endpoints are sampled independently with replacement. Assumptions include E max|θ|+E|h|<∞, e^θ=a(1+b∏fℓ), a strict product bound and $E(-b)^n \geq 0$, with the full factorization and independence conditions. This is not an unrestricted theorem for every diluted model.

Why it matters Finite mean degree leaves local environments random even as N grows. A hierarchy of distributions of local fields captures competing states beyond a single average field. Physical free energy is −pressure/β; keep track of signs and whether β is absorbed into θ.

Proposed proof roadmap Earlier interpolation provides hierarchical upper bounds; the claimed new step matches them from below. At fixed depth, bounded perturbations control finite multioverlaps. Delayed branching yields concentration averaged over branching depths. A finite-tree cavity expansion is then compared with independent hierarchical messages. N tends to infinity before depth increases; ultrametric physical Gibbs states are not assumed.

A first worked example For two spins with $H=-Js_1s_2$ and h=0, Z=4 cosh(βJ), and $\langle s_1 s_2 \rangle$=tanh(βJ). For N uncoupled spins, N^{-1}log Z=log 2. These are useful program checks, not proofs of the random diluted thermodynamic limit.

Learning and research extensions Study Ising partition functions and quenched versus annealed averages, then RSB, cavity methods and multioverlaps. Numerical message distributions for Viana-Bray, symmetric even-spin and soft even-K SAT models are possible projects. An infimum need not be attained at finite depth and does not guarantee polynomial-time optimization.

What to audit Audit independence when message roots are redrawn, control of all multioverlaps and the removal of truncation under first moments. This page does not solve the full hierarchical optimization problem.

Source locators

sections/introduction.tex: Introduction; final proof-strategy paragraphs

sections/model.tex: Equations model:integrability-model:positivity; main:theorem

Original PDF

271-01 • Quantum ferromagnets: spontaneous magnetization and Bloch's $T^{3/2}$ law

Manuscript claim The selected main paper claims that for a three-dimensional quantum Heisenberg ferromagnet at every fixed positive half-integer spin S, lim $\beta^{3/2}(S-m(\beta))=\zeta(3/2)/(8\pi^{3/2}\sqrt{\det D})$, where $D=(S/2)\Sigma z\, J(z)zz^{T}$.

Essential hypotheses Couplings are nonnegative, symmetric and finite range, and their support generates $\mathbb{Z}^3$. Magnetization is the right field derivative of infinite-volume pressure at zero. The order is volume→∞, then h↓0, then β→∞; zero-field magnetization of a finite symmetric box is not the same observable.

Why it matters Spin-wave theory suggests $T^{3/2}$, but passing from ideal magnons to the interacting quantum spin system requires control of the field derivative. A zero-field free-energy asymptotic alone cannot simply be differentiated to obtain magnetization. The paper also allows spatially anisotropic finite-range couplings.

Proposed proof roadmap Represent each spin by 2S spin-1/2 slots and use colored Poisson-exchange loops with up-spin pins to express down probabilities. Sublattice geometry and Nash estimates give rough bounds; late returns are controlled and compared with the full ideal jump kernel. Heat-kernel decay t^(−3/2) and Σn^(−3/2) produce the coefficient. A positive field is then represented as additional pins before taking the ordered limits.

A first worked example For nearest-neighbor J(±ej)=1, D=SI. At S=1/2 the coefficient is about 0.166 in the stated units. At T=0.01, the leading term predicts $S-m\approx 1.66\times 10^{-4}$; this is an asymptotic estimate without a certified finite-temperature error. Rescaling k in $\int d^3k/(2\pi)^3[e^{(\beta S|k|^2)}-1]^{-1}$ explains the β^(−3/2) power.

Learning and research extensions For spin-transport research, useful directions are anisotropic D, magnetization laws and effective magnons. Transport coefficients still need dynamics and scattering theory; this equilibrium theorem does not directly provide conductivity or device performance.

What to audit Audit uniformity under extra pins, late returns in a repeated environment and passage of pressure-increment bounds through h↓0. The technical lemmas have not been independently verified here.

Source locators

sections/01-introduction.tex: Model and result; thm:main; How returns determine the coefficient

Original PDF

362-01 • Relativistic Vlasov-Maxwell: can particles and fields lose smoothness in finite time?

Manuscript claim The manuscript claims unique global classical solutions for arbitrary smooth compatible large data in the three-space/three-momentum-dimensional, one-species relativistic Vlasov-Maxwell system without background charge, retaining smoothness and compact particle phase-space support on every finite interval.

Essential hypotheses Assumptions are $f_0 \geq 0$ with $f_0 \in C^\infty c(\mathbb{R}^6)$, $E_0, B_0 \in C^\infty b \cap L^2$ with all spatial derivatives bounded, div $E_0 = \int f_0 dv$ and div $B_0 = 0$. Nonzero total charge and a Coulomb tail are allowed; all field derivatives need not be square-integrable.

Why it matters Global weak existence, local classical existence and unique global smooth evolution are distinct. The velocity $u(v) = v/\sqrt{(1+|v|^2)}$ is subluminal, but momentum may grow without bound; bounded speed alone does not prove regularity.

Proposed proof roadmap The central estimate controls signed impulse $\int (E + u \times B) dt$ along characteristics before taking absolute values. Source intersections with backward light cones are grouped by momentum, distance and angles. Near-collinear retarded-force cancellation and integration by parts are combined with direction-change and occupation bounds. Lower bounds on momentum-doubling times have divergent sum, preventing finite-time momentum blow-up and enabling continuation.

A first worked example The teaching model is force-free transport $f(t,x,v) = f_0(x - tu(v), v)$, verified by direct differentiation. If $f_0 \neq 0$, setting $E = B = 0$ in the coupled system generally violates div $E = \rho$. Thus free transport is a decoupled exercise, not a nonzero-density exact solution of the charged system.

Learning and research extensions Study transport, Lorentz force and Maxwell constraints, then light-cone energy, bootstrap and continuation. Possible applications include charge-conserving particle-in-cell analysis. Regularity does not supply uniform-in-time bounds, scattering or fusion-device performance.

What to audit The critical checks are absorption of the receiver-acceleration term after signed cancellation and whether improved occupation bounds are established before they are used. The full PDE proof has not been independently reproduced.

Source locators

sections/introduction.tex: Equations eq:rvm, eq:data; thm:global; Proof strategy

Original PDF

102-01 • Unique Games and the limits of approximation

Manuscript claim The manuscript claims that for every fixed $\varepsilon,\delta\in(0,1/2)$, a deterministic polynomial-time reduction maps 3SAT to Unique Games over a fixed finite alphabet, with value$\geq 1-\varepsilon$ in the satisfiable case and value$\leq\delta$ otherwise. The graph is simple bipartite and constraints are translations over $\mathcal{F}_2^s$.

Essential hypotheses The two errors are fixed before choosing the alphabet; alphabet size and polynomial degree may depend on them. Constraints are explicit and unweighted. NP-hardness is not a proof that P≠NP.

Why it matters Many semidefinite approximation thresholds were conditional on Unique Games. If correct, this result removes that hypothesis through existing reductions. Worst-case hardness does not rule out effective algorithms for structured physical or industrial instances.

Proposed proof roadmap Starting from nearly satisfiable three-variable parity, a nonlinear latent-noise gadget keeps valid encodings stable while high-rank linear observations detect the noise. High test acceptance forces low-rank Fourier mass; the matrix-shortcode inverse theorem extracts local answers. Sparse questions and clean coordinates lead to parallel repetition, contradicting low agreement for NO instances.

A first worked example For binary variables, $x_2=x_1$, $x_3=x_2$ and $x_1=x_3+1$ mod 2 cannot all hold: summation gives 0=1, so at most 2/3 are satisfied. This illustrates permutation constraints, not the paper's simple bipartite reduction. Compare a triangle's Max-Cut value 2 to distinguish constraint value from approximation ratio.

Learning and research extensions Study SAT, graph cuts and SDP, then PCP, Fourier analysis and reduction quantifiers. These ideas guide choices between exact algorithms, approximation and structural assumptions; they do not yield a one-click solver for NP-hard problems.

What to audit Audit simultaneous near-one completeness and arbitrarily small soundness, information available to each decoder, conditional independence of clean questions and final output size. An integrality gap does not replace this audit.

Source locators

sections/00-introduction.tex: Theorem thm:ugc; Proofs, codes, and the quantifiers; The construction and proof

Original PDF

159-01 • Long arithmetic progressions inside sparse integer sets

Manuscript claim For every fixed k≥3, the manuscript claims $r_k(N) \leq Ck\ N \exp[-ck(\log N)^{\varepsilon k}]$, where $r_k(N)$ is the largest progression-free subset of [N]. It follows that every positive-integer set with divergent reciprocal sum contains arithmetic progressions of every finite length.

Essential hypotheses The common difference is positive, and constants are fixed only after k is fixed. This is not a single uniform bound for all k. The claimed advance is a summable bound at every fixed length, not an improvement of the best three-term exponent.

Why it matters Positive-density Szemerédi requires only $r_k(N)=o(N)$, whereas divergent reciprocal sums allow zero-density sets. A stronger quantitative decay is needed. The proposed argument controls the number of density increments, coordinates and precision losses together.

Proposed proof roadmap Absence of progressions forces higher-order structure, detected through inverse theorems and nilsequences, leading to density increments. Triangular polynomial cells preserve previous integer constraints while descending precision budgets control new losses. Total logarithmic length loss remains polynomial; density cannot increase indefinitely, giving the bound.

A first worked example One consequence can be checked fully: if A has no k-term progression, then on $[2^m,2^{m+1})$, $\Sigma 1/a \leq 2^{(-m)} r_k(2^m) \leq C \exp[-c(m \log 2)^{\varepsilon}]$. Since $m^{\varepsilon}/\log m \to \infty$, the latter is eventually at most m^{-2}; summation is finite. The contrapositive gives the reciprocal-sum consequence, conditional on the main bound.

Learning and research extensions Study progression counting, Roth and density increments, then Gowers norms and nilmanifolds. Connections include pseudorandomness and pattern detection; large unspecified constants do not guarantee practical finite-data thresholds.

What to audit Audit preservation of old cell equations, dependence only on fixed k, absence of same-level circular precision bounds and the scope of external inverse theorems. The dyadic consequence is elementary; the main increment theorem is not re-proved here.

Source locators

sections/00-introduction.tex: Theorem ap:main; Corollary ap:reciprocal; Preserving structure through density increments

Original PDF

107-01 • Matrix multiplication exponent 9/4: theory versus finite-size performance

Manuscript claim The paper claims $\omega \le 9/4$ over the complex numbers: for every $\varepsilon>0$, $n \times n$ matrix multiplication is possible with $O\varepsilon(n^{(9/4+\varepsilon)})$ scalar arithmetic operations.

Essential hypotheses This is asymptotic arithmetic complexity as $n \to \infty$; the algorithm and constants may depend on ε. It provides no competitive finite-size crossover, GPU implementation, floating-point stability or bit-complexity guarantee.

Why it matters The standard triple loop costs n^3, while Strassen's seven block products give exponent $\log_2 7 \approx 2.807$. Further improvements use tensor degenerations and asymptotic extraction, so a lower exponent need not be faster on a laptop.

Proposed proof roadmap A finite Fourier projection and weighted degeneration separate tensor blocks sharing one leg. Additive and multiplicative tensor characters measure polynomial multiplication; discrete concavity and threefold-scale growth imply $P(a,a) \ge a^{(4/3)}$. Comparing with an interpolation upper bound forces $t \le 3/4$ and hence $3t \le 9/4$.

A first worked example For $(a_0+a_1x)(b_0+b_1x)$, use $m_0=a_0b_0$, $m_2=a_1b_1$ and $m_1=(a_0+a_1)(b_0+b_1)$; the middle coefficient is $m_1-m_0-m_2$. Three multiplications replace four. This illustrates bilinear reorganization, not the claimed 9/4 algorithm.

Learning and research extensions Learn recurrence complexity and bilinear algorithms, then tensor rank, degeneration and asymptotic spectra. Applications need explicit finite constructions plus memory, communication and error benchmarks; a small matrix widget must not claim to implement the new exponent.

What to audit Audit full direct-sum separation, accounting for auxiliary dot-product costs, the detecting-character theorem and limit order. The tensor inequalities are not independently verified here.

Source locators

paper.tex: Introduction; thm:main; prop:separation; cor:entropy; Route to the exponent

Original PDF

113-01 • Perfect matchings: finding one versus approximately counting all

Manuscript claim The manuscript claims an FPRAS for perfect matchings of every finite simple undirected graph: relative error ε with success probability at least 1−δ, worst-case bit time polynomial in input length, 1/ε and log(1/δ), and exact zero output when no perfect matching exists.

Essential hypotheses The error and confidence parameters are rational inputs in the stated ranges. Relative differs from additive error, and approximation is not exact #P counting. Colored parallel edges occur internally and do not imply support for arbitrary compressed multiplicities.

Why it matters Existence of a perfect matching has a classical polynomial-time algorithm; counting all of them is much harder. Counting all matchings also differs from isolating perfect matchings, which may have tiny sampling mass.

Proposed proof roadmap Start with an easily counted complete graph, gradually suppress nonedges, estimate successive partition-function ratios and multiply them. Vertex scaling balances two-hole ratios. An enlarged graph adds controlled weak edges while keeping real matchings sufficiently probable. Sampling uses product matching spaces and alternating-cycle exchanges; replication and orthogonal decomposition lift a pair-energy comparison to a full spectral gap.

A first worked example K_4 has three perfect matchings: {12,34}, {13,24} and {14,23}; the cycle C_4 has two. If Z=3 and ε=0.1, an acceptable output lies in [2.7,3.3], with failure probability at most δ. The output need not be an integer.

Learning and research extensions Study matchings, partition functions and Monte Carlo, then mixing, conductance and annealing. Applications may involve pairing entropy and sampling, but the paper chooses constants for proof transparency rather than practical speed.

What to audit Audit sampling laws, worst-case runtime even after inaccurate balance estimates, accumulated relative error and failure probability, and the extension from additive pair functions to arbitrary functions.

Source locators

main.tex: Introduction; thm:main; Proof architecture; table tab:graph-laws

Original PDF

158-01 • Unit-distance coloring of the plane: why this is not the four-color theorem

Manuscript claim The paper claims that every five-coloring of the plane contains a monochromatic pair at distance exactly one, yielding $6 \leq \chi(\mathbb{R}^2) \leq 7$ with the classical seven-color construction. It does not decide between six and seven.

Essential hypotheses Color classes are arbitrary, without measurability or boundary regularity, in ZFC. Unit-distance edges can cross, so this is not the planar-map four-color problem.

Why it matters Finite unit-distance graphs provide lower bounds, but failure to find an obstruction does not establish a five-coloring of the entire plane. Arbitrary and measurable colorings differ substantially; the paper's key step is a bridge permitting analytic tools.

Proposed proof roadmap The proposed transfer turns an arbitrary proper coloring into a weak measurable coloring excluding monochromatic unit pairs for almost all positions and directions. Spectral decomposition and character-space rigidity remove discontinuous components. Palette transitions produce connected exclusion sets sharing a center; cycles on five labels have lengths 3, 4 or 5, each ruled out by angular, topological and Moser-configuration arguments.

A first worked example An equilateral unit triangle needs three colors, proving only $\chi \geq 3$; its coloring does not extend automatically to the plane. Exercise: explore finite point sets using a distance tolerance, but use exact distances or certified error for any proof.

Learning and research extensions Study graph coloring and finite configurations, then measure, harmonic analysis and group actions. This informs exclusion-distance design, but wireless interference usually concerns distance $\leq r$ rather than exactly r, so direct application is inappropriate.

What to audit Audit preservation of exclusions in the measurable transfer, null sets, rigidity and topological extraction. The finite obstruction is existential; no certified six-chromatic graph is generated here.

Source locators

sections/introduction.tex: thm:main; def:weak-coloring; thm:transfer; proof overview

Original PDF

179-01 • Circulant Hadamard matrices and Barker sequences: two autocorrelation boundaries

Manuscript claim The manuscript claims real circulant Hadamard matrices exist only at orders 1 and 4. Combined with prior odd-length results, it gives Barker-sequence lengths greater than one exactly 2, 3, 4, 5, 7, 11 and 13.

Essential hypotheses The matrix has ± 1 entries, $HH^T = nI$, and cyclic-shift rows. Periodic autocorrelation wraps around; Barker sequences use nonwrapping aperiodic $|C(t)| \leq 1$. The conclusion does not restrict all Hadamard matrices to orders 1 and 4.

Why it matters Sharp autocorrelation peaks aid synchronization and identification. Exact zero sidelobes are much stronger than small sidelobes. If correct, the result limits this special binary circulant design, not approximate or complex-valued codes.

Proposed proof roadmap Encode the first row as an integral group-ring element h with $hh^* = n$. First derive $n = 4u^2$ with u odd, then compare alternating products of character evaluations across prime directions. Local integrality and absolute values constrain these products to roots of unity. Modulo-two relations from the same sign coefficients give an incompatible alternating sum, forcing u=1.

A first worked example For h=(1,1,1,−1), P(1)=1+1−1−1=0, P(2)=1−1+1−1=0 and P(3)=0, giving a circulant Hadamard matrix of order four. Aperiodic correlations are C(1)=1, C(2)=0 and C(3)=−1, so it is also Barker. Every calculation is directly checkable.

Learning and research extensions Study correlation and the DFT, then cyclic difference sets, group rings and cyclotomic integers. Build code-search or pulse-compression demonstrations; longer designs may permit small sidelobes, noncirculant structure or nonbinary alphabets.

What to audit Audit divisibility in local descent, common coefficient origins of character values and careful modulo-two treatment of complex roots of unity. Verification of the small example does not prove exclusion at every order.

Source locators

sections/introduction.tex: thm:main; cor:barker; Group-ring notation; Proof strategy

Original PDF

266-01 • Mutually unbiased bases in dimension six: from numerical search to certificate

Manuscript claim Under its stated arithmetic/compiler contract and complete execution record, the paper claims there are no four pairwise mutually unbiased orthonormal bases in $\mathbb{C}^6$; an explicit construction of three yields N(6)=3.

Essential hypotheses Mutual unbiasedness means every cross-basis squared overlap is exactly 1/6. Exclusion covers arbitrary second bases, not just the Fourier family. The result is computer-assisted and needs rounding margins, exhaustive coverage and no unresolved records.

Why it matters A state definite in one basis has uniform measurement outcomes in an unbiased basis. Prime-power dimensions admit complete d+1 constructions; six is the first non-prime-power dimension, so those finite-field constructions do not transfer directly.

Proposed proof roadmap Fix the first basis to the standard basis and write the second as H/$\sqrt{}$6 with H complex Hadamard. Phase seeds, U(2) completions and certified tangent/normal bounds cover every H. Candidate unbiased vectors form graphs of possible orthogonality and possible unbiasedness. Two additional bases require two six-cliques and 36 cross-edges; the search excludes this pattern in every region.

A first worked example In $\mathbb{C}^2$ the Z, X and Y bases have squared cross overlaps 1/2. Choose three unbiased bases in $\mathbb{C}^3$ with squared overlaps 1/3. Tensor corresponding bases to obtain three in $\mathbb{C}^6$, since (1/2)(1/3)=1/6. This establishes the lower bound only, not exclusion of a fourth.

Learning and research extensions Study the Born rule and Hadamard/Fourier matrices, then interval arithmetic and exhaustive search. Possible applications include measurement and tomography teaching; noise robustness and hardware realization require separate work.

What to audit The paper explicitly retains failed certifications or exhausted search caps as unresolved regions, never as exclusions. This deliverable does not rerun the complete computation and must not label N(6)=3 as locally verified.

Source locators

sections/introduction.tex: Computer-assisted thm:main; Covering and exclusion strategy; Elementary lower bound

Original PDF

268-01 • The spin-one Haldane gap: how finite certificates control long chains

Manuscript claim The paper claims a uniform positive gap for the pure bilinear antiferromagnetic spin-one Heisenberg chain on even periodic rings. For even $L \geq 2304$, $\gamma L > \log(20)/784 \approx 0.00382$, hence the same lower bound for $\liminf \gamma L$.

Essential hypotheses The model is $H = \Sigma_j S_j \cdot S_{j+1}$ with coupling one, local space $\mathbb{C}^3$ and even periodic length. The gap is to the next distinct energy; uniqueness and a smaller bound are also claimed for $L \geq 60$. An AKLT result with biquadratic terms is not this pure model.

Why it matters Physical and numerical evidence supports a gap around 0.4105. A much smaller rigorous bound of 0.00382 is meaningful because it prevents closure as $L \to \infty$. Positive gaps at individual finite lengths do not establish a uniform lower bound.

Proposed proof roadmap Two representations of the same partition function provide physical Boltzmann weights and even moments of a self-adjoint spatial transfer operator. Purity near one means one weight dominates; squaring and renormalizing quadratically reduces the defect. Coupled doubling of inverse temperature and length starts from finite thermal-trace and matrix-product certificates, covers intermediate lengths by moment inequalities, and converts weight ratios to a gap.

A first worked example For two spin-one particles, $S_1 \cdot S_2 = (J^2 - 4)/2$ gives energies -2, -1 and 1 for total spin $j = 0,1,2$. This nine-dimensional problem is directly diagonalizable. Neighboring bond terms in a chain do not commute and cannot minimize independently, so a local gap of one does not establish a chain gap of one.

Learning and research extensions Learn angular-momentum coupling and exact diagonalization, then MPS, transfer operators and certified thermal traces. Connections include low-energy spin-chain models and topological phases; indices and open-boundary edge states require their own assumptions.

What to audit Audit the integer/rational interval certificates and a bootstrap covering all lengths. A floating-point DMRG curve is not the proof's premise. The full finite certificates have not been re-executed here.

Source locators

introduction.tex: Equations eq:main-H, eq:main-gap; thm:main; How the proof works

Original PDF

265-01 • Two-dimensional entanglement area laws and what PEPS existence means

Manuscript claim The main paper claims $S\Omega(A) \leq C|\partial A|$ for every region A in a finite induced square-lattice domain with bounded finite-range interactions, finite local dimension and a unique globally gapped ground state. C depends only on local dimension, range, strength and gap.

Essential hypotheses The gap is a uniform bound for the full Hamiltonian, without assuming every restricted Hamiltonian is gapped. Range uses intrinsic induced-graph distance, including holes and disconnected domains. Uniqueness, locality and fixed gap are essential.

Why it matters General n-site q-level systems can have O(n log q) entropy. Boundary scaling constrains the information structure of local gapped ground states and motivates tensor-network representation, but does not automatically make finding or contracting the tensors efficient.

Proposed proof roadmap The global gap first gives a subvolume entropy exponent. Positive quasilocal operators annihilating the ground state are constructed. Symmetric-copy interpolations compare information gain with excitation cost while scanning collars and amplifying remote mutual-information decay. Two ordered families of separated regions then cancel interior entropy terms through chain rules, leaving boundary cost and summable errors.

A first worked example For a Bell pair $(|00\rangle+|11\rangle)/\sqrt{2}$, one side has reduced state I/2 and entropy ln2; $|00\rangle$ has entropy zero. If b independent Bell pairs cross a cut, entropy is b ln2, illustrating boundary-controlled entanglement. General interacting ground states are not independent Bell pairs, so the theorem requires much more.

Learning and research extensions A companion claims PEPS existence on open $L \times L$ nearest-neighbor grids with bond dimension $\leq CL^{c}$ and global vector error $\leq L^{-1}$, using additional compression and routing estimates. This concerns representational power, not polynomial-time construction, contraction or training.

What to audit Audit uniform collar/entropy constants for arbitrary regions, absence of an implicit restricted-region gap and taking typical-subspace limits after fixing geometry. The full area law has not been independently verified here.

Source locators

sections/00-introduction.tex: Theorem thm:area; A tensor-network consequence; How the proof is organized

Original PDF

Manuscript abstracts and original sources

Papers are ordered by the official disciplines and result families. Each has its own identifier so distinct hypotheses or conclusions are not conflated into one theorem.

Number theory

58 manuscripts • Official subject grouping

001-01 • Result family 001

Milne's rationality conjecture for abelian varieties

We prove Milne's rationality conjecture for abelian varieties, including residue characteristic 2. After good reduction, the pairing of a rational Hodge class with any complementary product of divisor classes on the reduction is the same rational number in every prime-to-p realization and in crystalline cohomology. Using the Hodge theorem for CM abelian varieties, we also show that every such specialized Hodge class is represented by a single rational algebraic cycle in all these realizations.

Original PDF | Source files and supporting material

002-01 • Result family 002

Exact Birch-Swinnerton-Dyer Formula from Low Selmer Corank

We prove the full Birch-Swinnerton-Dyer leading-term formula for every elliptic curve over $\mathbb{Q}$ whose full q-power Selmer group has corank zero or one at some prime q. The analytic and Mordell-Weil ranks equal that corank, and the Tate-Shafarevich group is finite. The formula includes all prime factors and requires no additional hypotheses on reduction, rational torsion, isogenies, complex multiplication, or residual Galois representations.

Original PDF | Source files and supporting material

002-02 • Result family 002

The Selmer converse for elliptic curves at every prime

We prove the Selmer converse in coranks zero and one for every elliptic curve over $\mathbb{Q}$ and every prime p: if the full p-power Selmer group has $\mathbb{Z}$p-corank $r \in \{0, 1\}$, then the analytic and Mordell-Weil ranks both equal r, and the entire Tate-Shafarevich group is finite. As an application at the additive prime 3, we prove that for every prime $\ell \equiv 4, 7, 8 \pmod 9$, the cubic $X^3 + Y^3 = \ell Z^3$ has analytic and Mordell-Weil rank one and finite Tate-Shafarevich group. In particular, every such ℓ is a sum of two rational cubes.

Original PDF | Source files and supporting material

002-03 • Result family 002

The two-primary Birch-Swinnerton-Dyer formula in Selmer corank at most one

We prove the two-primary Birch and Swinnerton-Dyer leading-term formula for every elliptic curve over the rationals whose two-power Selmer group has corank at most one. In this range, the algebraic rank, analytic rank, and Selmer corank are equal, and the Tate-Shafarevich group is finite. Combined with the quadratic-twist Selmer distribution, this gives the exact two-primary formula for a density-one set of signed squarefree twists of each fixed curve, ordered by absolute value; the common rank is zero or one, with each value having density one half.

Original PDF | Source files and supporting material

003-01 • Result family 003

The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane $\Re s > 7/8$

We prove that all finite-order Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$ and all Dirichlet L-functions are zero-free in the half-plane $\Re s > 7/8$, with the principal pole at s = 1 allowed. In particular, the Riemann zeta function is zero-free in this half-plane, proving the quasi-Riemann hypothesis.

Original PDF | Source files and supporting material

003-02 • Result family 003

The Quasi-Riemann Hypothesis (alternate 11/12 proof)

We establish the quasi-Riemann hypothesis by proving that every Dirichlet L-function, including Riemann's zeta function, has no zeros in the half-plane Re $s > 11/12$. More generally, we prove the same zero-free half-plane for every finite-order Hecke L-function over $K = \mathbb{Q}(\sqrt{-3})$. In particular, this rules out the existence of Landau-Siegel zeros.

Original PDF | Source files and supporting material

003-03 • Result family 003

Uniform exclusion of Landau-Siegel zeros

We prove the uniform exclusion of Landau-Siegel zeros. There is an absolute constant c > 0 such that every real zero $\beta \in (0,1)$ of every primitive nonprincipal real Dirichlet L-function of conductor q $\geq$ 3 satisfies $(1-\beta)\log q \geq c$.

Original PDF | Source files and supporting material

004-01 • Result family 004

Hilbert’s tenth problem over the rational numbers

We give a negative answer to Hilbert's tenth problem over the rational numbers: no algorithm decides whether a polynomial with integer coefficients has a rational zero. The number of variables is part of the input.

Original PDF | Source files and supporting material

004-02 • Result family 004

A pointwise 2-converse for elliptic curves with rational two-torsion

We prove a pointwise 2-converse for elliptic curves over **Q** with nonzero rational two-torsion: if the 2^∞-Selmer corank is zero or one, then the analytic rank and Mordell-Weil rank equal that corank, and the Shafarevich-Tate group is finite. The result allows arbitrary reduction at 2.

Original PDF | Source files and supporting material

005-01 • Result family 005

Catalan's constant is irrational

We prove that Catalan's constant $G = \sum_{j\geq 0}(-1)^j/(2j+1)^2$ is irrational.

Original PDF | Source files and supporting material

006-01 • Result family 006

Goldfeld's analytic density conjecture and the 2-converse for elliptic curves

We prove Goldfeld's analytic density conjecture: for every elliptic curve E over $\mathbb{Q}$, the quadratic twists of E with analytic rank zero and one each have density 1/2 among signed squarefree twist parameters ordered by absolute value. We also prove the low-corank 2-converse: if the 2^∞-Selmer corank of E is zero or one, then it equals the analytic and Mordell-Weil ranks, and the Tate-Shafarevich group is finite.

Original PDF | Source files and supporting material

006-02 • Result family 006

The mean analytic rank of quadratic twists of elliptic curves

For every elliptic curve over $\mathbb{Q}$, we prove that the average analytic rank of its quadratic twists tends to 1/2 when signed squarefree twist parameters are ordered by absolute value. This resolves Goldfeld's mean analytic-rank conjecture in this counting convention.

Original PDF | Source files and supporting material

007-01 • Result family 007

Ordinary two-point correlations of multiplicative functions

We prove the ordinary two-point Chowla conjecture. For every fixed pair of nonproportional affine forms, the Liouville correlation has a power-of-logarithm saving at every cutoff, with an absolute exponent. We also prove the binary corrected Elliott conjecture for ordinary averages of complex multiplicative functions of modulus at most one, under uniform nonpretentiousness of at least one original factor. This qualitative conclusion holds in fixed residue classes and for fixed nonproportional affine forms.

Original PDF | Source files and supporting material

008-01 • Result family 008

The Deligne-Drinfeld conjecture

We prove the Deligne-Drinfeld conjecture: the rational Grothendieck-Teichmüller Lie algebra is freely generated by one element in every odd weight at least three, with the corresponding isomorphism after weight completion.

Original PDF | Source files and supporting material

009-01 • Result family 009

Reconstruction of Function Fields from Mod-ℓ Milnor K-Theory

We prove a mod-ℓ Bogomolov-Pop reconstruction theorem for function fields of transcendence degree at least two over arbitrary algebraically closed fields of characteristic different from ℓ. The groups K_1^{M}/ℓ and K_2^{M}/ℓ, together with their full bilinear product, determine the perfect closure and its constant field. Every compatible isomorphism of these data is induced by a field isomorphism up to a single scalar in $\mathbb{F}_\ell^\times$, with only Frobenius ambiguity in the field isomorphism in positive characteristic.

Original PDF | Source files and supporting material

009-02 • Result family 009

Reconstruction from Milnor K-theory modulo the characteristic

Let K and L be finitely generated extensions of transcendence degree at least two over algebraically closed fields of characteristic p. We prove that every isomorphism of their first Milnor K-groups modulo p preserving the degree-two Steinberg relations is a nonzero scalar multiple of the map induced by a unique field isomorphism. The proof recovers projective lines over the subfields of pth powers by an elementary calculation with derivations.

Original PDF | Source files and supporting material

009-03 • Result family 009

The Bogomolov-Pop reconstruction theorem

We prove the Bogomolov-Pop reconstruction conjecture for function fields of transcendence degree at least two over arbitrary algebraically closed fields of characteristic different from ℓ. The pro-ℓ abelian-by-central datum determines the perfect closure and its constant field, with precisely the Frobenius and ℓ-adic unit ambiguities.

Original PDF | Source files and supporting material

010-01 • Result family 010

Unrestricted pro-modularity at the prime two

Every continuous, odd, absolutely irreducible two-dimensional 2-adic representation of $G_{\mathbb{Q}}$ that is unramified outside finitely many finite primes occurs in the full completed Hecke algebra at some odd tame level. The level may contain auxiliary tame primes, and scalar and reducible residual representations are included. This is a completed-Hecke occurrence result, with no de Rham hypothesis; it does not assert classical modularity.

Original PDF | Source files and supporting material

010-02 • Result family 010

The Dimension of the Two-Adic Hecke Algebra at Odd Level

For every odd positive integer N, every irreducible component of the full two-adic Hecke algebra of level $\Gamma_1(N)$ has Krull dimension four. This proves the p = 2 case of Emerton's dimension conjecture, including all residual components.

Original PDF | Source files and supporting material

010-03 • Result family 010

Fontaine-Mazur modularity at the prime 2

We prove that every continuous, irreducible, odd two-dimensional 2-adic representation of $G_{\mathbb{Q}}$, unramified outside finitely many primes and de Rham at 2 with distinct Hodge-Tate weights, is modular up to Tate twist. This resolves the odd, regular two-dimensional Fontaine-Mazur conjecture over $\mathbb{Q}$ at 2, including all residual representations.

Original PDF | Source files and supporting material

011-01 • Result family 011

Weighted dilation graphs, smooth shifted primes and totient fibers

We prove Erdős's conjecture on the largest fibers of Euler's totient function: for every ε > 0, infinitely many positive integers n have more than $n^{1-\varepsilon}$ preimages. We also show that, for every fixed δ > 0, there are at least $x^{1-o(1)}$ primes p in $2x < p \leq 5x$ whose predecessors have no prime factor exceeding xδ.

Original PDF | Source files and supporting material

011-02 • Result family 011

The Poisson-Dirichlet law for prime predecessors

For a prime p chosen uniformly from $3 \leq p \leq x$, list the prime factors of $p-1$ in decreasing order, with multiplicity. As $x \to \infty$, their logarithms, divided by $\log(p-1)$, converge in every finite joint distribution to the Poisson-Dirichlet distribution with parameter one. This proves the conjecture of Ford, Konyagin and Luca.

Original PDF | Source files and supporting material

011-03 • Result family 011

Prime Predecessors with an Even Number of Prime Factors

We prove that there are infinitely many primes p for which $p-1$ is squarefree and has an even number of prime factors. Equivalently, there are infinitely many primes p with $\mu(p-1) = 1$.

Original PDF | Source files and supporting material

012-01 • Result family 012

The joint Dickman law for consecutive integers

Let $P^+(n)$ denote the largest prime factor of n. We prove that $\log P^+(n)/\log n$ and $\log P^+(n+1)/\log n$ are asymptotically independent in ordinary natural density, with Dickman marginals. This resolves the Erdős-Pomerance joint Dickman conjecture positively and implies that the ordering $P^+(n) < P^+(n+1)$ has natural density 1/2.

Original PDF | Source files and supporting material

013-01 • Result family 013

The additive indecomposability of the primes

We prove Ostmann's inverse Goldbach conjecture: no set differing from the primes by finitely many elements can be written as $A + B$, where A and B are sets of nonnegative integers with at least two elements each.

Original PDF | Source files and supporting material

014-01 • Result family 014

Global Arthur Enhancements of Cuspidal Excursion Parameters

Under the finite-level Ramanujan-Arthur decomposition stated in Theorem 1.1, we construct a global Arthur enhancement for every occurring cuspidal excursion parameter at a specified full finite level for a split connected semisimple group over a global function field. A single algebraic SL2 and a commuting Weil centralizer map recover the given parameter by diagonal specialization on the entire Weil group, including inertia. The result does not assert ellipticity, Arthur-packet classification, or a multiplicity formula.

Original PDF | Source files and supporting material

014-02 • Result family 014

Rationality of the Canonical Unramified Arthur Filtration

Under the characteristic hypotheses of restricted geometric Langlands theory, we prove that the canonical Arthur filtration on finitely supported unramified automorphic functions is defined over $\mathbb{Q}$ for split connected semisimple groups. The result includes the noncuspidal part and every closed invariant nilpotent support, establishing the rationality conjecture of Gaitsgory-Lafforgue-Raskin in this setting.

Original PDF | Source files and supporting material

014-03 • Result family 014

Ramanujan-Arthur Decompositions of Cuspidal Functions at Full Finite Level

We prove rational and $\overline{\mathbb{Q}}_\ell$ decompositions of cuspidal automorphic functions for split semisimple groups over global function fields into subspaces indexed by nilpotent orbits of the dual group. The decompositions hold at every full finite level, with arbitrary divisor multiplicities. Each summand is governed by the same orbit at every unramified place and every complex embedding. At trivial level this proves Conjectures 3.4.5 and 3.4.6 of Gaitsgory-Lafforgue-Raskin. For split connected adjoint absolutely simple groups, we also prove that a cuspidal automorphic representation with one generic unramified local component is tempered at every unramified place. Thus globally generic

cuspidal representations in this scope satisfy the unramified part of the generalized Ramanujan conjecture.

Original PDF | Source files and supporting material

014-04 • Result family 014

Temperedness at ramified places for globally generic exceptional groups

We prove the generalized Ramanujan conjecture for globally generic cuspidal automorphic representations of split connected adjoint exceptional groups over global function fields. Every local component is tempered, without restrictions on characteristic or ramification depth. The proof extends the unramified Ramanujan theorem of a companion paper to all ramified places.

Original PDF | Source files and supporting material

014-05 • Result family 014

The Restricted Geometric Langlands Equivalence in Positive Characteristic

We prove the $\overline{\mathbb{Q}}_\ell$-linear restricted geometric Langlands equivalence for smooth projective connected curves and connected reductive groups in characteristic p > 0, with $\ell \neq p$, in two regimes. Over $\overline{\mathbb{F}}_q$, we assume the four characteristic conditions of the restricted theory stated below. Over arbitrary algebraically closed fields, we assume these conditions and additionally that p is very good and does not divide the Weyl-group order. This proves Gaitsgory-Raskin's full-support conjecture [[12, Conjecture 1.3.10]](https://arxiv.org/abs/2508.02237v1) in these regimes.

Original PDF | Source files and supporting material

014-06 • Result family 014

Constructible tame Hecke eigensheaves in positive characteristic

We construct nonzero locally constructible perverse Hecke eigensheaves with Borel level at one marked point on a smooth projective curve of genus at least two over an algebraic closure of a finite field. The parameter is a Zariski-dense geometric ℓ-adic local system with tame regular-unipotent monodromy. The group is simple and simply connected and satisfies four explicit Lie-theoretic characteristic hypotheses. The eigensheaves have parabolic nilpotent singular support, and their eigenisomorphisms are compatible with tensor products, permutations, and fusion.

Original PDF | Source files and supporting material

014-07 • Result family 014

Tame Hecke Eigensheaves with Several Marked Points

For SLn in characteristic p > n, we construct nonzero locally constructible perverse Hecke eigensheaves with Borel level at two or more marked points on a smooth projective curve of genus at least two over an algebraic closure of a finite field. The parameter is a Zariski-dense geometric ℓ-adic PGLn-local system with unipotent tame monodromy; its nilpotent logarithms may have any Jordan type, including zero. The eigensheaves have parabolic nilpotent singular support, and their eigenisomorphisms retain the full tensor and fusion structure.

Original PDF | Source files and supporting material

014-08 • Result family 014

Frobenius Structures on Tame Hecke Eigensheaves

We construct nonzero perverse Weil Hecke eigensheaves for SLn with Borel level at one marked point, after a finite extension of the field of constants. The parameter is a geometrically dense arithmetic PGLn-local system with tame regular-unipotent monodromy on a once-punctured curve of genus at least two over a finite field of characteristic p > n. The full multi-leg eigenstructure is Frobenius compatible with the prescribed arithmetic eigenvalue, and the geometric sheaf has parabolic nilpotent singular support.

Original PDF | Source files and supporting material

015-01 • Result family 015

Equidistribution of Prime-Degree Torus Packets with Arbitrary Local Type

We prove the packet form of the higher-dimensional Duke equidistribution problem for totally real fields of any fixed prime degree at least five, allowing arbitrary local homothety types of full lattices. As the multiplier-order discriminant tends to infinity, the volume-weighted packet measures converge to Haar probability measure, with no escape of mass.

Original PDF | Source files and supporting material

015-02 • Result family 015

Equidistribution of primitive quartic torus packets for arbitrary orders

Let K range over totally real quartic fields with no proper intermediate field, and let $\mathcal{O}$ be any order in K. We prove that the packets of periodic diagonal orbits attached to invertible $\mathcal{O}$-ideal classes, weighted by orbit volume, equidistribute with no escape of mass as $|\mathrm{Disc}(\mathcal{O})|$ tends to infinity. The proof combines measure rigidity with a cubic-resolvent estimate that remains uniform at primes dividing the order index.

Original PDF | Source files and supporting material

015-03 • Result family 015

Equidistribution of Primitive Sextic Torus Packets

We prove that the complete ideal-class packets of maximal orders in totally real sextic fields with no proper intermediate fields become equidistributed, with no escape of mass, in the space of unimodular lattices as their field discriminants tend to infinity. The limit is Haar probability measure. Each packet includes all coordinate-sign translates and is weighted by diagonal-orbit volume.

Original PDF | Source files and supporting material

016-01 • Result family 016

The abelian Zilber-Pink conjecture

We prove the abelian Zilber-Pink conjecture over $\overline{\mathbb{Q}}$. Every irreducible subvariety of an abelian variety has only finitely many maximal atypical subvarieties, where atypicality is measured inside its smallest containing torsion coset.

Original PDF | Source files and supporting material

016-02 • Result family 016

The E×CM component of Zilber-Pink for curves in A2

We prove the $E \times$ CM component of Zilber-Pink for Hodge-generic algebraic curves in $\mathcal{A}_2$ over $\overline{\mathbb{Q}}$. Each such curve contains only finitely many points whose abelian surface is isogenous to a product of elliptic curves with at least one factor having complex multiplication.

Original PDF | Source files and supporting material

016-03 • Result family 016

Quaternionic division points on curves in the Siegel threefold

We prove the quaternionic-division component of Zilber-Pink for curves in $\mathcal{A}_2$ over $\overline{\mathbb{Q}}$. A Hodge-generic algebraic curve contains only finitely many points whose full geometric rational endomorphism algebra is an indefinite quaternion division algebra over $\mathbb{Q}$. No boundary or reduction hypothesis is required.

Original PDF | Source files and supporting material

016-04 • Result family 016

Elliptic squares and Zilber-Pink for curves in A2

We prove that every Hodge-generic algebraic curve in $\mathcal{A}_2$ over $\overline{\mathbb{Q}}$ contains only finitely many points whose abelian surface is isogenous to the square of an elliptic curve without complex multiplication (CM). Combining this result with the companion $E \times$ CM and quaternionic-division finiteness theorems, we prove the curve case of Zilber-Pink in $\mathcal{A}_2$ over $\overline{\mathbb{Q}}$, without a boundary hypothesis.

Original PDF | Source files and supporting material

017-01 • Result family 017

The irrationality exponent of pi is 2

We prove the conjecture that the irrationality exponent of π is 2. As a consequence, the classical Flint-Hills series $\sum_{n\geq 1} 1/(n^3 \sin^2 n)$ converges, with angles in radians.

Original PDF | Source files and supporting material

018-01 • Result family 018

The Margulis-Platonov conjecture over global function fields

We prove the Margulis-Platonov conjecture over every global function field, including characteristic two. For an absolutely almost simple simply connected algebraic group, every noncentral abstract normal subgroup of its rational points is the inverse image of an open normal subgroup of the finite product of its anisotropic local groups.

Original PDF | Source files and supporting material

018-02 • Result family 018

The Margulis-Platonov conjecture over number fields

We prove the Margulis-Platonov conjecture over number fields. For an absolutely almost simple simply connected group, every noncentral abstract normal subgroup of its rational points is the inverse image of an open normal subgroup of the product of its anisotropic nonarchimedean local groups.

Original PDF | Source files and supporting material

019-01 • Result family 019

Étale covers with a prescribed exterior sheet

Let X be a smooth proper hyperbolic curve over an algebraic closure of a p-adic local field. Given finitely many disjoint small open disks, we construct one connected finite étale cover with a sheet isomorphic to their entire exterior and with degree divisible by p on every connected component above each disk.

Original PDF | Source files and supporting material

019-02 • Result family 019

The p-adic section conjecture

We prove the local p-adic section conjecture for smooth proper geometrically connected curves of genus at least two, over every finite extension of $\mathbb{Q}$p. Combined with established finite-descent theorems, this also yields the global section conjecture for smooth proper geometrically connected curves of genus at least two over number fields when their finite-cover descent locus equals their rational points; this includes $X_0(N)$ and $X_1(N)$ of genus at least two over $\mathbb{Q}$.

Original PDF | Source files and supporting material

020-01 • Result family 020

Squarefree values of quartics and power-free values of polynomials

We prove that every irreducible integer quartic with no fixed prime-square divisor takes squarefree values with the predicted positive Euler-product density. More generally, we obtain the corresponding $(d-2)$-power-free density in degrees $4 \le d \le 8$. The proof combines number-field factorization, determinant estimates with adaptive auxiliary primes, and explicit low-degree geometry. Together with Browning's theorem for higher degrees, this gives the $(d-2)$-power-free density for every d $\geq$ 4.

Original PDF | Source files and supporting material

021-01 • Result family 021

A quadratic bound for Jacobsthal's function

Let $h(k)$ be the least integer such that every interval of $h(k)$ consecutive integers contains an integer coprime to any prescribed positive integer having at most k distinct prime divisors. We prove $h(k) \ll k^2/(\log\log(3k))^2$, giving an affirmative answer to Jacobsthal's quadratic-bound question.

Original PDF | Source files and supporting material

022-01 • Result family 022

The Weak Inhomogeneous Duffin-Schaeffer Conjecture

We prove the weak inhomogeneous Duffin-Schaeffer conjecture. For every fixed γ ∈ $\mathbb{R}$ and finite-valued $\psi : \mathbb{N} \to [0, \infty)$, divergence of $\sum_{q \geq 1} \phi(q)\psi(q)/q$ implies $\|qx - \gamma\| < \psi(q)$ for infinitely many q, for almost every x. Here ϕ is Euler's totient and $\| \cdot \|$ is distance to the nearest integer. Numerators need not be reduced, and the shift satisfies no Diophantine restriction.

Original PDF | Source files and supporting material

023-01 • Result family 023

An unconditional first moment for cubic Gauss sums

We prove Patterson's first-moment conjecture for normalized cubic Gauss sums over all primary Eisenstein primes, unconditionally. The sharp-cutoff main term is $(6/5)c_* X^{5/6}/\log X$, where $c_* = (2\pi)^{2/3}/(3\Gamma(2/3))$. For every fixed nonzero angular Fourier mode of the prime argument, we also prove cancellation at the first-moment scale.

Original PDF | Source files and supporting material

024-01 • Result family 024

An asymptotic formula for the number of totients

Let $V(x)$ count the distinct values of Euler's totient function up to x. We give an explicit asymptotic equivalent for $V(x)$. Its coefficient is a uniform limit of functions defined from finite arithmetic data. We also prove that $V(cx)/V(x) \to c$ as $x \to \infty$ for every fixed c > 0, answering a question of Erdős and Hall.

Original PDF | Source files and supporting material

025-01 • Result family 025

Short Egyptian fractions

We prove a conjecture of Erdős: for every sufficiently large integer b, every rational number a/b with $1 \leq a < b$ is a sum of $O(\log \log b)$ distinct positive unit fractions, with an absolute implied constant. This order is best possible when the numerator varies. We also show that both the number of expansions of 1 with exactly k distinct terms and the least integer at least 2 that never occurs as a denominator in such an expansion grow doubly exponentially in k: their double logarithms have order k.

Original PDF | Source files and supporting material

026-01 • Result family 026

Positive lower density of large prime gaps

For every fixed C > 0, we prove that a positive proportion of consecutive prime gaps exceed $C \log p$, where p is the smaller prime. The proportion is bounded below for every sufficiently large initial segment of the prime sequence, with a constant depending on C. It follows that the indices at which p_n/n increases have positive lower asymptotic density, answering a question of Erdős and Prachar.

Original PDF | Source files and supporting material

027-01 • Result family 027

Integral points on character varieties of curves

We prove potential Zariski density of integral points on SLr-character varieties of smooth complex curves in every rank. The result allows prescribed quasi-unipotent boundary monodromy, including exact nonsemisimple conjugacy classes, and holds on every component over the full ring of integers of one finite extension. Here integrality is measured in the ambient character variety, while the exact boundary conditions are imposed on the complex representation.

Original PDF | Source files and supporting material

028-01 • Result family 028

Bounded-Step Walks on Gaussian Primes

We prove the Gaussian moat conjecture: no infinite walk through distinct Gaussian primes can have bounded steps. More strongly, for each fixed finite step bound, the connected components of the Gaussian-prime graph have uniformly bounded size. This bound applies to every starting prime, including primes on the coordinate axes, and is nonexplicit. The proof constructs a finite periodic sieve obstruction using geometric sampling and information-theoretic estimates.

Original PDF | Source files and supporting material

029-01 • Result family 029

Primitive roots for every admissible integer base

We prove the infinitude assertion in Artin's primitive root conjecture: for every integer a that is neither −1 nor a square, there are at least $c_a x/(\log x)^2$ primes in $(x, 2x)$ with primitive root a, for some $c_a > 0$ and every sufficiently large x.

Original PDF | Source files and supporting material

029-02 • Result family 029

Simultaneous primitive roots: a conditional lower bound for prime bases

For every fixed finite set of distinct positive primes, we prove that at least $cx/(\log x)^2$ primes in $(x, 2x)$ have every member of the set as a primitive root, for some c > 0 and all sufficiently large x. The result assumes four explicitly stated analytic and sieve inputs from the companion paper on primitive roots for admissible integer bases.

Original PDF | Source files and supporting material

030-01 • Result family 030

Modularity of elliptic curves over imaginary quadratic fields

We prove the modularity conjecture for elliptic curves over imaginary quadratic fields: every such curve is modular, with matching local parameters at every place.

Original PDF | Source files and supporting material

031-01 • Result family 031

Open Homomorphisms of Global Solvably Closed Galois Groups

We prove Uchida's conjecture on open homomorphisms of Galois groups. Every continuous open homomorphism between Galois groups of solvably closed Galois extensions of number fields is induced by a unique equivariant field embedding in the opposite direction. No restriction on the kernel or additional compatibility hypothesis is required.

Original PDF | Source files and supporting material

Algebraic and complex geometry

86 manuscripts • Official subject grouping

032-01 • Result family 032

The rational Hodge conjecture for CM abelian varieties

We prove the rational Hodge conjecture for complex abelian varieties with complex multiplication: every rational Hodge class on such a variety is a rational linear combination of algebraic cycle classes. As consequences, we obtain the generalized Hodge conjecture for CM abelian varieties, the Tate conjecture for abelian varieties over finite fields, and the Hodge standard conjecture for abelian varieties in arbitrary characteristic.

Original PDF | Source files and supporting material

032-02 • Result family 032

Algebraic Kuga-Satake correspondences and Hodge conjectures on a K3 quadratic locus

We prove the rational Hodge and generalized Hodge conjectures in every cohomological degree of every self-power of a projective complex K3 surface whose transcendental quadratic space, with its cup-product form, admits a rational isometric embedding in $\mathbb{U}_{\mathbb{Q}}^{\oplus 2} \perp \langle -1 \rangle^4$. This includes every ample P-polarized K3 surface, including all Picard jumps, for $P = \mathbb{U} \oplus D_8(-1) \oplus D_4(-1)$. On this locus we construct algebraic correspondences inducing every prescribed standard even-Clifford Kuga-Satake tensor. More generally, for any projective complex K3 surface, algebraicity of one exact standard even-Clifford Kuga-Satake tensor implies both conjectures for every self-power.

Original PDF | Source files and supporting material

032-03 • Result family 032

Weil classes and Hodge classes on abelian powers

We prove the rational Hodge conjecture in every codimension on every self-power of a complex abelian sixfold with an imaginary-quadratic action and a compatible polarization whose rational homological Hermitian form is hyperbolic of signature $(3, 3)$. We also prove it in every codimension on every self-power of a complex abelian variety of dimension at most five admitting an imaginary-quadratic action. Both results include nonsimple varieties and special periods with additional endomorphisms. The proof uses the companion theorem on the rational Hodge conjecture for CM abelian varieties.

Original PDF | Source files and supporting material

032-04 • Result family 032

Abelian covers, Gale correspondences, and the Hodge conjecture for powers

We prove the rational Hodge conjecture on every self-power of the Jacobian at each tensor Hodge-generic point of the full marked variation of a connected abelian cover of curves. This holds in every base genus and for every compatible branching pattern. For any CM abelian variety, the conclusion also holds for every self-power of its product with the Jacobian, on the same Hodge-generic locus. We also prove the conjecture on every self-power of a very general member of the full smooth

labelled family of diagonal complete intersections cut out by at most two equations of a common degree, in every dimension and every degree at least two.

Original PDF | Source files and supporting material

032-05 • Result family 032

A Conditional Reduction for Algebraic Kuga-Satake Correspondences

For a polarized K3 surface whose primitive cohomology has full orthogonal Hodge group, one algebraic correspondence inducing a nonzero map from that cohomology to the second cohomology of an abelian variety suffices to recover the prescribed full Kuga-Satake correspondence. We give an equivalent condition using holomorphic one-forms on a generically finite surface cover. If this input holds very generally in a polarized component, the prescribed correspondence is algebraic throughout that component, for every choice of standard data on the transcendental part. The existence of the initial correspondence remains a hypothesis.

Original PDF | Source files and supporting material

033-01 • Result family 033

Orbifold and logarithmic Iitaka subadditivity

We prove Campana's orbifold Iitaka subadditivity conjecture for rational simple normal crossing boundaries on compact manifolds in Fujiki class $\mathcal{C}$, including coefficient one. Ordinary and logarithmic subadditivity follow.

Original PDF | Source files and supporting material

033-02 • Result family 033

Logarithmic Kodaira dimension and whole-fiber variation

We prove the logarithmic Iitaka-Viehweg inequality for projective surjective morphisms with connected fibers between smooth complex quasi-projective varieties whose base has nonnegative logarithmic Kodaira dimension. The variation measures the birational field of definition of the whole geometric generic fiber. This resolves Popa's logarithmic variation conjecture positively.

Original PDF | Source files and supporting material

033-03 • Result family 033

The reverse logarithmic Kodaira inequality and additivity

We prove the reverse logarithmic Kodaira inequality for a surjective connected-fiber morphism $f : (X, E) \to (Y, D)$ of smooth projective reduced simple-normal-crossing pairs, with $\mathrm{Supp}(f^*D) \subseteq \mathrm{Supp}\, E$, such that X and every boundary stratum are smooth over $Y \setminus \mathrm{Supp}\, D$. Together with logarithmic subadditivity, the inequality gives additivity, including both negative-infinity cases. This resolves Popa's logarithmic additivity conjecture positively in the projective reduced-SNC, stratum-smooth setting.

Original PDF | Source files and supporting material

033-04 • Result family 033

Projective Hodge lines and ordinary Iitaka subadditivity

We prove the ordinary Iitaka subadditivity conjecture for surjective projective morphisms with connected fibers between smooth connected projective varieties over algebraically closed fields of characteristic zero. If F is the geometric generic fiber of $f : X \to Z$, then $\kappa(X) \geq \kappa(F) + \kappa(Z)$.

Original PDF | Source files and supporting material

033-05 • Result family 033

B-semiampleness for compact log-smooth Kähler fibrations

We prove the compact log-smooth Kähler case of b-semiampleness. Let $f : Y \to X$ be a surjective holomorphic map with connected fibers between smooth compact connected Kähler manifolds, and let Δ be an effective rational divisor with simple normal crossing support and coefficients in $[0, 1]$, with $K_Y + \Delta \sim_{\mathbb{Q}} f^*L$ for $L \in \mathrm{Pic}(X)_{\mathbb{Q}}$. There is a smooth compact Kähler modification $S \to X$ for which the threshold-moduli line satisfies $M_{S_1} = \nu^* M_S$ in $\mathrm{Pic}(S_1)_{\mathbb{Q}}$ for every smooth compact Kähler modification $\nu : S_1 \to S$, and some positive multiple of MS is represented by a holomorphic line bundle generated by global sections. Horizontal components of coefficient one are allowed; neither projectivity nor a Campana orbifold Iitaka hypothesis is assumed.

Original PDF | Source files and supporting material

034-01 • Result family 034

Log abundance for compact Kähler spaces under logarithmic Iitaka subadditivity

Assume logarithmic Iitaka subadditivity for surjective morphisms with connected fibers between smooth projective complex varieties with compatible reduced simple normal crossing boundaries. We prove log abundance for normal irreducible compact Kähler spaces in every dimension: for a log canonical pair (X, Δ) with effective rational boundary and $K_X + \Delta$ $\mathbb{Q}$-Cartier, analytic nefness of $K_X + \Delta$ implies semiampleness.

Original PDF | Source files and supporting material

034-02 • Result family 034

Uniform indices for semi-log-canonical log Calabi-Yau pairs

We prove a uniform index theorem for connected projective semi-log-canonical log Calabi-Yau pairs in every fixed dimension at least four over an algebraically closed field of characteristic zero. For boundary coefficients in a fixed finite rational set, a single multiple of the log canonical divisor is Cartier and linearly trivial. The multiple depends only on the dimension and coefficient set, not on the number of irreducible components. Together with the established theorem in dimensions at most three, this resolves the finite-rational-coefficient semi-log-canonical index conjecture.

Original PDF | Source files and supporting material

034-03 • Result family 034

Conditional good minimal models for compact Kähler fourfolds

Assuming orbifold Iitaka subadditivity and abundance for nef fourfold adjoints of nonnegative Kodaira dimension, we prove the existence of good minimal models for globally strongly $\mathbb{Q}$-factorial compact Kähler klt fourfold pairs with effective rational boundary and analytically pseudo-effective actual $\mathbb{Q}$-Cartier adjoint. The ordinary fourfold minimal model program supplies the nef endpoint. We prove nonvanishing by fibration arguments and, in algebraic dimension zero, by singular metrics, holomorphic foliations, and extension from a reduced boundary. The projective abundance argument used in the proof is included in full.

Original PDF | Source files and supporting material

034-04 • Result family 034

Log abundance in characteristic zero

We prove the rational-boundary log abundance conjecture in every dimension over algebraically closed fields of characteristic zero: every nef $\mathbb{Q}$-Cartier log canonical divisor on a projective log canonical pair with effective rational boundary is semiample. Over $\mathbb{C}$, the proof also establishes canonical nonvanishing for smooth projective varieties in every dimension.

Original PDF | Source files and supporting material

034-05 • Result family 034

Minimal metrics and interior injectivity for nef adjoints

Let (H, Θ) be a projective complex klt pair with effective rational boundary and nef $\mathbb{Q}$-Cartier adjoint. On any projective log resolution, minimal semipositive metrics on the pulled-back adjoint exist and have zero Lelong numbers everywhere. On smooth projective complex varieties, we also prove an H1-injectivity theorem for rational interior boundaries with simple-normal-crossing support when both endpoint bundles carry zero-Lelong semipositive metrics.

Original PDF | Source files and supporting material

034-06 • Result family 034

Fourfold nonvanishing by minimal metrics and moving jets

We prove canonical nonvanishing for smooth connected projective complex fourfolds: if KX is pseudo-effective, then $H^0(X, mK_X) \neq 0$ for some positive integer m.

Original PDF | Source files and supporting material

034-07 • Result family 034

Lifting sections from the reduced support of an adjoint

For a projective $\mathbb{Q}$-factorial dlt pair with effective rational boundary over an algebraically closed field of characteristic zero, we prove that the adjoint has positive Iitaka dimension whenever a nonzero effective Cartier multiple is supported on the coefficient-one boundary and restricts to a semiample line bundle on its whole reduced support. This gives log abundance after nonvanishing in dimension at most four over $\mathbb{C}$.

Original PDF | Source files and supporting material

034-08 • Result family 034

Schnell fiber spaces and good canonical models

We prove the Campana-Peternell inequality $\kappa(X) \geq \kappa(D)$ for a smooth connected projective complex variety X and an effective Cartier divisor D whenever $m_0 K_X - D$ is pseudo-effective for some positive integer m0. For an algebraic fiber space $f\colon X \to Y$ between smooth connected projective complex varieties, the hypothesis that $m_0 K_X - f^*H$ is pseudo-effective with H ample Cartier gives $\kappa(X) = \kappa(F) + \dim Y$ for a very general smooth fiber F, as well as nonzero sections of $mK_X - f^*H$ for all sufficiently large divisible m.

Original PDF | Source files and supporting material

034-09 • Result family 034

Uniform log Iitaka fibrations and bounded moduli denominators

For normal projective log canonical pairs over algebraically closed fields of characteristic zero, of fixed dimension d ≥ 5 and with effective boundary coefficients in a fixed finite rational set, we prove that one complete rounded pluricanonical system generates the full Iitaka field whenever the $\mathbb{Q}$-Cartier log canonical divisor has nonnegative Kodaira dimension. Its degree depends only on the dimension and coefficient set. For the paper's normalized canonical bundle formulae over $\mathbb{C}$, we also bound the Cartier denominators of the moduli divisors on smooth projective determining models in terms of the dimension and coefficient set.

Original PDF | Source files and supporting material

034-10 • Result family 034

Uniform Pluricanonical Iitaka Fibrations

We prove the effective Iitaka fibration conjecture in characteristic zero. For each dimension, one pluricanonical degree defines the Iitaka fibration of every smooth integral projective variety of that dimension and nonnegative Kodaira dimension over an algebraically closed field. The associated sections generate the full Iitaka function field.

Original PDF | Source files and supporting material

034-11 • Result family 034

Relative denominators and effective systems for log Calabi-Yau fibrations

We prove uniform denominator and effective-system bounds for log Calabi-Yau fibrations from projective log canonical complex pairs of dimension at most four onto positive-dimensional bases, with boundary coefficients in a fixed finite rational set. The bounds give a uniform trivializing degree and a uniform b-Cartier multiple of the moduli b-divisor. When the base divisor is big, a uniform complete rounded adjoint system has section ratios generating the full base function field.

Original PDF | Source files and supporting material

034-12 • Result family 034

Arithmetic Stein-degree bounds for log Calabi-Yau pairs

Fix d ≥ 1 and t > 0. Let (X, B) be an ordinary projective log canonical $\mathbb{Q}$-pair of dimension d over a characteristic-zero field k, with X normal and integral, $H^0(X, \mathcal{O}_X) = k$, B effective, and $K_X + B \sim_{\mathbb{Q}} 0$. We prove that every prime component S of B with coefficient at least t satisfies $[k_S : k] \leq N(d, t)$,

where kS is the relative algebraic closure of k in $k(S)$. This also bounds the Stein degree of S over k, proving the contraction-to-a-point formulation of Birkar's Stein-degree conjecture for ordinary $\mathbb{Q}$-pairs.

Original PDF | Source files and supporting material

034-13 • Result family 034

Uniform effective log Iitaka fibrations for fourfolds

We prove the finite-rational-coefficient case of the effective log Iitaka conjecture in dimension four. For normal projective log canonical complex fourfolds with boundary coefficients in a fixed finite rational set and pseudo-effective rational Cartier adjoint, one uniform degree makes the complete rounded reflexive system nonempty and its section ratios generate the full Iitaka field. We also prove a uniform canonical index bound for projective klt complex fourfolds with rationally trivial canonical divisor.

Original PDF | Source files and supporting material

034-14 • Result family 034

Abundance after nonvanishing for compact Kähler fourfolds

We prove semiampleness of the actual $\mathbb{Q}$-Cartier adjoint $K_X + \Delta$ of a normal connected compact Kähler klt fourfold whenever it is analytically nef and some positive Cartier multiple has a nonzero section. The boundary is effective and rational; the fourfold need not be projective or $\mathbb{Q}$-factorial. In Iitaka dimension zero, a positive Cartier multiple is the trivial holomorphic line bundle.

Original PDF | Source files and supporting material

035-01 • Result family 035

Log abundance in numerical dimension one for threefolds in positive characteristic

We prove the numerical-dimension-one case of log abundance for threefolds over algebraically closed fields of characteristic p > 3. If (X, B) is a projective log canonical threefold pair with effective rational boundary, and $K_X + B$ is $\mathbb{Q}$-Cartier, nef, and of numerical dimension one, then $K_X + B$ is semiample. Neither terminality nor $\mathbb{Q}$-factoriality of X is required.

Original PDF | Source files and supporting material

035-02 • Result family 035

Abundance in numerical dimension one for terminal threefolds in positive characteristic

Let X be a projective $\mathbb{Q}$-factorial terminal threefold over an algebraically closed field of characteristic p > 3. We prove the numerical-dimension-one case of abundance: if KX is nef with $\nu(K_X) = 1$, then KX is semiample and $\kappa(X, K_X) = 1$.

Original PDF | Source files and supporting material

036-01 • Result family 036

Numerical semiampleness of nef adjoint classes on compact Kähler manifolds

Let X be a smooth connected compact Kähler manifold, let B be an effective rational simple normal crossing divisor with coefficients less than one, and let M be a nef rational holomorphic line bundle on X. If $K_X + B$ is pseudo-effective and $K_X + B + M$ is nef, we prove that its first Chern class in real Bott-Chern cohomology is represented by a semiample rational line bundle. This numerical statement allows a flat change of line bundle; it does not assert semiampleness of the original adjoint.

Original PDF | Source files and supporting material

036-02 • Result family 036

Numerical Semiampleness of Nef Adjoint Divisors

We prove the Generalised Abundance Conjecture: if (X, B) is a projective klt $\mathbb{Q}$-pair over an algebraically closed field of characteristic zero, $K_X + B$ is pseudo-effective, M is a nef $\mathbb{Q}$-Cartier divisor on X, and $K_X + B + M$ is nef, then $K_X + B + M$ is numerically equivalent to a semiample $\mathbb{Q}$-Cartier divisor on X.

Original PDF | Source files and supporting material

036-03 • Result family 036

Minimal models and Mori fibre spaces for generalized log canonical Q-pairs

We resolve the existence form of the minimal-model conjecture for projective generalized log canonical $\mathbb{Q}$-pairs over algebraically closed fields of characteristic zero. Such a pair admits a minimal model when its adjoint divisor is pseudo-effective, and a Mori fibre space otherwise, while keeping the nef $\mathbb{Q}$-Cartier b-divisor fixed. Taking the nef part to be zero gives the corresponding result for ordinary log canonical $\mathbb{Q}$-pairs.

Original PDF | Source files and supporting material

036-04 • Result family 036

Minimal models in numerical dimension one

We resolve the numerical-dimension-one case of the minimal-model conjecture for smooth connected complex projective varieties of dimension at least three. If KX is pseudo-effective and $\kappa_\sigma(X, K_X) = 1$, with κσ defined by section growth with a fixed ample twist, then X admits a projective $\mathbb{Q}$-factorial terminal minimal model.

Original PDF | Source files and supporting material

037-01 • Result family 037

The ordinary-double-point gap in every dimension

We prove the ordinary-double-point gap conjecture for boundary-zero complex algebraic klt germs in every dimension: a singular n-dimensional germ has normalized volume at most $2(n-1)^n$, with equality precisely at an analytic ordinary double point.

Original PDF | Source files and supporting material

037-02 • Result family 037

The normalized-volume gap in dimension four

We prove that every singular complex algebraic klt fourfold germ with zero boundary has normalized volume at most 162, with equality precisely for an analytic ordinary double point. This resolves the ordinary-double-point volume-gap conjecture in dimension four.

Original PDF | Source files and supporting material

038-01 • Result family 038

Fujita's freeness conjecture

We prove Fujita's freeness conjecture. If X is a smooth complex projective variety of dimension n and L is an ample line bundle, then $K_X + mL$ is globally generated for every integer $m \geq n + 1$.

Original PDF | Source files and supporting material

039-01 • Result family 039

Nagata's conjecture for plane curves

We prove that a nonzero effective plane curve of degree d at r $\geq$ 10 very general complex points has total multiplicity strictly less than $d\sqrt{r}$. The inequality holds simultaneously for all curves, including reducible and nonreduced curves, and establishes Nagata's conjecture in its strict, nonhomogeneous form.

Original PDF | Source files and supporting material

039-02 • Result family 039

Maximal Seshadri constants on arbitrary polarized surfaces

We prove that, for every smooth integral complex projective surface S and every ample line bundle L, the multipoint Seshadri constant at r very general points equals $\sqrt{L^2/r}$ for every sufficiently large integer r. This resolves positively the qualitative Nagata-Biran conjecture for surfaces.

Original PDF | Source files and supporting material

039-03 • Result family 039

Maximal Multipoint Seshadri Constants in Higher Dimensions

Let L be an ample line bundle on a smooth integral complex projective variety X of dimension n $\geq$ 3. We prove that there is a threshold $r_0 = r_0(X, L)$ such that for every integer $r \geq r_0$, the ordinary multipoint Seshadri constant at r very general points equals the volume bound $(L^n/r)^{1/n}$. This establishes the qualitative Nagata-Biran-Szemberg assertion in these dimensions.

Original PDF | Source files and supporting material

039-04 • Result family 039

Maximal multipoint Seshadri constants in positive characteristic

Let L be an ample line bundle on a smooth integral projective variety X over an algebraically closed field of positive characteristic, with dim $X = n \geq 3$. We prove that there is a threshold $r_0 = r_0(X,$

$L)$ such that for every integer $r \geq r_0$, the ordinary multipoint Seshadri constant at the geometric generic tuple of r points equals the volume bound $(L^n/r)^{1/n}$. The same conclusion holds in dimension two. This establishes the positive-characteristic form of the qualitative Nagata-Biran-Szemberg assertion at geometric generic tuples.

Original PDF | Source files and supporting material

040-01 • Result family 040

Bloch’s conjecture for surfaces with p_g=0

We prove Bloch's conjecture for smooth connected projective complex surfaces with $p_g = 0$: the Albanese homomorphism on integral degree-zero zero-cycles is an isomorphism.

Original PDF | Source files and supporting material

041-01 • Result family 041

Projective-space bases of Lagrangian fibrations

We prove that the normal projective base of a projective Lagrangian fibration from a compact irreducible holomorphic symplectic Kähler manifold is projective space. This resolves the projective-space base conjecture for such fibrations in every dimension and deformation type.

Original PDF | Source files and supporting material

041-02 • Result family 041

The strong hyperkähler SYZ conjecture

We prove the strong hyperkähler SYZ conjecture: every holomorphic line bundle with nonzero nef isotropic first Chern class on a compact irreducible holomorphic symplectic Kähler manifold is semiample.

Original PDF | Source files and supporting material

042-01 • Result family 042

Every complex K3 surface is Oka

We prove that every complex K3 surface X is Oka, resolving the K3 Oka conjecture. Equivalently, for every m ≥ 1, every holomorphic map to X from a neighborhood of a compact convex set in $\mathbb{C}$m can be approximated uniformly on that set by entire maps $\mathbb{C}^m \to X$.

Original PDF | Source files and supporting material

043-01 • Result family 043

P=W in composite rank for fixed determinant

We prove the P = W conjecture on the full rational cohomology of fixed-determinant, trace-free Higgs moduli spaces in composite rank and coprime degree, for smooth projective complex curves of genus at least two. Together with the established prime-rank cases, this gives the equality in every coprime rank.

Original PDF | Source files and supporting material

044-01 • Result family 044

The equivariant cohomological Hikita conjecture for arbitrary quivers

We prove the equivariant cohomological Hikita conjecture for arbitrary finite quivers, including loops and multiple arrows. For any dimension and framing vectors, any commuting flavor torus, and a stability character whose semistable locus is stable and has free gauge action, the equivariant cohomology of the Nakajima variety is canonically isomorphic to the coordinate ring of the scheme-theoretic fixed locus of the stability cocharacter on the flavor-deformed Coulomb branch. This is an isomorphism of graded algebras over the common coefficient ring, retaining nilpotents and including the empty case.

Original PDF | Source files and supporting material

046-01 • Result family 046

A projective fourfold with large fundamental group and non-Stein universal cover

We construct a smooth projective complex fourfold with large fundamental group whose universal cover is not Stein. The universal cover contains no positive-dimensional compact complex-analytic subvariety, so this disproves Shafarevich's holomorphic-convexity conjecture even under the large-fundamental-group hypothesis.

Original PDF | Source files and supporting material

046-02 • Result family 046

A surface counterexample to Shafarevich holomorphic convexity

We construct a smooth connected projective complex surface whose universal cover is not holomorphically convex. This disproves the Shafarevich conjecture on holomorphic convexity in complex dimension two.

Original PDF | Source files and supporting material

047-01 • Result family 047

An explicit failure of complex affine-space cancellation

We construct an explicit integral complex affine fourfold X with $X \times \mathbb{A}^1 \cong \mathbb{A}^5$ but $X \ncong \mathbb{A}^4$. This gives a negative answer to Zariski's affine-space cancellation problem over $\mathbb{C}$ in dimension four. The same construction disproves the Stable Coordinate Conjecture in ambient dimension five and yields smooth $\mathbb{A}3$-fibrations over $\mathbb{A}1$ and $\mathbb{A}2$ that are not Zariski-locally trivial. These fibrations disprove the Dolgachev-Weisfeiler affine-fibration conjecture over these bases.

Original PDF | Source files and supporting material

048-01 • Result family 048

A singular normal affine surface with free tangent sheaf

We construct a singular normal affine complex surface whose tangent sheaf is free of rank two. This disproves the Lipman-Zariski conjecture in characteristic zero.

Original PDF | Source files and supporting material

049-01 • Result family 049

A stable coordinate that is not a coordinate in four variables

We construct an explicit degree-five polynomial over **C** that is not a coordinate in four variables but becomes one after adjoining a single variable. This gives a counterexample to the stable coordinate conjecture in four variables. Every fiber is isomorphic to affine three-space, yet its embedding in affine four-space is not rectifiable. Thus the example also disproves the Abhyankar-Sathaye embedding conjecture in ambient dimension four.

Original PDF | Source files and supporting material

049-02 • Result family 049

An explicit noncoordinate polynomial with affine three-space zero fibre

We construct an explicit counterexample to the Abhyankar-Sathaye conjecture: a noncoordinate polynomial in four complex variables whose zero fibre is affine three-space. Adjoining variables gives counterexamples in every ambient dimension at least four.

Original PDF | Source files and supporting material

050-01 • Result family 050

Ample rank-two bundles on the quadric surface without Griffiths-positive metrics

We give counterexamples to Griffiths' conjecture in rank two on the quadric surface $\mathbb{P}^1 \times \mathbb{P}^1$. We construct an explicit bundle G whose coordinatewise power pullbacks, tensored with $\mathcal{O}(1,1)$, are ample for every positive power, but admit no smooth strictly Griffiths-positive Hermitian metric for all sufficiently large powers.

Original PDF | Source files and supporting material

051-01 • Result family 051

Canonical ampleness of compact hyperbolic Kähler manifolds

We prove that every compact connected Kähler manifold of positive complex dimension containing no nonconstant entire curve has ample canonical bundle and is projective. This resolves positively Kobayashi's canonical-ampleness conjecture in the smooth compact Kähler category.

Original PDF | Source files and supporting material

052-01 • Result family 052

Universal-cover splitting for compact Kähler manifolds

We prove the two-summand form of Beauville's compatible splitting conjecture. If the tangent bundle of a compact connected Kähler manifold decomposes into two integrable holomorphic subbundles of positive rank, then its ordinary universal cover admits a product decomposition whose factor tangent bundles are the lifted specified summands.

Original PDF | Source files and supporting material

052-02 • Result family 052

Integrability of split tangent bundles on rationally connected manifolds

We prove that both summands of every specified holomorphic splitting of the tangent bundle of a smooth rationally connected projective complex manifold into two positive-rank subbundles are integrable. This proves Höring's conjecture on rationally connected projective manifolds. Höring's product theorem then gives a product decomposition compatible with the specified splitting.

Original PDF | Source files and supporting material

053-01 • Result family 053

A high-arity counterexample to Pixton completeness in Chow

We disprove the Chow and rational-cohomological forms of Pixton's completeness conjecture for his original relation system. We construct a formal tautological class outside the original Pixton relation span whose image is zero in the Chow ring and in rational cohomology. The example has genus 10^{60} and $3\binom{10^{60}}{3}$ markings.

Original PDF | Source files and supporting material

054-01 • Result family 054

Irrational cubic fourfolds with geometric K3 categories

For every sufficiently large admissible Hassett discriminant, we prove that a very general cubic fourfold of that discriminant is irrational, although its Kuznetsov component is equivalent to the ordinary derived category of a projective K3 surface. The discriminant threshold is ineffective. This disproves Kuznetsov's rationality conjecture. The same cubics have associated untwisted polarized K3 surfaces in the Hodge-theoretic sense, so they also disprove the sufficiency direction of the associated-K3 rationality prediction.

Original PDF | Source files and supporting material

055-01 • Result family 055

A Gepner stability condition on every smooth quintic threefold

We prove Toda's normalized quintic Gepner conjecture. On every smooth complex quintic threefold, we construct a numerical Bridgeland stability condition for which tensoring by the hyperplane bundle, followed by the spherical twist at the structure sheaf, increases phase by 2/5.

Original PDF | Source files and supporting material

055-02 • Result family 055

Prescribed large-volume charges on threefolds with trivial canonical bundle

Let X be a smooth projective complex threefold with trivial canonical bundle. We construct numerical Bridgeland stability conditions with the exact ordinary and square-root-Todd central charges at every sufficiently large volume. One volume threshold works on an open set of real twists and ample directions. The resulting stability conditions have the support property on the full numerical Grothendieck group and stable point sheaves. Separately, on every smooth projective complex threefold we prove a strong tilt inequality above a volume threshold uniform in the object and twist along a fixed polarization.

Original PDF | Source files and supporting material

056-01 • Result family 056

Termination of generalized log canonical flips on compact Kähler fourfolds

Every sequence of generalized log canonical flips on a normal irreducible globally Weil $\mathbb{Q}$-factorial compact Kähler fourfold terminates, provided the flips are projective small diagrams with the stated opposite ample signs. The boundary is rational, and the nef b-divisor is fixed and represented by an analytically nef $\mathbb{Q}$-Cartier divisor on a projective modification. No scaling rule or pseudo-effectivity assumption is required. The theorem concerns existing flip sequences; it does not assert the existence of all contractions or flips.

Original PDF | Source files and supporting material

056-02 • Result family 056

Termination of generalized-canonical flips on compact Kähler fourfolds

Every sequence of generalized-canonical flips on compact Kähler fourfolds with rational boundary and fixed rational analytically nef b-divisor terminates when the small morphisms are projective. We prove this for normal globally Weil $\mathbb{Q}$-factorial models with boundary coefficients less than one, allowing exceptional log discrepancies equal to one. No scaling rule or pseudo-effectivity assumption is required. The theorem applies to existing projective small diagrams with the specified opposite ample signs.

Original PDF | Source files and supporting material

056-03 • Result family 056

Termination for projective log canonical fourfolds with rational boundary

We prove that every permitted minimal model program for a projective log canonical fourfold with rational boundary over an algebraically closed field of characteristic zero terminates. The result allows arbitrary negative extremal rays and mixed birational steps on the given models, without pseudo-effectivity or initial $\mathbb{Q}$-factoriality.

Original PDF | Source files and supporting material

056-04 • Result family 056

Finite ordinary minimal model programs on compact Kähler fourfolds

We prove that every maximal ordinary negative-ray program starting from a compact Kähler klt fourfold pair with effective rational boundary in the global Weil-divisor $\mathbb{Q}$-factorial category terminates. It ends at a nef model when the adjoint is pseudo-effective and at a projective Mori fibre space otherwise.

Original PDF | Source files and supporting material

056-05 • Result family 056

Finite ordinary minimal model programs on compact Kähler fourfolds

We prove that a globally Weil-$\mathbb{Q}$-factorial compact Kähler klt fourfold pair with effective rational boundary and canonical rational line bundle admits a finite ordinary minimal model program starting on the given pair. It ends at a nef model when the adjoint class is pseudo-effective and at a Mori fibre space otherwise.

Original PDF | Source files and supporting material

057-01 • Result family 057

The abelianity conjecture for special compact Kähler manifolds

We prove that the fundamental group of every special compact Kähler manifold is virtually abelian, resolving Campana's abelianity conjecture in all dimensions. In particular, every smooth compact connected Kähler manifold of Kodaira dimension zero has virtually abelian ordinary fundamental group.

Original PDF | Source files and supporting material

057-02 • Result family 057

Two-step monodromy of special quasi-projective varieties

We give an independent proof that every complex linear representation of the ordinary fundamental group of a connected smooth special complex quasi-projective variety has virtually nilpotent image of class at most two. This conclusion was previously announced by Cao-Deng-Hacon-Păun. We also construct special open surfaces whose general quasi-Albanese fibres are not special.

Original PDF | Source files and supporting material

057-03 • Result family 057

A conditional abelianity theorem for special fourfold pairs with a half-weight divisor

Using the abelianity theorem for special compact Kähler manifolds in every dimension, we prove virtual abelianity of the entire orbifold fundamental group of the special order-two root orbifold associated with a pair $(X, \frac{1}{2}D)$, where X is a smooth projective fourfold and D is a nonempty smooth connected divisor. We construct a special smooth projective eightfold whose fundamental group surjects onto the required group.

Original PDF | Source files and supporting material

058-01 • Result family 058

Semialgebraic universal covers of normal projective varieties

We prove the Kollár-Pardon conjecture: the universal cover of a connected normal projective complex variety is biholomorphic to a semialgebraic open subset of a projective variety if and only if it is a product of a bounded symmetric domain, a complex affine space, and a simply connected normal projective variety.

Original PDF | Source files and supporting material

058-02 • Result family 058

Symmetry of semialgebraic bounded domains with compact quotient

Every nonempty connected semialgebraic bounded open subset of a complex affine variety admitting a properly discontinuous cocompact group of biholomorphisms is smooth and biholomorphic to a bounded symmetric domain. This answers the bounded-domain question of Kollár and Pardon affirmatively.

Original PDF | Source files and supporting material

059-01 • Result family 059

Ambiently homeomorphic isolated hypersurfaces of multiplicities two and three

We give a negative answer to the embedded Zariski multiplicity conjecture. We construct two reduced hypersurface germs that are ambiently homeomorphic but have multiplicities two and three. Both have isolated singularities and lie in a common complex affine space of dimension divisible by eight. Their defining function germs are also topologically right equivalent, and the same examples answer Arnold's corank problem negatively for ambient topological equivalence.

Original PDF | Source files and supporting material

059-02 • Result family 059

Ambiently homeomorphic isolated hypersurface germs in $\mathbb{C}^4$ with multiplicities four and five

We give a negative answer to Zariski's multiplicity question in four complex variables. We construct two reduced convergent holomorphic function germs with isolated critical points and multiplicities four and five whose zero-set germs are ambiently homeomorphic.

Original PDF | Source files and supporting material

060-01 • Result family 060

Global Spherical Shells on Minimal Surfaces of Class VII

We prove the Global Spherical Shell conjecture: every connected minimal compact complex surface of class VII with positive second Betti number contains a global spherical shell. This is a holomorphically embedded neighborhood of the standard three-sphere in $\mathbb{C}^2 \setminus \{0\}$ whose complement is connected.

Original PDF | Source files and supporting material

062-01 • Result family 062

Contact Fano manifolds and the LeBrun-Salamon conjecture

We resolve the contact-Fano homogeneity conjecture and the Riemannian LeBrun-Salamon conjecture positively. Every smooth connected complex projective contact Fano manifold of complex dimension at least three, with its given contact distribution, is contact-isomorphic to the adjoint variety of a simple complex Lie algebra. Consequently, every closed connected smooth positive quaternionic-Kähler manifold of real dimension $4m \geq 8$ is homothetic to a compact symmetric Wolf space.

Original PDF | Source files and supporting material

063-01 • Result family 063

The generalized Mukai conjecture

We prove the generalized Mukai conjecture: every positive-dimensional smooth complex Fano manifold of dimension n, Picard number ρ, and pseudoindex ι satisfies $\rho(\iota-1)\leq n$. Equality holds precisely for the product of ρ copies of $\mathbb{P}^{\iota-1}$.

Original PDF | Source files and supporting material

064-01 • Result family 064

Topological triviality of mu-constant families of surface singularities

We prove that every holomorphic one-parameter family of isolated hypersurface singularities in $\mathbb{C}3$ with constant Milnor number is topologically right-trivial. This gives a positive answer to the surface case of the μ-constant problem. The trivialization fixes the parameter and the origin section, and preserves the defining functions.

Original PDF | Source files and supporting material

065-01 • Result family 065

Virasoro Constraints under Projectivization

We prove that full ordinary descendant Virasoro constraints pass from a smooth projective complex base to the projectivization of any algebraic vector bundle of rank at least two. The bundle need not split and satisfies no positivity requirement. Assuming the full constraints on the base, the conclusion includes every genus, each individual integral curve class, and all cohomology insertions, including primitive and odd classes. The result also applies successively to towers of projective bundles.

Original PDF | Source files and supporting material

065-02 • Result family 065

Virasoro Constraints for Projective Complete Intersections

We prove the Virasoro conjecture for the ordinary descendant Gromov-Witten theory of smooth complete intersections in projective space, in every genus and curve class and with arbitrary cohomology insertions. This includes primitive and odd cohomology classes, with no semisimplicity assumption.

Original PDF | Source files and supporting material

066-01 • Result family 066

Bounded klt complements for Fano contractions

For fixed dimension d and positive rational ϵ, we prove that every ϵ-log-canonical Fano contraction over an algebraically closed field of characteristic zero admits, near each closed base point, a klt complement of index bounded only by d and ϵ. Over $\mathbb{C}$, we also obtain monotone klt complements for Fano type pairs with nef anti-log-canonical divisor and coefficients in a fixed finite ra-

tional set. This proves the finite-rational-coefficient form of Shokurov's bounded-klt-complement conjecture.

Original PDF | Source files and supporting material

066-02 • Result family 066

Uniform Cartier sections for Fano type contractions

We prove the Cartier-divisor conjecture of Birkar and Shokurov for rational boundaries in characteristic zero and for real boundaries over $\mathbb{C}$. For an ϵ-lc Fano type contraction with positive-dimensional base and nef negative log canonical class, a Cartier divisor through any prescribed base point can be chosen with pullback log canonical threshold bounded below in terms of the dimension and ϵ alone.

Original PDF | Source files and supporting material

067-01 • Result family 067

The Campana-Peternell conjecture in dimension six

We prove that every smooth connected complex projective Fano sixfold with nef tangent bundle is rational homogeneous, resolving the Campana-Peternell conjecture in complex dimension six. As a consequence, a connected compact Kähler manifold X with nef holomorphic tangent bundle and $\dim_{\mathbb{C}} X - \tilde{q}(X) \leq 6$ has ordinary universal cover $F \times \mathbb{C}^{\tilde{q}(X)}$, where F is a rational homogeneous manifold and $\tilde{q}(X)$ is the maximal irregularity of a connected finite étale cover.

Original PDF | Source files and supporting material

068-01 • Result family 068

Anticanonical nonvanishing from smooth semipositivity

Every smooth connected projective complex variety with smoothly semipositive anticanonical bundle has a nonzero section of some positive anticanonical power.

Original PDF | Source files and supporting material

068-02 • Result family 068

Invariant anticanonical indices and conversion of twisted differentials

For a holomorphic action of a compact torus on a compact complex manifold, we prove that the invariant Euler characteristic of naturally linearized anticanonical powers is polynomial on a divisible progression, with its actual value at exponent zero. Independently, on a smooth projective variety with smoothly semipositive anticanonical bundle, we remove a fixed pseudoeffective error from an unbounded sequence of effective twists. These results convert invariant cohomology into twisted differential forms and prove anticanonical nonvanishing on smooth projective varieties with smoothly semipositive anticanonical bundle, by descending the forms before conversion.

Original PDF | Source files and supporting material

068-03 • Result family 068

Bounded anticanonical metrics on klt pairs and torus quotients

Let (W, D) be a projective $\mathbb{Q}$-factorial klt pair with effective rational boundary. We prove that a positive Cartier multiple of $-(K_W + D)$ has a nonzero section if this divisor is nef, its pullback to a resolution admits a semipositive metric with locally bounded weights, and the resolution has nonzero structure-sheaf Euler characteristic. Consequently a smooth rationally connected projective variety with smoothly semipositive anticanonical bundle has a nonzero naturally invariant anticanonical plurisection for every algebraic torus action.

Original PDF | Source files and supporting material

068-04 • Result family 068

Cohomological transfer and equivariant anticanonical sections

Let X be a smooth projective complex variety with smoothly semipositive anticanonical bundle. For any torus linearization of that bundle, we show that invariant sections in unbounded degrees with one fixed negative pseudoeffective error yield an invariant section in a positive untwisted degree. Combining the natural invariant index with compact-monodromy structure, we deduce that every such X has a nonzero section of some positive anticanonical power.

Original PDF | Source files and supporting material

068-05 • Result family 068

Metric descent and rank-preserving contractions

From a smooth semipositive anticanonical metric on a smooth projective complex variety, we construct an unweighted integrable semipositive metric on a corrected anticanonical line of a smooth base, using a normal equidimensional toroidal model and full generic adjoint rank one at exponent zero. The explicit rational boundary correction pulls back to an exceptional divisor. On varieties without positive-degree holomorphic forms, two-metric transfer gives sections with prescribed boundary poles from a finite-volume log-anticanonical metric and a target line with bounded semipositive weights. In particular, a smooth projective complex variety with a smoothly semipositive anticanonical bundle and no such forms has a nonzero invariant section of a positive anticanonical multiple for every torus action with its natural linearization. On this class of varieties, generic section and adjoint-rank hypotheses give contractions of invariant fibrations that preserve both conditions unless an invariant global section already exists.

Original PDF | Source files and supporting material

068-06 • Result family 068

Integrable metrics and effectivity with controlled boundary

Let Y be smooth projective and let C be an effective integral divisor. If $-K_Y + C$ admits a metric with a global strict curvature lower bound for which the canonical section of C is locally square integrable, every pseudoeffective rational divisor becomes rationally effective after an effective correction supported on C. For a smooth projective X with smoothly semipositive $L = -K_X$, sections of $m_j L - P$ at unbounded positive exponents, with any fixed pseudoeffective Cartier error P, yield a section of a positive multiple of L.

Original PDF | Source files and supporting material

068-07 • Result family 068

Exact orders and invariant anticanonical linear systems

On a smooth connected projective complex variety with smoothly semipositive anticanonical bundle, we prove that asymptotically zero valuation on the curvature-null face is attained by an effective rational anticanonical divisor. The fixed auxiliary line bundle is arbitrary.

Original PDF | Source files and supporting material

069-01 • Result family 069

Global quantum geometric Langlands at irrational level

We prove the unramified quantum geometric Langlands equivalence for every connected simple complex algebraic group G and every level $c \in \mathbb{C} \setminus \mathbb{Q}$, including non-real levels. For every smooth projective connected complex curve X, it identifies the full twisted D-module categories on $\mathrm{Bun}_G(X)$ and $\mathrm{Bun}_{G^\vee}(X)$ at the dual shifted levels c and $-1/(rc)$, where r is the lacing number. The equivalence uses the given global forms and includes all connected components.

Original PDF | Source files and supporting material

Real and complex analysis

26 manuscripts • Official subject grouping

071-01 • Result family 071

Removable Boundaries and Rigidity of Circle Domains

We prove that a circle domain with conformally removable boundary is conformally rigid, with no restriction on the number of complementary components. This establishes the removability-to-rigidity direction of the He-Schramm Conjecture.

Original PDF | Source files and supporting material

071-02 • Result family 071

Koebe's Circle-Domain Conjecture

We prove that every domain in the Riemann sphere is conformally equivalent to a circle domain, resolving Koebe's circle-domain conjecture positively.

Original PDF | Source files and supporting material

072-01 • Result family 072

Brennan's conjecture and sharp inverse-square integral means

We prove Brennan's conjecture: if a simply connected plane domain admits a conformal bijection ϕ onto the unit disk, then $|\varphi'|^s$ is area-integrable for every $4/3 < s < 4$. We also prove the sharp inverse-square integral-means exponent $B_{\mathcal{S}}(-2) = 1$ for the normalized schlicht class $\mathcal{S}$.

Original PDF | Source files and supporting material

072-02 • Result family 072

A strict inverse-first-power bound for univalent functions

We prove a uniform upper bound for inverse-first-power integral means of normalized univalent disk maps with exponent strictly below 1/4. Consequently, the bounded universal integral-means spectrum satisfies $B_b(-1) < 1/4$, disproving Kraetzer's conjectured spectrum at p = −1.

Original PDF | Source files and supporting material

073-01 • Result family 073

The Falconer distance conjecture in all dimensions

We resolve the Falconer distance conjecture in every dimension. For every integer d ≥ 2, a compact subset of ℝd with Hausdorff dimension greater than $d/2$ determines a set of Euclidean distances of positive Lebesgue measure.

Original PDF | Source files and supporting material

074-01 • Result family 074

The Kakeya maximal conjecture in three dimensions

We prove the Kakeya maximal conjecture in three dimensions. For every ε > 0, the maximal average over unit tubes of radius δ maps $L^3(\mathbb{R}^3)$ to $L^3(S^2)$ with norm at most $C_\varepsilon \delta^{-\varepsilon}$.

Original PDF | Source files and supporting material

074-02 • Result family 074

Every four-dimensional Kakeya set has full Hausdorff dimension

We prove the four-dimensional Hausdorff-dimension Kakeya conjecture: every subset of ℝ4 containing a unit line segment in every direction has Hausdorff dimension four. No compactness or regularity assumption is imposed on the set or its witnessing line family.

Original PDF | Source files and supporting material

075-01 • Result family 075

Almost-everywhere Fourier convergence in L log L

We prove that the ordinary symmetric Fourier partial sums of every complex-valued function in $L \log L$ on the circle converge almost everywhere to the function along the full sequence. This establishes the classical $L \log L$ sufficiency conjecture.

Original PDF | Source files and supporting material

076-01 • Result family 076

Ultraflat real Littlewood polynomials

For every $\varepsilon \in (0,1)$ and every sufficiently large integer N, there is a polynomial of length N with coefficients in $\{-1,1\}$ whose modulus lies between $(1-\varepsilon)\sqrt{N}$ and $(1+\varepsilon)\sqrt{N}$ everywhere on the unit circle. Thus real Littlewood polynomials can be ultraflat through every sufficiently large integer length. The signs may be chosen separately at each length.

Original PDF | Source files and supporting material

076-02 • Result family 076

Nearly minimal maxima and positive minima of Littlewood polynomials

For every η > 0 and every sufficiently large integer N, there is a polynomial with N consecutive coefficients in $\{-1,1\}$ whose modulus lies between $\sqrt{N}/16$ and $(1+\eta)\sqrt{N}$ everywhere on the unit circle.

Original PDF | Source files and supporting material

076-03 • Result family 076

Asymptotically minimal maxima of real Littlewood polynomials

We prove that the minimum possible maximum modulus on the unit circle of a polynomial with N consecutive real coefficients in $\{-1,1\}$ is $(1+o(1))\sqrt{N}$, as N tends to infinity through all integers. This disproves the real-sign analogue of Erdős's fixed relative-gap conjecture. As a consequence,

the largest binary merit factor at length N tends to infinity through all integer lengths, disproving Turyn's conjecture.

Original PDF | Source files and supporting material

077-01 • Result family 077

Elliptic capacity propagation and Fourier restriction to the sphere

We prove the bounded-data Fourier restriction conjecture for the sphere in three dimensions: the Fourier extension operator maps $L^\infty(S^2)$ boundedly into $L^p(\mathbb{R}^3)$ for every p > 3. This is the full conjectured open range, and the threshold p = 3 is sharp.

Original PDF | Source files and supporting material

077-02 • Result family 077

Diagonal Fourier extension for positively curved surfaces in three dimensions

We prove the diagonal Fourier extension conjecture for compact smooth positively curved surfaces $\Sigma \subset \mathbb{R}^3$, including surfaces with smooth boundary. For every $3 < p < \infty$, the extension operator maps $L^p(\Sigma)$ boundedly into $L^p(\mathbb{R}^3)$. The compact paraboloid case also yields free Schrödinger local smoothing in two spatial dimensions for every $3 < p < \infty$ and Sobolev order $s > 2 - 6/p$.

Original PDF | Source files and supporting material

078-01 • Result family 078

Bochner-Riesz multipliers in three dimensions

We prove the three-dimensional Bochner-Riesz conjecture in its strict-order formulation. The main result is boundedness of the Bochner-Riesz multipliers on $L^3(\mathbb{R}^3)$ for every positive order. Interpolation and duality then give the full conjectured strict range.

Original PDF | Source files and supporting material

079-01 • Result family 079

Critical local smoothing for the three-dimensional wave equation

We prove the critical L3 local smoothing estimate for the wave equation in three spatial dimensions, with every positive Sobolev loss. This resolves Sogge's Euclidean local smoothing conjecture in dimension three.

Original PDF | Source files and supporting material

080-01 • Result family 080

Endpoint convergence for the planar Schrodinger equation

We resolve the planar Sobolev endpoint of Carleson's convergence problem: for initial data in $H^{1/3}(\mathbb{R}^2)$, the Schrödinger evolution converges almost everywhere to the initial data as $t \downarrow 0$. The evolution is defined by taking the Gaussian regularization limit first; on one full-measure spatial set, this limit exists for every $0 < t < 1$, and the resulting evolution is continuous at t = 0.

Original PDF | Source files and supporting material

080-02 • Result family 080

Endpoint pointwise convergence for the Schrodinger equation in higher dimensions

We resolve the Sobolev endpoint of Carleson's pointwise convergence problem in every dimension n ≥ 3. For initial data in $H^{n/(2(n+1))}(\mathbb{R}^n)$, the free Schrödinger evolution converges almost everywhere to the initial data as $t \downarrow 0$. The evolution is defined by removing Gaussian regularization on one full-measure spatial set, uniformly over the interval $0 < t < 1$.

Original PDF | Source files and supporting material

081-01 • Result family 081

Riesz transforms and uniform rectifiability in higher codimension

We prove that an n-Ahlfors-David regular Radon measure on ℝd is uniformly n-rectifiable if its n-dimensional Riesz transform is uniformly bounded from scalar $L^2(\mu)$ to vector-valued $L^2(\mu)$ over all positive hard truncations, for integers d ≥ 4 and $2 \leq n \leq d-2$. This gives a positive answer to the David-Semmes Riesz-transform question in its remaining higher-codimension range. The conclusion gives uniform big pieces of Lipschitz images of Euclidean balls.

Original PDF | Source files and supporting material

082-01 • Result family 082

Annular variation of the triangular Hilbert transform at the symmetric point

We prove the annular r-variation estimate for the triangular Hilbert transform from complex $L^3 \times L^3$ to L3/2 for every r > 2. The partitions may depend on the output point and range over all positive scales. The estimate yields the two-endpoint maximal bound and joint almost-everywhere and norm principal values, and resolves the symmetric scalar triangular Hilbert transform problem.

Original PDF | Source files and supporting material

082-02 • Result family 082

An L^3 bound for the dyadic triangular Hilbert form

We prove a uniform $L^3 \times L^3 \times L^3$ estimate for the dyadic triangular Hilbert form with unrestricted real inputs. The sum of the absolute local contributions over any finite set of scales is bounded by forty times the product of the input norms. In particular, the bound allows coefficients of modulus at most one to vary independently among admissible interval triples.

Original PDF | Source files and supporting material

082-03 • Result family 082

The maximal triangular Hilbert transform at the symmetric point

For arbitrary complex inputs in $L^3(\mathbb{R}^2)$, we prove the pointwise maximal $L^3 \times L^3 \to L^{3/2}$ estimate for the triangular Hilbert transform, with the supremum over both hard truncation endpoints. The estimate yields joint almost-everywhere and L3/2 convergence as the lower endpoint tends to zero and the upper endpoint tends to infinity. This also proves the conjectured scalar estimate at the symmetric point.

Original PDF | Source files and supporting material

083-01 • Result family 083

A uniform Hilbert transform estimate for Lipschitz directions

We prove a uniform L2 bound for the Hilbert transform along a Lipschitz unit vector field in the plane, with integration restricted to a fixed absolute multiple of the reciprocal Lipschitz seminorm. The bound is uniform in the inner truncation and holds for fields depending on both coordinates. This gives an affirmative answer to Stein's weak-$(2,2)$ conjecture.

Original PDF | Source files and supporting material

084-01 • Result family 084

The geometric case of the Erdős similarity conjecture

We prove the geometric-progression case of the Erdős similarity conjecture. For every fixed $q \in (0,1)$ and every $\eta \in (0,1)$, we construct a compact set $E_{q,\eta} \subseteq [0,1]$ of measure greater than $1-\eta$ containing no nontrivial affine copy of $\{q^n : n \geq 1\}$, for any translation and either sign of nonzero dilation. The set may depend on q; the result makes no simultaneous assertion for different ratios.

Original PDF | Source files and supporting material

084-02 • Result family 084

The dyadic case of the Erdős similarity conjecture

We construct a compact subset of the unit interval, of measure arbitrarily close to one, that contains no affine copy of the dyadic sequence $\{2^{-n} : n \geq 1\}$. The conclusion holds for every translation and every nonzero real dilation, of either sign, proving the dyadic case of the Erdős similarity conjecture.

Original PDF | Source files and supporting material

085-01 • Result family 085

An Endpoint Gradient Bound for the Centered Disk Maximal Operator

We prove the endpoint gradient bound $\|\nabla Mf\|_1 \leq C\|\nabla f\|_1$ for every real-valued $f \in W^{1,1}(\mathbb{R}^2)$, where $Mf(x)$ is the supremum of the averages of $|f|$ over disks centered at x, and C is an absolute constant. The maximal function belongs to $W^{1,1}_{\mathrm{loc}}(\mathbb{R}^2)$ and has a globally integrable weak gradient. This gives a positive resolution of the planar centered-disk case of the endpoint question of Hajłasz and Onninen.

Original PDF | Source files and supporting material

086-01 • Result family 086

An L3 bound for the trilinear Hilbert transform

We prove that the trilinear Hilbert transform with fixed slopes 1, 2, 3 maps $L^3(\mathbb{R}) \times L^3(\mathbb{R}) \times L^3(\mathbb{R})$ to $L^1(\mathbb{R})$, resolving this case of the trilinear Hilbert transform conjecture.

Original PDF | Source files and supporting material

Convex and metric geometry

30 manuscripts • Official subject grouping

087-01 • Result family 087

The symmetric Mahler conjecture and its equality cases

We resolve the symmetric Mahler conjecture positively, including its equality classification. Every origin-symmetric convex body in $\mathbb{R}$n has volume product at least $4^n/n!$, with equality exactly for invertible linear images of Hanner polytopes.

Original PDF | Source files and supporting material

087-02 • Result family 087

The Mahler Conjecture for General Convex Bodies

We resolve the Mahler conjecture for general convex bodies positively. For every convex body $K \subset \mathbb{R}^n$, n $\geq$ 1, with Santaló point $s(K)$, $|K|\,|(K - s(K))^\circ| \geq (n+1)^{n+1}/(n!)^2$, with equality exactly for simplices.

Original PDF | Source files and supporting material

087-03 • Result family 087

Symplectic Balls in Symmetric Polar Products

For every integer n $\geq$ 2 and every origin-symmetric convex body $K \subset \mathbb{R}^n$, we prove that the Gromov width of int $K \times$ int K° is 4. We construct smooth symplectic embeddings of standard balls of every capacity $0 < c < 4$ into this polar product. Volume preservation then resolves the symmetric Mahler conjecture positively in every dimension.

Original PDF | Source files and supporting material

088-01 • Result family 088

Petty’s projection-volume conjecture in dimensions at least four

We prove that ellipsoids uniquely minimize the volume of the projection body among convex bodies of fixed volume in every dimension at least four. This proves Petty's projection-volume conjecture in these dimensions.

Original PDF | Source files and supporting material

088-02 • Result family 088

A product counterexample to the simplex maximum for projection-body volume

The product of two ten-dimensional simplices has larger normalized projection-body volume than a twenty-dimensional simplex. This gives a counterexample to Brannen's proposed simplex maximum.

Original PDF | Source files and supporting material

089-01 • Result family 089

Planar Graph Metrics Embed into L1 with Constant Distortion

We prove that every finite connected planar graph with arbitrary positive real edge lengths embeds into real L1 with a universal distortion bound. This resolves the planar embedding conjecture positively.

Original PDF | Source files and supporting material

089-02 • Result family 089

L1 Embeddings of Graphs of Bounded Treewidth

For every fixed treewidth bound, the shortest-path metrics of finite connected graphs with arbitrary positive real edge lengths embed into real L1 with uniformly bounded distortion. This resolves the bounded-treewidth case of the Gupta-Newman-Rabinovich-Sinclair conjecture positively.

Original PDF | Source files and supporting material

090-01 • Result family 090

An atomic certificate for triangular-lattice universal optimality

We prove that the density-one triangular lattice minimizes the lower energy per particle for every nonnegative completely monotone function of squared distance, among all locally finite planar configurations of centered disk density one. The comparison includes infinite energies. The proof constructs sharp Gaussian Fourier minorants using an atomic interpolation certificate.

Original PDF | Source files and supporting material

090-02 • Result family 090

Universal optimality of the triangular lattice

We prove universal energy minimality for the triangular lattice in the plane. Among locally finite configurations of centered density one, it minimizes the lower limit of centered-ball energy averages for every nonnegative completely monotone function of squared distance, including when the energy is infinite. We also prove triangular minimality for planar logarithmic and Riesz renormalized energies, with $0 < s < 2$ in the Riesz case, and the corresponding jellium minima. The proof uses sharp Gaussian Fourier bounds, positive mixtures, and heat-kernel comparison.

Original PDF | Source files and supporting material

090-03 • Result family 090

A sharp Fourier certificate for planar circle packing

We resolve the planar Cohn-Elkies sharpness conjecture: the two-point Fourier bound attains the optimal circle-packing density $\pi/(2\sqrt{3})$. We construct a radial Schwartz certificate and prove its global sign conditions using rigorous interval arithmetic and analytic estimates. The same certificate recovers the classical uniqueness of the triangular packing among periodic equality cases.

Original PDF | Source files and supporting material

090-04 • Result family 090

Triangular minimality for planar Coulomb renormalized energy

We prove the Sandier-Serfaty conjecture: the triangular lattice of covolume one minimizes planar Coulomb renormalized energy over all admissible curl-free fields with a unit uniform background. Combined with Bétermin and Sandier's asymptotic formula, this also proves the Brauchart-Hardin-Saff conjecture for the linear term of optimal ordered-pair logarithmic energy on the unit two-sphere. The proof uses a direct Voronoi-cell comparison with rigorous interval arithmetic.

Original PDF | Source files and supporting material

091-01 • Result family 091

The logarithmic Brunn-Minkowski conjecture

We prove the logarithmic Brunn-Minkowski conjecture for arbitrary origin-symmetric convex bodies in every dimension. The theorem also gives the symmetric Lp Brunn-Minkowski inequality for every $0 < p < 1$. Combined with Saroglou's transfer theorem and a support-subspace reduction, it yields the logarithmic inequality for every even log-concave Radon measure and the scalar-dilation (B)-conjecture.

Original PDF | Source files and supporting material

092-01 • Result family 092

A single-lattice covering bound of order n log n

Every convex body in $\mathbb{R}$n, n $\geq$ 2, admits a covering by translates along one full-rank lattice with density at most $Cn \log n$, for an absolute constant C. No symmetry or boundary regularity is assumed.

Original PDF | Source files and supporting material

092-02 • Result family 092

Translative covering densities of order n log n

For every sufficiently large dimension n, we construct a centrally symmetric convex body whose translative covering density exceeds $cn \log n$, where c > 0 is absolute. This disproves the existence of a universal linear upper bound and matches the order of Rogers' upper bound.

Original PDF | Source files and supporting material

093-01 • Result family 093

A dimension-free logarithmic Sobolev inequality for subgaussian log-concave measures

We prove that every centered log-concave probability measure with a Lebesgue density on $\mathbb{R}$n and linear subgaussian parameter a satisfies $\mathrm{Ent}_\mu(f^2) \leq Ca^2 \int |Df|^2 \, d\mu$ for compactly supported smooth f, with one universal constant C. This resolves positively the dimension-free logarithmic Sobolev conjecture for subgaussian log-concave measures.

Original PDF | Source files and supporting material

094-01 • Result family 094

Subpolynomial dimension reduction in Lp

For every fixed $1 < p < \infty$ and D > 1, every n-point subset of a real Lp space embeds into ℓ_p^d with distortion at most D and subpolynomial dimension $d = n^{o(1)}$. The target has the same exponent p, and the embedding need not be linear.

Original PDF | Source files and supporting material

095-01 • Result family 095

Hyperbolicity Cones Without Semidefinite Lifts

We prove that not every hyperbolicity cone is a spectrahedral shadow: some closed hyperbolicity cones admit no finite affine semidefinite lift, even with arbitrary real coefficients and any finite number of auxiliary variables. This disproves the Projected Lax Conjecture and hence the generalized Lax conjecture.

Original PDF | Source files and supporting material

095-02 • Result family 095

A nonspectrahedral hyperbolicity cone

We construct a homogeneous polynomial of degree 16 in 23 real variables whose hyperbolicity cone has no representation by a finite homogeneous real symmetric linear matrix inequality. This disproves the geometric Generalized Lax conjecture.

Original PDF | Source files and supporting material

095-03 • Result family 095

An Exact Semidefinite Lift of a Nonspectrahedral Hyperbolicity Cone

We construct an exact semidefinite lift of the explicit nonspectrahedral hyperbolicity cone in twenty-three variables defined in the companion paper. The lift is a homogeneous real symmetric pencil of size 100 with 307 auxiliary variables and represents the entire closed cone, including every point with singular X. Thus, although this cone has no semidefinite representation in its original coordinates, it admits one when auxiliary variables are allowed. The stated sizes are not claimed to be minimal.

Original PDF | Source files and supporting material

096-01 • Result family 096

The Gaussian propeller bound in every dimension

We prove the Gaussian propeller conjecture: for every finite measurable partition of a Euclidean space, the sum of the squared lengths of its Gaussian first moments is at most $9/(8\pi)$. For dimension at least two and at least three cells, three planar sectors of angle $2\pi/3$, extended by an orthogonal Euclidean factor, attain the bound.

Original PDF | Source files and supporting material

097-01 • Result family 097

The Euclidean Steinitz-Bergström theorem

Every prescribed-order finite sequence of vectors in the Euclidean unit ball of $\mathbb{R}$d has one signing for which every signed prefix has norm at most $C\sqrt{d}$, with C absolute and independent of the sequence length. Consequently, every indexed zero-sum family of unit-ball vectors admits an ordering with the same bound for its unsigned partial sums. This determines the Euclidean Steinitz constant up to absolute factors, $S_2(d) = \Theta(\sqrt{d})$, and resolves the Euclidean Steinitz-Bergström conjecture.

Original PDF | Source files and supporting material

098-01 • Result family 098

A doubling Hilbert subset with no finite-dimensional bi-Lipschitz embedding

Every infinite-dimensional real Banach space contains a compact doubling subset that admits no bi-Lipschitz embedding into any finite-dimensional real normed space. The doubling constant has a universal bound, independent of the ambient Banach space. This answers the Lang-Plaut problem negatively, even for compact subsets of Hilbert space.

Original PDF | Source files and supporting material

099-01 • Result family 099

Edit Distance in l1: Matching Bounds up to Constants in the Exponent

We determine the exponential scale of the least $\ell 1$ distortion of unit-cost edit distance on all strings of length at most d. For every sufficiently large d, uniformly over finite alphabets of size at least two, the distortion lies between $\exp(c\sqrt{\log d \, \log\log d})$ and $\exp(C\sqrt{\log d \, \log\log d})$ for absolute constants $c, C > 0$. The lower bound already holds on binary strings of one common length. Thus the order of logarithmic distortion is sharp up to absolute constants.

Original PDF | Source files and supporting material

099-02 • Result family 099

Finite-Circle Obstructions, Binary Codes, and Histogram Embeddings for Edit Distance

We give two finite-circle constructions of binary strings whose least $\ell 1$ distortion is $\exp(\Omega(\sqrt{\log d \, \log\log d}))$, where d bounds their length. Both constructions supply words of one common length for every sufficiently large cap. Two direct binary coding arguments transfer the constructions with absolute distortion and logarithmic block width. We also develop the overlapping-substring method of Ostrovsky and Rabani into a complete finite histogram embedding at the same exponential scale, uniformly over all finite alphabets and all words of length at most d, including the empty word.

Original PDF | Source files and supporting material

099-03 • Result family 099

Tree Constructions for the l1 Distortion of Binary Edit Distance

We give two independent constructions of binary words of one length at most d whose ordinary edit-distance metrics require $\ell 1$ distortion $\exp(\Omega(\sqrt{\log d \, \log\log d}))$ for every sufficiently large d. We also prove a constant-distortion binary conversion for one prescribed input length. Together

with the companion upper embedding theorem, these lower bounds determine the order of logarithmic distortion uniformly over finite alphabets with at least two symbols.

Original PDF | Source files and supporting material

100-01 • Result family 100

Finite angular cylinder covers below the half-area bound

A regular tetrahedron admits a finite cylinder covering with compact triangular perpendicular bases whose total area is less than half its minimum orthogonal projection area. This disproves the half-area cylinder-covering conjecture. By affine invariance, the same construction gives a counterexample to the directionwise normalized half-bound for every nondegenerate tetrahedron.

Original PDF | Source files and supporting material

100-02 • Result family 100

Finite cylinder approximation of ruled sets

We approximate compact ruled families of segments by finitely many cylinders with square intercept tiles and perpendicular-base area at most their integral projection cost plus any positive error. The velocity field is C1, and its differential has opposite real eigenvalues whose magnitudes are strictly below the inverse segment half-length; both eigenvalues may vanish. The result includes square-zero differentials with unrestricted shear and fields that pass between the two regimes. It holds for arbitrary compact label sets.

Original PDF | Source files and supporting material

100-03 • Result family 100

Slope-field perturbations of the two-cylinder covering

The two-cylinder covering of a regular tetrahedron can be perturbed to give finite covers with total perpendicular base area strictly below half its minimum projection area. These covers give negative answers to both the half-area question and the directionwise normalized half-bound conjecture. By affine invariance, the directionwise conclusion holds for every nondegenerate tetrahedron.

Original PDF | Source files and supporting material

100-04 • Result family 100

Finite triangular approximation of radial sweeps

We prove that radially aligned segment sweeps admit finite cylinder covers with one triangular base for each interval of any tagged partition. As the mesh tends to zero, the total perpendicular base area converges to a weighted parameter area. An explicit application covers every regular tetrahedron with total base area below half its minimum projection area, giving negative answers to the half-area question and the directionwise normalized 1-Codimensional Cylinder Covering Conjecture.

Original PDF | Source files and supporting material

101-01 • Result family 101

A sharp entropy bound and the simplex inequality for isotropic constants

We prove the strong isotropic constant conjecture: in each dimension, simplices are the unique maximizers of the isotropic constant among convex bodies. We also prove the sharp entropy bound $h(f) \geq m + \frac{1}{2} \log \det \mathrm{Cov}(f)$ for every log-concave probability density on $\mathbb{R}$m, with equality precisely for invertible affine images of products of one-sided exponential laws.

Original PDF | Source files and supporting material

Theoretical computer science

73 manuscripts • Official subject grouping

102-01 • Result family 102

The Unique Games Theorem

We prove the Unique Games Conjecture. For every fixed $\varepsilon, \delta \in (0, 1/2)$, we give a deterministic polynomial-time reduction from 3SAT to Unique Games over a fixed finite alphabet, with completeness at least $1 - \varepsilon$ and soundness at most δ.

Original PDF | Source files and supporting material

102-02 • Result family 102

A Direct Proof of Optimal Max-Cut Hardness

We prove that approximating Max-Cut on simple unweighted graphs within any fixed factor greater than the Goemans-Williamson constant is NP-hard.

Original PDF | Source files and supporting material

102-03 • Result family 102

The Factor-Two Hardness Threshold for Vertex Cover

We prove that minimum Vertex Cover is NP-hard to approximate within every fixed factor below two, even on simple unweighted graphs.

Original PDF | Source files and supporting material

102-04 • Result family 102

Constant-factor hardness of Min-UnCut

For every fixed C > 1, approximating Min-UnCut within factor C is NP-hard, even on simple undirected unweighted graphs.

Original PDF | Source files and supporting material

102-05 • Result family 102

Constant-factor hardness of directed feedback vertex set

Approximating minimum directed feedback vertex set within any fixed constant factor is NP-hard, even on unweighted digraphs.

Original PDF | Source files and supporting material

103-01 • Result family 103

Exact derandomization of logarithmic space: L = RL = BPL

We prove L = RL = BPL, resolving the derandomization problem for polynomial-time randomized logarithmic space.

Original PDF | Source files and supporting material

104-01 • Result family 104

Turn-Based Stochastic Mean-Payoff Games in Deterministic Quasipolynomial Time

We give a uniform deterministic quasipolynomial-time algorithm for finite turn-based stochastic mean-payoff games with signed integer rewards and rational chance-transition probabilities encoded in binary. It computes exactly the vertices of nonnegative value, including value zero, for the expectation of the pathwise liminf mean payoff. The algorithm uses exact rational arithmetic and $2^{O((\log(L+2))^2)}$ bit operations, where L is the complete binary input length.

Original PDF | Source files and supporting material

104-02 • Result family 104

Mean-payoff parity games in quasipolynomial time

We give a uniform deterministic quasipolynomial-time algorithm for mean-payoff parity games. It computes all vertices from which a player can enforce both nonnegative liminf mean payoff and the parity condition, with arbitrary signed binary rewards and unrestricted binary priorities. The running time is $2^{O((\log(L+2))^2)}$ bit operations, where L is the complete input length.

Original PDF | Source files and supporting material

104-03 • Result family 104

Deterministic quasipolynomial-time mean-payoff games

We give a deterministic algorithm that computes the complete zero-threshold winning set of a finite mean-payoff game with arbitrary signed integer edge weights encoded in binary. For total explicit input length L, it uses $2^{O((\log(L+2))^2)}$ bit operations. A reduction also computes the exact rational value at every vertex and globally optimal positional strategies for both players within the same quasipolynomial bound.

Original PDF | Source files and supporting material

104-04 • Result family 104

Randomized quasipolynomial-time mean-payoff games

We give a randomized algorithm that computes the complete zero-threshold winning set of a finite mean-payoff game with arbitrary signed integer edge weights encoded in binary. For total explicit input length L, it uses $2^{O((\log(L+2))^2)}$ bit operations on every random tape and is correct with probability at least 7/8. A polynomial-time check certifies the winning regions and positional strategies for both players or reports failure. Independent repetition therefore gives an always-correct algorithm with the same expected quasipolynomial bit bound.

Original PDF | Source files and supporting material

105-01 • Result family 105

Perfect completeness for 2-to-1 games

We prove the 2-to-1 Games Conjecture with perfect completeness. For every fixed rational $\delta \in (0, 1)$, it is NP-hard to distinguish satisfiable 2-to-1 games from games of value at most δ, with a fixed alphabet and an explicitly listed unweighted multiset of constraints.

Original PDF | Source files and supporting material

106-01 • Result family 106

Hardness of finding large independent sets in three-colorable graphs

We prove that, for every fixed $0 < \delta < 1/3$, it is NP-hard to distinguish three-colorable graphs from graphs in which every independent set has fewer than δ times the number of vertices. Consequently, for every fixed integer c ≥ 3, finding a proper c-coloring of a three-colorable graph is NP-hard.

Original PDF | Source files and supporting material

107-01 • Result family 107

An Upper Bound of 9/4 for the Matrix Multiplication Exponent

We prove that the exponent of matrix multiplication over the complex numbers is at most 9/4.

Original PDF | Source files and supporting material

107-02 • Result family 107

Complex Matrix Multiplication Below 2.258 and Rectangular Bounds

Over every field of characteristic zero, we prove that the square matrix-multiplication exponent satisfies ω < 2.258, the dual exponent satisfies α > 0.465, and $\omega(1, 0.709, 1) < 2.092$. The strict square and k = 0.709 rectangular bounds also hold over every field except possibly in one finite set of positive characteristics, in the arithmetic-operation model.

Original PDF | Source files and supporting material

107-03 • Result family 107

Staggered extraction for exact matrix multiplication over every field

We prove that the arithmetic exponent of square matrix multiplication over every fixed field satisfies ω < 2.371054886006746. This includes every positive characteristic.

Original PDF | Source files and supporting material

108-01 • Result family 108

A cubic lower bound for border determinantal complexity of the permanent

We prove that the complex border determinantal complexity of the $m \times m$ permanent is $\Omega(m^3)$, allowing arbitrary affine-linear determinant representations and coefficientwise limits. It also gives cubic lower bounds for exact determinantal complexity and for the numbers of vertices and edges in affine-linear algebraic branching programs, including coefficientwise limits with a fixed vertex or edge budget.

Original PDF | Source files and supporting material

109-01 • Result family 109

Integer multiplication below n log n

We give a deterministic algorithm that multiplies two n-bit integers in $O(n(\lg n)^{1-\kappa})$ worst-case time, with $\kappa = 2^{-182}$, on one fixed finite-alphabet Turing machine with a fixed finite number of one-dimensional tapes. The algorithm is exact for every input length and disproves the $n \log n$ optimality conjecture of Schönhage and Strassen in this model.

Original PDF | Source files and supporting material

110-01 • Result family 110

Squared-logarithmic randomized k-server on arbitrary metrics

We prove that randomized k-server has competitive ratio $O((\log(k+1))^2)$ on every metric space against oblivious request sequences, matching the known worst-case lower bound. For each metric and initial configuration, one policy works for all finite request sequences, including on infinite and unbounded spaces. When the initial server positions are distinct, no additive term is needed.

Original PDF | Source files and supporting material

110-02 • Result family 110

Uniform computation of the squared-logarithmic k-server bound

We construct a uniform randomized k-server algorithm on finite rational metrics with competitive ratio $O(\log^2(k+1))$ against oblivious request sequences. Preprocessing is polynomial in the input length, and per-request bit complexity is polynomial in that length and the binary request-counter length. The additive movement constant is finite and instance-dependent, but may be enormous. The construction uses the companion squared-logarithmic existence theorem.

Original PDF | Source files and supporting material

111-01 • Result family 111

One Sample Suffices for Matroid Prophet Inequalities against an Almighty Adversary

We prove that one independent sample per element suffices for a constant-competitive prophet inequality on every finite matroid. The guarantee holds even when the arrival-order adversary observes all samples, all online values, and the algorithm's entire random seed. The rule needs no description of the value distributions and achieves the absolute competitive ratio 2^{-310}.

Original PDF | Source files and supporting material

112-01 • Result family 112

Beyond the Square-Root Exponent for Depth-Three Boolean Circuits

We construct a language in deterministic polynomial time whose n-bit membership function requires $2^{\omega(\sqrt{n})}$ gates in an unbounded-fan-in OR-AND-OR circuit. The bound holds at every sufficiently large input length and counts all gates, including the bottom layer.

Original PDF | Source files and supporting material

113-01 • Result family 113

A Fully Polynomial Randomized Approximation Scheme for Perfect Matchings in General Graphs

We give a fully polynomial randomized approximation scheme (FPRAS) for counting perfect matchings in arbitrary finite simple undirected graphs, resolving the general-graph perfect-matching approximation problem. The algorithm returns zero with certainty when no perfect matching exists. Otherwise, it achieves relative error ε with failure probability at most δ in worst-case bit time polynomial in the input length, ε^{-1}, and $\log \delta^{-1}$.

Original PDF | Source files and supporting material

113-02 • Result family 113

Entropy and Face Dimension of the Perfect-Matching Polytope

We prove the perfect-matching entropy conjecture of Anari, Oveis Gharan, and Vinzant. For every feasible vector x of perfect-matching edge marginals in a loopless labelled multigraph on $2m \geq 2$ vertices, the maximum entropy $H(x)$ of a matching law with marginals x satisfies $F(x) - (2 - 2/m)B(x) \leq H(x) \leq F(x)$, where $F(x) = -\sum_e x_e \log x_e$ and $B(x) = -\sum_e (1 - x_e) \log(1 - x_e)$. This bound holds throughout the polytope, including its boundary. We also prove the sharp bound $|\operatorname{supp} x| - \dim F_x \leq 3m - 2$, where Fx is the minimal face of the perfect-matching polytope containing x.

Original PDF | Source files and supporting material

114-01 • Result family 114

An FPRAS for Common Integer Polymatroid Bases with Binary Capacities

We give a fully polynomial randomized approximation scheme for counting common integer bases of two polymatroids with the same total rank, supplied by exact rank-value oracles. The total rank and capacities are encoded in binary, and each integer vector is counted once. On every execution, the number of oracle calls and the bit work outside the oracles are bounded by a fixed polynomial in the ground-set size, the binary input length, the inverse relative-error tolerance, and the logarithm of the inverse failure probability. The algorithm handles binary capacities directly, without expanding them into labelled copies.

Original PDF | Source files and supporting material

114-02 • Result family 114

Approximate counting of common bases of two matroids

We give a fully polynomial randomized approximation scheme for counting the common bases of two arbitrary matroids of the same rank, supplied by independence oracles. The algorithm requires no explicit representation of either matroid and has polynomial bounds on both oracle calls and bit operations on every execution.

Original PDF | Source files and supporting material

115-01 • Result family 115

Exact Uniform Sampling of Contingency Tables with Arbitrary Margins

We give an exact uniform sampler for nonnegative integer contingency tables with arbitrary prescribed margins. It terminates almost surely and has expected bit complexity polynomial in both dimensions and the binary length of the margins. No positivity, balance, sparsity, or fixed-dimension assumption is required.

Original PDF | Source files and supporting material

115-02 • Result family 115

An FPRAS for Cell-Bounded Contingency Tables

We give a fully polynomial randomized approximation scheme for counting nonnegative integer matrices with prescribed row sums, column sums, and individual entry bounds. Both dimensions vary, all numerical data are encoded in binary, and zero bounds are allowed. The algorithm uses only unbiased random bits and has a polynomial bound on its bit operations on every execution.

Original PDF | Source files and supporting material

116-01 • Result family 116

Uniform Matrix Hitting Points in Every Positive Characteristic

We construct a single matrix substitution that detects every nonzero size-s division-free noncommutative formula in n variables over every field of a given positive characteristic. One deterministic machine, given a promised prime p in binary and n, s in unary, outputs matrices over $\mathbb{F}$p of dimension $O(n^3 s^6)$ in polynomial bit time. The same tuple works with arbitrary extension-field coefficients, including in characteristic two. The construction also applies to the stated acyclic algebraic path programs.

Original PDF | Source files and supporting material

116-02 • Result family 116

One Rational Matrix Hitting Point for Noncommutative Formulas

We construct, in deterministic polynomial bit time, one tuple of rational matrices that detects every nonzero polynomial computed by a noncommutative division-free formula of a prescribed size. The matrices have dimension $O(ns^2)$ for n variables and formula size s, and the same tuple works over every field of characteristic zero.

Original PDF | Source files and supporting material

116-03 • Result family 116

Polynomial Hitting Lists for Noncommutative Rational Formulas

We construct, in deterministic polynomial bit time, a polynomial-size list of rational matrix tuples for noncommutative rational formulas over $\mathbb{Q}$ of bounded tree size. Every nonzero admissible formula has a defined, invertible value at one tuple, with no separate bounds on inverse nesting or rational constant heights. Matrix dimensions, entry bit lengths, and total output length are polynomially bounded.

Original PDF | Source files and supporting material

117-01 • Result family 117

Constant-factor hardness of uniform sparsest cut

We prove that, for every fixed C > 1, approximating Uniform Sparsest Cut within factor C is NP-hard. The output graphs have nonnegative rational capacities and unit demand between every pair of distinct vertices.

Original PDF | Source files and supporting material

117-02 • Result family 117

Near-square-root logarithmic integrality gaps for uniform sparsest cut

We construct uniform sparsest-cut instances whose Goemans-Linial semidefinite integrality gap is at least $c\sqrt{\log n}/(\log\log n)^3$ along a sequence $n \to \infty$. The demand is one between every pair of distinct vertices, and the capacities are nonnegative real numbers. This matches the Arora-Rao-Vazirani upper bound up to a power of $\log\log n$.

Original PDF | Source files and supporting material

118-01 • Result family 118

Additive hardness and unbounded configuration gaps in bin packing

We disprove the Modified Integer Round-Up Conjecture for bin packing by constructing instances with arbitrarily large additive gaps between the configuration-LP value and the integral optimum. We also prove that, for every fixed nonnegative integer c, distinguishing instances that fit in B bins from those requiring more than $B + c$ bins is NP-hard. Both results hold with rational item sizes greater than 1/6, so each bin contains at most five items.

Original PDF | Source files and supporting material

119-01 • Result family 119

Sharp binary-information contraction on the discrete cube

We prove sharp contraction of the information carried by a binary channel under independent symmetric noise on a uniform discrete cube. At fixed initial information, a noisy coordinate retains the most information. The Boolean specialization resolves the Courtade-Kumar conjecture and gives an output-entropy refinement. We also establish a stronger mean-dependent entropy-production bound. The proof combines an explicit three-point optimizer for the local joining problem, two entropy capacities, and a common-output thinning inequality, followed by dimension induction and integration along the noise semigroup.

Original PDF | Source files and supporting material

119-02 • Result family 119

Hellinger contraction with arbitrary Boolean output bias

We prove the Hellinger conjecture for Boolean functions on the uniform discrete cube, with arbitrary output bias. For a Boolean function of mean m and every $\rho \in [-1, 1]$, the loss $\sqrt{1-m^2} - \mathbb{E}\sqrt{1-(T_\rho f)^2}$ is at most $1 - \sqrt{1-\rho^2}$, with equality for signed coordinates. The proof combines asymmetric dimension induction, a calibrated noise-semigroup energy estimate, and finite ex-

act arithmetic certificates. The Hellinger inequality also yields the Courtade-Kumar information bound.

Original PDF | Source files and supporting material

120-01 • Result family 120

Almost-Linear-Time Maximum-Cardinality Matching in General Graphs

We prove that maximum-cardinality matching in a simple undirected graph with n vertices and m edges can be found by one uniform randomized algorithm in $(n + m)^{1+o(1)}$ time. The bound holds on every computation path in a logarithmic-word model, and the algorithm returns an explicit maximum matching with probability at least 2/3. An explicit reduction gives the same time and probability guarantees for deciding whether a simple host graph has a spanning subgraph with prescribed valid vertex degrees, and for finding one when it exists.

Original PDF | Source files and supporting material

121-01 • Result family 121

An Almost-Linear Approximation Scheme for Edit Distance

We give a uniform randomized approximation scheme for unit-cost edit distance. For every fixed rational $\varepsilon \in (0, 1)$, it estimates the distance between arbitrary explicitly stored strings of total length N within a factor $1 + \varepsilon$ with probability at least 2/3, in worst-case expected time $N^{1+o(1)}$ on a logarithmic-word RAM. The algorithm supports polynomially bounded integer alphabets and returns zero deterministically on equal strings.

Original PDF | Source files and supporting material

122-01 • Result family 122

Uniform quasipolynomial-time trace reconstruction

We give a uniform algorithm that reconstructs every binary string from independent deletion traces when its length and rational retention probability are known. For each fixed retention probability, both the number of traces and the bit complexity are quasipolynomial in the string length. More generally, we give an explicit sample bound uniform over all rational retention probabilities, with running time polynomial in the sample budget and the binary input length. If the deletion probability is at most $n^{-\varepsilon}$ for fixed ε > 0, the sample and running-time bounds are polynomial. Reconstruction succeeds with probability at least 2/3 for each input string.

Original PDF | Source files and supporting material

122-02 • Result family 122

A latest-anchor induction with spectrally compact masks for worst-case trace reconstruction

We give an improved worst-case sample bound for reconstructing a string from independent deletion traces, with its length and retention probability known. For each fixed retention probability, the number of traces is quasipolynomial: the logarithm of the sample budget is $O((\log n)^3(1 + \log\log(2n))^6)$. If the deletion probability is at most $n^{-\varepsilon}$ for fixed ε > 0, polynomially many traces suffice. These bounds apply to binary strings and to strings of general symbols observed exactly. They concern sample complexity and do not assert an efficient reconstruction algorithm or matching optimality.

Original PDF | Source files and supporting material

122-03 • Result family 122

Quantitative lower bounds for trace reconstruction

Exact worst-case reconstruction of a binary word from independent deletion traces requires $n^{\Omega(\log\log n)}$ samples for every fixed deletion probability $q \in (0,1)$, even with unrestricted computation and any fixed positive success probability. This gives a negative answer to the polynomial-sample question for binary trace reconstruction. More generally, when $q^3 \log n \to \infty$, we prove a lower bound of $n^{c\log(q^3\log n)}$ samples for every fixed $0 < c < 1/(4\log 2)$. Here q is the known deletion probability, and all logarithms are natural.

Original PDF | Source files and supporting material

124-01 • Result family 124

A Polynomial-Time Algorithm for Three-Machine Unit-Job Scheduling

We give a uniform deterministic polynomial-time algorithm for scheduling unit-length jobs with arbitrary precedence constraints on three identical parallel machines. The algorithm constructs a schedule of minimum makespan and decides exactly whether all jobs can finish by a specified deadline. The proof reorganizes feasible schedules into intervals whose job sets have descriptions of bounded size. A dynamic program searches a family containing polynomially many such descriptions. Global boundary conditions and simplification of inherited information keep the descriptions bounded throughout the decomposition.

Original PDF | Source files and supporting material

125-01 • Result family 125

Single-exponential recovery and bounded-price strictness for metric k-median

We give an exact-budget recovery algorithm for metric k-median with single-exponential dependence on the number of comparison clusters without accurate, distinct proxies in a supplied anchor solution. On positive integral metrics of polynomially bounded diameter, a sufficiently small total proxy error and logarithmically many such clusters yield a $(1+2/e+\varepsilon)$ approximation in polynomial time with arbitrarily high success probability. We also prove bounded-price strictness for one compatible execution of the logarithmic-surplus construction. Together the recovery and payment arguments give a randomized $(2-\sigma)$ approximation, for an absolute σ > 0, on arbitrary finite rational metrics, both with high probability and in expectation, while opening at most k facilities on every output.

Original PDF | Source files and supporting material

125-02 • Result family 125

The approximation threshold for metric k-median

For every fixed ε > 0, we give a deterministic polynomial-time $(1+2/e+\varepsilon)$-approximation for finite rational metric k-median with specified candidate facilities, opening at most k facilities. Under $P \neq NP$, the infimum approximation factor in this model is therefore $1+2/e$.

Original PDF | Source files and supporting material

126-01 • Result family 126

Exponential PSD rank of positively shifted matching matrices

For every fixed $0 < \rho < 1$, the matrix indexed by odd vertex sets U and perfect matchings M of Kn, with entries $|M \cap \delta(U)| - 1 + \rho$, has real positive semidefinite rank $2^{\Omega(n)}$ as even n tends to infinity. Here $\delta(U)$ is the edge cut of U. Consequently, every exact semidefinite lift of the perfect matching polytope has exponential size.

Original PDF | Source files and supporting material

127-01 • Result family 127

Average sensitivity of polynomial threshold functions

For every n $\geq$ 1 and $1 \leq d \leq n$, we prove that a polynomial threshold function of degree at most d on the uniform Boolean cube has average sensitivity at most $8d\sqrt{n}$. This proves the asymptotic form of the Gotsman-Linial conjecture. The bound is uniform in both parameters and uses the convention $\mathrm{sgn}(0) = 1$.

Original PDF | Source files and supporting material

128-01 • Result family 128

A Polynomial-Time 2-Approximation for Shortest Common Superstring

We give a deterministic algorithm that, for every finite family of explicitly represented ordinary strings, outputs a common superstring of length at most twice the optimum in time polynomial in the total encoded input size, including symbol labels. The guarantee applies to the algorithm constructed here, not the classical maximum-overlap Greedy procedure.

Original PDF | Source files and supporting material

129-01 • Result family 129

An exponential state lower bound for two-way nondeterministic complementation

We prove that two-way nondeterministic finite automata cannot be complemented with a polynomial number of states independent of the alphabet. For each n $\geq$ 4 we construct an n-state automaton over a finite alphabet whose complement requires at least $\frac{1}{2}2^{\lfloor (n-4)/127 \rfloor} - 1$ states.

Original PDF | Source files and supporting material

129-02 • Result family 129

An exponential two-way deterministic state lower bound for one-way liveness

One-way liveness on h points accepts a word of binary relations when their ordered product is nonempty. For every h $\geq$ 2, it has a nondeterministic automaton with $h+3$ states and no left moves, whereas every equivalent s-state two-way deterministic automaton satisfies $4(s+2)^2 \geq 2^{\lfloor (h-2)/31 \rfloor}$. Partial transition rules, stay moves, and nonaccepting infinite computations are allowed. The alphabets are finite and grow with h, so the result rules out an alphabet-independent polynomial state bound for deterministic two-way simulation, already for one-way nondeterministic sources.

Original PDF | Source files and supporting material

130-01 • Result family 130

An explicit power saving for the exact discrete Fourier transform

We give a deterministic algorithm that computes the discrete Fourier transform at every length n in $O(n(\log n)^{1-10^{-13}})$ operations. The model uses exact complex arithmetic, unrestricted coefficients, specified Fourier roots, and unit-cost logarithmic-size indexing; scalar preparation and array organization are included.

Original PDF | Source files and supporting material

130-02 • Result family 130

Finite tensor savings and exact Fourier circuits

We construct exact nonuniform Fourier circuits of size $o(n \log n)$ along an unbounded sequence of lengths, counting every addition, subtraction, and scalar multiplication. This refutes the $\Omega(n \log n)$ lower bound in the unrestricted complex linear-circuit model. The construction uses a finite tensor saving: a tensor power of some invertible nonmonomial complex matrix can be computed with fewer matrix calls than the standard tensor-axis algorithm on the same coordinates, when invertible monomial maps are allowed freely between calls.

Original PDF | Source files and supporting material

131-01 • Result family 131

Polynomial mixing of the switch chain for every graphical degree sequence

We prove the simple-undirected form of the Kannan-Tetali-Vempala conjecture: the switch chain on simple undirected graphs mixes in polynomial time for every graphical degree sequence. For a lazy chain that proposes switches uniformly on four vertices, the total-variation mixing time at distance 1/4 is at most $2n^8$. We also give an exactly uniform sampler for every graphical labeled degree vector. It uses unbiased random bits, terminates almost surely, and has expected polynomial bit running time.

Original PDF | Source files and supporting material

132-01 • Result family 132

A superquadratic separation between sensitivity and block sensitivity

We disprove the quadratic strengthening of the Sensitivity Conjecture by constructing nonconstant total Boolean functions whose block sensitivity grows faster than any constant multiple of sensitivity squared. In fact, for some fixed α > 2, our examples have unbounded block sensitivity and satisfy $\mathrm{bs}(f) \geq s(f)^{\alpha}$.

Original PDF | Source files and supporting material

133-01 • Result family 133

Parity lifts and bounded-treewidth witnesses for Weisfeiler-Leman equivalence

For k $\geq$ 4, we construct two uncolored graphs that are k-dimensional Weisfeiler-Leman equivalent exactly when a prescribed finite-domain choice system has no compatible choice. The system has $k+1$ domains for joint refinement and k for separate-coordinate refinement. A successful choice is detected after two joint rounds or one separate round. Applied to sparse satisfiability, the reduction gives fixed-dimension $n^{\Omega(k)}$ time exclusions under positive-rate ETH, even for deciding equality of these early histograms.

Original PDF | Source files and supporting material

133-02 • Result family 133

The complexity of identifying a graph by Weisfeiler-Leman refinement

We prove that deciding whether Weisfeiler-Leman refinement of an input dimension identifies a given graph is EXPTIME-complete. The input is a nonempty finite simple uncolored graph in adjacency-matrix form and a positive binary-encoded dimension. Identification quantifies over every comparison graph.

Original PDF | Source files and supporting material

133-03 • Result family 133

Unconditional time lower bounds for Weisfeiler-Leman equivalence

For every sufficiently large fixed k, deciding whether two n-vertex graphs are k-Weisfeiler-Leman equivalent requires $n^{\Omega(k)}$ deterministic sequential time in the worst case. The bound holds at every sufficiently large graph order, even for simple connected uncolored graphs of diameter at most two, without a complexity assumption. Inputs are explicit adjacency matrices; the models are multitape Turing machines and sequential logarithmic-word RAMs with fixed polynomial-bit-time instructions. Both joint and separate replacement conventions are covered.

Original PDF | Source files and supporting material

133-04 • Result family 133

Variable-dimension Weisfeiler-Leman equivalence on general and subcubic graphs

Deciding joint-update k-dimensional Weisfeiler-Leman equivalence is EXPTIME-complete when the two graphs are given by explicit adjacency matrices and k $\geq$ 2 is encoded in binary. The result holds even for connected simple uncolored graphs of equal positive order and maximum degree at most three.

Original PDF | Source files and supporting material

134-01 • Result family 134

Finite Monoid Computations and a Uniform Generalized Star-Height Bound

Every regular language over a finite alphabet has a generalized regular expression of star height at most thirteen over that same alphabet. We prove this uniform bound by representing finite

monoid computations as affine updates and recovering them through twelve successive split constructions.

Original PDF | Source files and supporting material

134-02 • Result family 134

Generalized Star Height at Most Four

Every regular language over a finite alphabet has generalized star height at most four over that same alphabet. We give a complete construction using an affine correction that hides one interval product, a finite clock, and several scales for moving boundaries through periodic words.

Original PDF | Source files and supporting material

134-03 • Result family 134

Generalized Star Height at Most Three

Every regular language over a finite alphabet has generalized star height at most three, with complement taken in the same free monoid. We express finite-monoid computations using a prefix code of word pieces.

Original PDF | Source files and supporting material

135-01 • Result family 135

Homogeneous depth-five lower bounds for iterated matrix multiplication

Let $\mathrm{IMM}_{n,n}$ be the $(1,1)$ entry of a product of n independent $n \times n$ matrices of variables. Over every field of characteristic zero, every syntactically homogeneous $\Sigma\Pi\Sigma\Pi\Sigma$ circuit computing $\mathrm{IMM}_{n,n}$ has at least $n^{\sqrt{n}/400}$ gates for all sufficiently large n, with an absolute threshold independent of the field. Bottom linear forms may have arbitrary support, and arbitrary finite fan-in, fan-out, and gate sharing are allowed. Over every field, a block expansion gives such circuits with at most $n^{\sqrt{n}+4}$ gates for $n \geq 2$. Thus the gate complexity over characteristic-zero fields is $n^{\Theta(\sqrt{n})}$.

Original PDF | Source files and supporting material

136-01 • Result family 136

The PCP-for-PPAD conjecture: a quasilinear reduction

We prove the quasilinear-size PCP-for-PPAD conjecture of Babichenko, Papadimitriou, and Rubinstein. There are fixed positive rational constants ε and δ and a deterministic polynomial-time reduction that transforms an End-of-Line instance of binary length N into a generalized circuit of total binary length $N(\log N)^{O(1)}$. From any rational assignment of polynomial encoding length that ε-satisfies all but a δ fraction of the gates, a solution to the original End-of-Line instance can be recovered in polynomial time, regardless of which gates fail. Such assignments always exist, with one fixed polynomial bound on their encoding length.

Original PDF | Source files and supporting material

137-01 • Result family 137

Simulating One-Tape Time in Two-Fifths-Power Space

We show that a fixed deterministic Turing machine with one writable tape and head can be simulated in $O(T^{2/5}\,\mathrm{polylog}(T+2))$ work-space bits when a binary time cap T $\geq$ 2 is supplied. The simulator computes the finite-control and halting outcome by time T; its running time is unrestricted. The result allows a fixed number of read-only input heads and requires a fixed accessor that supplies every initial writable and read-only symbol within distance T of the relevant head origin in polylogarithmic space. This improves the square-root space exponent for one-tape machines, answering Williams's question for this model.

Original PDF | Source files and supporting material

138-01 • Result family 138

Subset Sum in Time $O(2^{0.49n})$

We give a uniform randomized classical algorithm for Subset Sum with bounded error and worst-case running time $O(2^{0.49n})$ on polynomial-bit inputs in a word-RAM model, where n is the number of input integers. The time bound holds on every random execution.

Original PDF | Source files and supporting material

138-02 • Result family 138

A Low-Space Algorithm for Worst-Case Subset Sum

We give a uniform classical randomized decision algorithm for worst-case Subset Sum. Under every fixed polynomial bound on input-integer bit length, it uses $\mathrm{poly}(n)2^{n/2}$ time and ordinary $O(2^{n/5})$ writable words of $O(n+b)$ bits, where b is the largest input bit length. Both resource bounds hold on every execution. The error is one-sided: the algorithm always rejects unsolvable instances and accepts each solvable instance with probability at least 2/3.

Original PDF | Source files and supporting material

139-01 • Result family 139

Subpolynomial query complexity for well-conditioned log-concave sampling

For every fixed ε > 0, we give a sampling algorithm using at most $C_\varepsilon d^\varepsilon$ exact first-order queries on every execution for C2 potentials on $\mathbb{R}$d with a known minimizer and Hessian between Id and $2I_d$. The output has total-variation distance at most 1/10 from the target Gibbs law. Computation between queries is unrestricted. We also prove an $\Omega(\log d)$ query lower bound for arbitrary randomized adaptive algorithms, determining the optimal dimension exponent to be zero.

Original PDF | Source files and supporting material

140-01 • Result family 140

Memory and precision in noiseless Gaussian regression

For every fixed A > 0, a learner that retains at most Ad^2 bits between fresh exact Gaussian linear measurements needs $\Omega_A(d\log(1/\epsilon))$ measurements to estimate a uniformly random unit vector to angular error at most ϵ, for any $0<\epsilon\leq 1/10$, with probability at least 2/3. The constant is absolute for $o(d^2)$ memory.

Original PDF | Source files and supporting material

140-02 • Result family 140

Posterior replicas and conditional information in Gaussian regression

For a signal with density bounded by L relative to uniform probability on S^{d-1}, we bound the information in a finite message W formed from exact Gaussian measurements, conditional on an independent projection revealed only to the analyst. For explicit row counts proportional to d, the bound is $O(H(W)/d + d + \log(2 + \log L))$. Consequently, a finite-state learner with $o(d^2)$ persistent bits and a deterministic sample horizon needs $\Omega(d \log(1/\epsilon))$ fresh noiseless Gaussian measurements for constant-probability angular accuracy $0 < \epsilon \leq 1/10$ under the uniform spherical prior.

Original PDF | Source files and supporting material

140-03 • Result family 140

Localization costs and information growth for exact Gaussian observations

For the image of a uniform cube under a spherical coordinate map, we prove that finite messages from t blocks of $\Theta(d)$ exact Gaussian measurements reveal only $O_A(dt)$ information when each message has at most $\exp(Ad^2)$ values, for fixed A. The same bound holds when each message is supplemented with a nested cell that restores the required geometric spread.

Original PDF | Source files and supporting material

140-04 • Result family 140

Projection moments, positive cap domination, and Riesz estimates on the sphere

We prove moment estimates for exact random projections of finite measures whose mass is controlled on Euclidean balls, and derive positive domination by countable sums of spherical cap measures. For learners with $M = o(d^2)$ bits of memory, these estimates give three proofs that uniform-sphere average success at least 2/3 at angular accuracy $0 < \epsilon \leq 1/10$ requires $\Omega(d \log(1/\epsilon))$ noiseless Gaussian observations. The three proofs keep their different stopping and accuracy costs explicit.

Original PDF | Source files and supporting material

140-05 • Result family 140

Replacing Gaussian observations in memory-constrained inference

Replacing the Gaussian rows used to select a finite message by independent rows increases the remaining conditional information by at most Cd, for a uniform spherical signal, message entropy at most d2, and the specified row dimensions proportional to d. As an application, we prove that learners with $M = o(d^2)$ persistent bits need $T = \Omega(d \log(1/\epsilon))$ exact observations to attain uniform-sphere angular success at least 3/5, for $0 < \epsilon \leq 1/10$ and a deterministic finite horizon.

Original PDF | Source files and supporting material

140-06 • Result family 140

Subsphere methods for memory-sample lower bounds in noiseless Gaussian regression

Let a finite-state streaming learner estimate a uniformly random unit vector from independent exact Gaussian linear measurements. We prove that $o(d^2)$ bits of persistent memory and angular success probability at least 2/3 require at least $2^{-16} d \log_2(1/\epsilon)$ samples for all sufficiently large d, uniformly for $0 < \epsilon \le 1/10$. The proof conditions each batch on its observed projection and controls the resulting random residual subsphere.

Original PDF | Source files and supporting material

141-01 • Result family 141

Existential-universal real sentences in the counting hierarchy

We prove that the existential theory of the reals lies in the counting hierarchy. More generally, we show that the truth of existential-universal sentences over the reals can be decided in a fixed level of the counting hierarchy, even when the integer polynomials are given by arithmetic circuits.

Original PDF | Source files and supporting material

142-01 • Result family 142

Deterministic Polynomial Factorization over Prime Fields

We give a uniform deterministic polynomial-time algorithm for complete factorization over prime fields. For a prime p in binary and a nonzero polynomial $f \in \mathbf{F}_p[x]$ given by its dense coefficient list, the algorithm computes the irreducible factors and their multiplicities using a number of bit operations polynomial in $(\deg f + 1) \log p$. The proof uses the uniform Hecke zero-free theorem from the companion paper *Primitive roots for every admissible integer base*.

Original PDF | Source files and supporting material

Dynamical systems and ergodic theory

19 manuscripts • Official subject grouping

143-01 • Result family 143

Uniform bounds for planar polynomial limit cycles

For every degree, we prove that the number of isolated periodic orbits of a real planar polynomial vector field is bounded by a finite constant depending only on that degree. This establishes the uniform boundedness assertion in the second part of Hilbert's sixteenth problem. The proof uses separation of asymptotic expansions on nested complex domains and a finite-dimensional counting argument.

Original PDF | Source files and supporting material

143-02 • Result family 143

Two limit cycles for quintic Liénard systems

Every classical Liénard system $\dot{x} = y - F(x), \dot{y} = -x$, with F an arbitrary real polynomial of degree at most five, has at most two geometrically distinct isolated periodic orbits, and the bound is attained. This proves the degree-five case of the Lins Neto-de Melo-Pugh conjecture.

Original PDF | Source files and supporting material

144-01 • Result family 144

A smooth three-torus diffeomorphism with simple Lebesgue spectrum

We solve the probability-preserving form of Banach's simple Lebesgue-spectrum problem in smooth dynamics. We construct a C∞ diffeomorphism of the three-torus that preserves standard volume and whose Koopman operator has simple Lebesgue spectrum on the entire mean-zero L2 space.

Original PDF | Source files and supporting material

145-01 • Result family 145

Rokhlin's multiple-mixing problem for one transformation

Every invertible mixing probability-preserving transformation is mixing of every finite order. Thus Rokhlin's multiple-mixing problem for a single transformation has an affirmative answer.

Original PDF | Source files and supporting material

146-01 • Result family 146

Positive Metric Entropy for the Standard Map at Large Parameters

We prove that the standard sine map of the two-dimensional torus has positive metric entropy with respect to normalized area for every sufficiently large positive parameter. This gives a full parameter tail, and hence answers Sinai's positive-parameter-measure conjecture affirmatively.

Original PDF | Source files and supporting material

147-01 • Result family 147

Continuous Phase Foliations Create Analytic Caustic Collars

For a smooth strictly convex planar billiard of positive curvature, a continuous foliation of a full grazing annulus by individually invariant essential curves forces an analytic boundary and a jointly analytic collar of smooth strictly convex caustics. The physical-collar rigidity theorem proved in the companion article then implies that the table is an ellipse.

Original PDF | Source files and supporting material

147-02 • Result family 147

Rigidity of Smooth Billiards with a Continuous Caustic Collar

We prove that a smooth strictly convex planar billiard with positive curvature is an ellipse whenever a full neighborhood of its boundary is continuously foliated by smooth closed convex caustics. No differentiability across the leaves is assumed. This resolves the near-boundary Birkhoff conjecture in the stated smooth class.

Original PDF | Source files and supporting material

148-01 • Result family 148

The entropy-rate dimension formula for self-similar measures on the line

We prove that the Hausdorff dimension of every finite real self-similar measure equals the minimum of one and its random-walk entropy rate divided by its Lyapunov exponent. Exact overlaps are allowed, and the contraction ratios may be unequal and negative. This resolves the entropy-rate dimension conjecture.

Original PDF | Source files and supporting material

149-01 • Result family 149

Uniform Permanence in Weakly Reversible Mass-Action Systems

We prove the permanence conjecture for finite weakly reversible mass-action systems with fixed positive reaction rates. Every positive stoichiometric compatibility class admits one compact convex forward-invariant set that every positive trajectory in that class enters in finite time, even when the class is unbounded. The set depends only on the network, the rates, and the class; the entry time may depend on the initial state. Thus, after its entry time, every concentration satisfies the same positive lower and finite upper bounds throughout that class.

Original PDF | Source files and supporting material

149-02 • Result family 149

Boundedness and persistence of weakly reversible mass-action systems

We prove the boundedness and persistence conjectures for finite weakly reversible mass-action systems with positive constant reaction rates. For every positive initial condition, the solution exists for all forward time, and every concentration remains bounded above and bounded away from zero. The bounds may depend on the initial condition, and no boundedness assumption is imposed on its stoichiometric compatibility class.

Original PDF | Source files and supporting material

150-01 • Result family 150

Weak mixing of triangular billiards with an irrational angle

We prove that the unit-speed billiard flow in every nondegenerate Euclidean triangle with at least one angle irrational relative to π is weakly mixing for normalized Liouville measure. Equivalently, the product of the flow with itself is ergodic.

Original PDF | Source files and supporting material

150-02 • Result family 150

Ergodicity of triangular billiards with an irrational angle

We prove that the unit-speed billiard flow in every nondegenerate Euclidean triangle with at least one angle irrational relative to π is ergodic for normalized area times uniform angular measure. No genericity or Diophantine condition is required. The flow is considered outside the null set of trajectories that hit a vertex.

Original PDF | Source files and supporting material

151-01 • Result family 151

A C^1 Counterexample to the Entropy Conjecture

We disprove the general C1 self-map formulation of Shub's entropy conjecture. We construct a noninvertible C1 self-map of a compact smooth manifold without boundary whose topological entropy is zero, while its action on second real homology has eigenvalue 2. Thus the topological entropy is strictly smaller than the logarithm of the homological spectral radius.

Original PDF | Source files and supporting material

152-01 • Result family 152

A zero-entropy system without a smooth positive-volume model

We construct an ergodic invertible transformation of a standard nonatomic probability space with zero Kolmogorov-Sinai entropy that has no smooth positive-volume model. More precisely, it is not measurably conjugate to any C∞ diffeomorphism preserving a strictly positive smooth probability density on a compact finite-dimensional manifold. A single example excludes every finite dimension, including models on nonorientable manifolds and manifolds with smooth boundary.

Original PDF | Source files and supporting material

153-01 • Result family 153

Arithmetic classification and non-Pisot singularity for Bernoulli convolutions

We give an arithmetic classification of singular and absolutely continuous unbiased Bernoulli convolutions for every parameter $\lambda \in (0,1)$. The criterion is expressed through one-sided approximation by explicitly defined finite sets of algebraic units; it is an infinite approximation condition, not a finite membership algorithm. We also establish singular examples beyond reciprocals of Pisot numbers: singularity holds at the reciprocal of every quartic Salem number in $(1,2)$ and at the reciprocal of a specified non-Pisot root of an explicit degree-31 polynomial.

Original PDF | Source files and supporting material

154-01 • Result family 154

Pointwise Multiple Ergodic Averages for Mixing Transformations

Let T be an invertible mixing probability-preserving transformation. For every integer n $\geq$ 2 and every fixed tuple of bounded measurable functions, we prove that the consecutive multiple ergodic averages of length n converge almost everywhere to the product of the integrals, as the averaging length tends to infinity through all positive integers. The probability space need not be standard, and no rate of mixing is required.

Original PDF | Source files and supporting material

154-02 • Result family 154

Pointwise convergence of fourfold ergodic averages for mixing transformations

We prove that fourfold ergodic averages along the times $n, 2n, 3n, 4n$ converge almost everywhere to the product of the integrals for every invertible, bimeasurable, mixing probability-preserving transformation and every fixed choice of bounded measurable inputs. Convergence holds along all positive integer averaging lengths. No quantitative mixing rate is assumed, and the probability space need not be standard.

Original PDF | Source files and supporting material

154-03 • Result family 154

Triple ergodic averages with distinct integer slopes

For every invertible mixing probability-preserving transformation, triple ergodic averages of bounded measurable functions along any three pairwise distinct nonzero integer slopes converge almost everywhere to the product of their integrals. The slopes may be positive or negative.

Original PDF | Source files and supporting material

154-04 • Result family 154

Pointwise convergence of triple ergodic averages for mixing transformations

We prove pointwise convergence of triple ergodic averages for every invertible mixing probability preserving transformation T of an arbitrary probability space $(X, \mathcal{F}, \mu)$, with measurable inverse. For every triple of bounded measurable functions $f_1, f_2, f_3 : X \to \mathbb{C}$, $\frac{1}{N}\sum_{n=1}^{N} f_1(T^n x) f_2(T^{2n} x) f_3(T^{3n} x) \longrightarrow \prod_{j=1}^{3} \int_X f_j \, d\mu$ for μ-almost every x as $N \to \infty$ through all positive integers.

Original PDF | Source files and supporting material

Combinatorics

50 manuscripts • Official subject grouping

155-01 • Result family 155

A translational tile with no fully periodic tiling in dimension three

We construct a finite translational tile in $\mathbb{Z}3$ that admits tilings but no fully periodic tiling. Its unit-cube thickening has the same property in $\mathbb{R}3$, even when arbitrary real translations are allowed. This gives a negative resolution of the periodic tiling conjecture in dimension three.

Original PDF | Source files and supporting material

156-01 • Result family 156

A nine-dimensional counterexample to Borsuk's covering assertion

The compact set of rank-one orthogonal projectors on $\mathbb{R}4$, with the Frobenius metric, cannot be covered by ten sets of strictly smaller diameter. It therefore gives a counterexample to Borsuk's conjecture in dimension nine.

Original PDF | Source files and supporting material

157-01 • Result family 157

A counterexample to Hadwiger's conjecture

We disprove Hadwiger's conjecture by constructing arbitrarily large graphs whose chromatic number exceeds their Hadwiger number. The examples have independence number at most two, and even their ordinary fractional chromatic number exceeds their Hadwiger number. Thus they also disprove the fractional-coloring weakening discussed by Reed and Seymour.

Original PDF | Source files and supporting material

157-02 • Result family 157

A counterexample to the Colin de Verdière chromatic conjecture

We disprove the Colin de Verdière chromatic conjecture by constructing graphs whose chromatic number exceeds their Colin de Verdière invariant by more than one. The examples have independence number at most two. In fact, their ordinary fractional chromatic number also exceeds their Colin de Verdière invariant by more than one.

Original PDF | Source files and supporting material

157-03 • Result family 157

A linear list-coloring bound in terms of the Hadwiger number

We prove that every finite nonempty graph G satisfies $\chi_{\mathrm{list}}(G) \le Ch(G)$ for an absolute integer C, where $h(G)$ is the largest order of a clique minor. This resolves the Linear List Hadwiger conjecture affirmatively.

Original PDF | Source files and supporting material

158-01 • Result family 158

The Euclidean plane is not five-colorable

We prove that every coloring of the Euclidean plane with five colors has a monochromatic unit-distance pair, with no regularity assumption on the color classes. Consequently, the chromatic number of the plane is either six or seven.

Original PDF | Source files and supporting material

159-01 • Result family 159

Quasipolynomial Bounds for Arithmetic Progressions

We prove Erdős's conjecture that every set of positive integers with divergent reciprocal sum contains arithmetic progressions of every finite length. More quantitatively, for every fixed k ≥ 3, we show $r_k(N) \le C_k N \exp(-c_k(\log N)^{\varepsilon_k})$ with $C_k, c_k, \varepsilon_k > 0$, where $r_k(N)$ is the largest size of a subset of $\{1, \dots, N\}$ with no nonconstant k-term arithmetic progression.

Original PDF | Source files and supporting material

160-01 • Result family 160

Quantitative Superexponential Bounds for van der Waerden Numbers

We prove that there are absolute constants c > 0 and K0 such that $W_r(k) > k^{ck^{\lfloor \log_2 r \rfloor}}$ for every $k \ge K_0$ and r ≥ 2. Consequently $W_r(k)^{1/k} \to \infty$ for each fixed r ≥ 2, giving a quantitative positive resolution of Erdős's superexponential-growth question, including the two-color case.

Original PDF | Source files and supporting material

161-01 • Result family 161

A counterexample to Sidorenko's conjecture

We disprove Sidorenko's conjecture with a bipartite graph on 35 vertices and 66 edges: its homomorphism density in some finite simple graph is smaller than the conjectured lower bound. The same connected graph also disproves the forcing conjecture: at one fixed density, asymptotically matching the edge and pattern densities does not imply quasirandomness.

Original PDF | Source files and supporting material

162-01 • Result family 162

Balanced counterexamples to Ryser's conjecture at prime orders

For every sufficiently large prime q, we construct a finite intersecting $(q+1)$-partite $(q+1)$-uniform hypergraph with covering number $q+1$ and exactly $q+1$ nonisolated vertices in each part. This disproves Ryser's covering conjecture, even for intersecting hypergraphs with equally sized parts.

Original PDF | Source files and supporting material

162-02 • Result family 162

A counterexample to Ryser's covering conjecture

For every sufficiently large prime $s \equiv 2 \pmod 3$ and every sufficiently large odd integer n, with the threshold depending on s, we construct an intersecting (s^n+1)-partite (s^n+1)-uniform hypergraph with covering number s^n+1. This disproves Ryser's covering conjecture in its intersecting case.

Original PDF | Source files and supporting material

164-01 • Result family 164

Monochromatic finite sums and products in the positive integers

We prove Hindman's finite sums and products conjecture: for every finite coloring of the positive integers and every positive integer k, there is a k-element set whose nonempty subset sums and nonempty subset products all have the same color.

Original PDF | Source files and supporting material

165-01 • Result family 165

The crossing number of complete graphs

We prove the Harary-Hill conjecture: for every positive integer n, the ordinary crossing number of the complete graph Kn is $\frac{1}{4}\lfloor\frac{n}{2}\rfloor\lfloor\frac{n-1}{2}\rfloor\lfloor\frac{n-2}{2}\rfloor\lfloor\frac{n-3}{2}\rfloor$.

Original PDF | Source files and supporting material

165-02 • Result family 165

The crossing number of complete bipartite graphs

We prove the Zarankiewicz crossing-number conjecture, resolving Turán's brickyard problem. For all positive integers m, n, the ordinary crossing number of the complete bipartite graph $K_{m,n}$ is $\lfloor\frac{m}{2}\rfloor\lfloor\frac{m-1}{2}\rfloor\lfloor\frac{n}{2}\rfloor\lfloor\frac{n-1}{2}\rfloor$.

Original PDF | Source files and supporting material

166-01 • Result family 166

The higher-dimensional Erdős distinct-distances conjecture

For every fixed integer d $\geq$ 3, we prove that every set of n $\geq$ 2 distinct points in $\mathbb{R}$d determines at least $c_d n^{2/d}$ distinct distances, where $c_d > 0$ depends only on d. This resolves the higher-dimensional Erdős distinct-distances conjecture positively.

Original PDF | Source files and supporting material

167-01 • Result family 167

The weak pinned planar distance theorem

We prove the weak pinned Erdős distinct-distance conjecture. For every fixed ε > 0, all but $o(n)$ points of any n-point planar set determine at least $n^{1-\varepsilon}$ distinct nonzero distances.

Original PDF | Source files and supporting material

167-02 • Result family 167

A power saving for planar unit distances

We prove a power saving for the planar unit-distance problem: for some absolute $\beta < 4/3$, every set of n points in the Euclidean plane determines $O(n^{\beta})$ unordered pairs at unit distance.

Original PDF | Source files and supporting material

168-01 • Result family 168

Combinatorial invariance of Kazhdan-Lusztig polynomials

We prove that an isomorphism of Bruhat intervals in arbitrary Coxeter systems preserves their equal-parameter Kazhdan-Lusztig polynomials. This resolves the full combinatorial invariance conjecture positively.

Original PDF | Source files and supporting material

169-01 • Result family 169

Elementary positivity of chromatic quasisymmetric functions

We prove that the chromatic quasisymmetric function of every natural unit interval graph is elementary-positive over $\mathbb{N}[q]$. This resolves the elementary-positivity part of the Shareshian-Wachs conjecture.

Original PDF | Source files and supporting material

170-01 • Result family 170

The sharp logarithmic exponent of r(5,t)

We determine the sharp logarithmic exponent of the off-diagonal Ramsey number $r(5,t)$: $r(5,t) = \frac{t^4}{(\log t)^{3+o(1)}} \qquad (t \longrightarrow \infty)$.

Original PDF | Source files and supporting material

170-02 • Result family 170

Sharp logarithmic exponents for fixed off-diagonal Ramsey numbers

For every fixed integer s $\geq$ 6, we determine the sharp logarithmic exponent of the off-diagonal Ramsey number: $r(s,t) = \frac{t^{s-1}}{(\log t)^{s-2+o(1)}} \qquad (t \longrightarrow \infty)$.

Original PDF | Source files and supporting material

171-01 • Result family 171

The hypercube Ramsey number has linear order

We prove that the two-color Ramsey number of the n-dimensional binary cube is at most $C2^n$, where C is an absolute constant. This resolves positively the hypercube Ramsey conjecture of Burr and Erdős.

Original PDF | Source files and supporting material

172-01 • Result family 172

A classification of finite Euclidean Ramsey configurations

We classify finite Euclidean Ramsey configurations by a necessary and sufficient tensor condition over their coordinate fields. The Ramsey property here concerns monochromatic congruent copies at the original scale under arbitrary finite colorings. The criterion shows that every nonempty subtransitive set and every nonempty set of at most five points on a circle is Ramsey. In particular, some Ramsey cyclic quadrilaterals are not subtransitive, disproving the necessity direction of the Leader-Russell-Walters conjectured characterization.

Original PDF | Source files and supporting material

173-01 • Result family 173

A proof of Seymour’s second-neighborhood conjecture

We prove that every nonempty finite oriented graph has a vertex with at least as many vertices at directed distance two as at directed distance one. This resolves Seymour's second neighborhood conjecture positively.

Original PDF | Source files and supporting material

174-01 • Result family 174

The strong thin tree conjecture

We prove that every finite loopless k-edge-connected multigraph on at least two vertices, with k $\geq$ 1, has a spanning tree meeting each cut in at most C/k times the size of the cut, where C is a universal constant. This resolves the strong thin tree conjecture.

Original PDF | Source files and supporting material

174-02 • Result family 174

A polynomial-time construction of strong thin trees

We give a deterministic polynomial-time construction of strong thin trees. Given a finite k-edge-connected loopless multigraph on at least one vertex, the algorithm constructs a spanning tree meeting every cut in at most a C/k fraction of its edges, for a universal constant C. The running time is polynomial in the binary input length, including when parallel-edge multiplicities are encoded in binary.

Original PDF | Source files and supporting material

175-01 • Result family 175

Graph Decompositions at the Integral Expectation Threshold

We prove the graph-decomposition conjecture of Ascoli, He, Park, and Talagrand. Every graph admits a partition into a universally bounded number of fixed edge pieces, each having ordinary containment threshold at most a universal constant times the original graph's integral expectation threshold. The partition is chosen before sampling the random host, and the separate embeddings of the pieces need not agree on shared vertices.

Original PDF | Source files and supporting material

175-02 • Result family 175

Integral and fractional expectation thresholds are equivalent

We prove Talagrand's conjecture that integral and fractional expectation thresholds are within a universal constant factor, with the same covering budget.

Original PDF | Source files and supporting material

175-03 • Result family 175

Talagrand’s discrete-convexity conjecture

We prove Talagrand's discrete-convexity conjecture. There is a universal integer k such that, whenever an arbitrary family has Bernoulli product measure at least $1 - 1/k$, the sets not contained in a union of k members admit a cover of total cost at most 1/2 at the same density.

Original PDF | Source files and supporting material

176-01 • Result family 176

The second Kahn-Kalai conjecture

We prove the second Kahn-Kalai conjecture. For every finite simple graph H with h $\geq$ 1 edges and at most n vertices, the threshold for $G(n,p)$ to contain an ordinary copy of H is at most $Cp_{\mathsf{E}}(n,H)(1+\log_2 h)$, where C is universal. Here $p_{\mathsf{E}}(n,H)$ is the least density at which every subgraph of H has expected copy count at least one half.

Original PDF | Source files and supporting material

177-01 • Result family 177

Bounded-degree coboundary expanders in every dimension

For every integer d $\geq$ 3, we construct arbitrarily large finite d-dimensional simplicial complexes with uniformly bounded vertex degrees and uniform $\mathbb{F}2$ coboundary expansion in every degree below d. Together with the known graph and two-dimensional cases, this establishes the existence of bounded-degree $\mathbb{F}2$ coboundary expanders in every positive dimension.

Original PDF | Source files and supporting material

178-01 • Result family 178

Deterministic nonbipartite Ramanujan graphs in every fixed degree

For every fixed integer d $\geq$ 3, we give a deterministic algorithm that constructs a simple nonbipartite d-regular Ramanujan graph on every sufficiently large even number n of vertices. It outputs the full adjacency list in polynomially many bit operations, with an exponent that may depend on d. Every nonconstant adjacency eigenvalue lies strictly between $-2\sqrt{d-1}$ and $2\sqrt{d-1}$.

Original PDF | Source files and supporting material

179-01 • Result family 179

The circulant Hadamard conjecture

We prove the circulant Hadamard conjecture: a real circulant Hadamard matrix has order 1 or 4. As a consequence, Barker sequences of length greater than one exist exactly at lengths 2, 3, 4, 5, 7, 11, 13, proving the Barker-sequence conjecture.

Original PDF | Source files and supporting material

180-01 • Result family 180

Paired states and Hamiltonian cycles in cubic bipartite planar graphs

We prove Barnette's conjecture: every finite simple cubic bipartite planar 3-vertex-connected graph has a Hamiltonian cycle.

Original PDF | Source files and supporting material

181-01 • Result family 181

A linear cycle-and-edge decomposition of every graph

We prove that every finite simple undirected graph on n vertices has an edge partition into at most Cn simple cycles and single edges, for an absolute constant C. This resolves the Erdős-Gallai cycle decomposition conjecture positively.

Original PDF | Source files and supporting material

182-01 • Result family 182

A power saving for intersective polynomial differences with an exponent depending only on the degree

An integer polynomial is intersective if it has a root modulo every positive integer. For each degree k $\geq$ 2, we prove that there is an exponent $c_k > 0$ such that every set $A \subseteq \{1, \dots, N\}$ whose differences avoid all nonzero values $h(1), h(2), \dots$, where h is an intersective polynomial of degree k with positive leading coefficient, satisfies $|A| = O_h(N^{1-c_k})$. The implied constant may depend on h, but the power-saving exponent depends only on its degree.

Original PDF | Source files and supporting material

182-02 • Result family 182

A Power Saving for Polynomial Differences at Prime Arguments

Let h be a fixed integer polynomial of degree at least two with positive leading coefficient, having a unit root modulo every positive integer. We prove that any set $A \subseteq \{1, \dots, N\}$ whose differences avoid all nonzero values $h(p)$ at primes satisfies $|A| \leq C_h N^{1-c_h}$, where $c_h > 0$ and $C_h \geq 1$ depend only on h. Thus the local unit-root condition gives a fixed power saving even when polynomial arguments are restricted to primes. The proof uses the companion zero-free half-plane theorem for Dirichlet L-functions to obtain the required prime-distribution estimates.

Original PDF | Source files and supporting material

182-03 • Result family 182

A power saving for square-difference-free sets

We prove that there are absolute constants c > 0 and C < ∞ such that every set $A \subseteq \{1, \dots, N\}$ with no nonzero square difference satisfies $|A| \leq CN^{1-c}$. This answers the fixed-power question posed by Green and Sawhney.

Original PDF | Source files and supporting material

183-01 • Result family 183

A power saving for planar halving lines

There are absolute constants ε > 0 and C such that every sufficiently large even n-point set in the plane with no three collinear has at most $Cn^{4/3-\varepsilon}$ unordered halving pairs. This gives a power saving over the classical $O(n^{4/3})$ bound for planar halving lines. The proof is nonquantitative and does not supply explicit constants.

Original PDF | Source files and supporting material

184-01 • Result family 184

Correspondence coloring graphs with a forbidden clique

For every fixed integer r ≥ 4, we prove that every Kr-free graph of sufficiently large maximum degree Δ has correspondence chromatic number $O_r(\Delta/\log\Delta)$. This resolves the Alon-Krivelevich-Sudakov coloring conjecture in the stronger correspondence-coloring form. The same bound, with a constant depending on F, holds when any fixed graph F is excluded as an ordinary subgraph. Ordinary and list coloring satisfy the same bounds.

Original PDF | Source files and supporting material

184-02 • Result family 184

A logarithmic independence bound for clique-free graphs

For every fixed integer r ≥ 4, every Kr-free graph on n vertices with average degree d ≥ 2 has an independent set of size at least $c_r n \log d/d$, where $c_r > 0$ depends only on r. This proves the fixed-clique-size independence conjecture of Ajtai, Erdős, Komlós and Szemerédi.

Original PDF | Source files and supporting material

185-01 • Result family 185

A Counterexample to the Infinite Matroid Packing/Covering Conjecture

We construct in ZFC two self-dual partitional matroids on a countably infinite common ground set that admit neither a packing/covering partition nor an intersection witness. This disproves the unrestricted infinite matroid packing/covering and intersection conjectures and answers Joó's question for two partitional matroids negatively. The examples are neither finitary nor cofinitary.

Original PDF | Source files and supporting material

186-01 • Result family 186

A uniform influence bound for hypergraph properties

For every fixed integer r ≥ 3, we prove that every relabeling-invariant Boolean property of simple r-uniform hypergraphs on n vertices satisfies $\mathrm{Var}_p(f) \le C_r I_p(f)/(\log n)^{r/(r-1)}$. The constant depends only on r, and the bound holds uniformly for all $0 < p < 1$ without a monotonicity assumption. For increasing properties, it gives the corresponding threshold-width bound with exponent $r/(r-1)$, proving the hypergraph threshold-width conjecture of Friedgut and Kalai.

Original PDF | Source files and supporting material

186-02 • Result family 186

A Sharp Threshold Bound for Monotone Graph Properties

We prove the Friedgut-Kalai sharp-threshold conjecture. For every integer n ≥ 2, every nontrivial increasing family of graphs on n vertices invariant under all vertex permutations, and every $0 < \varepsilon < 1/2$, the edge probabilities at which its probability equals ε and $1 - \varepsilon$ differ by at most $C \log(1/(2\varepsilon))/(\log n)^2$, for a universal constant C.

Original PDF | Source files and supporting material

187-01 • Result family 187

Snaky in 21 Maker moves

We prove that Maker can achieve the Snaky hexomino within 21 actual Maker moves against arbitrary legal Breaker play on the initially empty infinite square board. The same bound holds on a 17×17 square; in fact, Maker can confine its claims to a fixed 251-cell board.

Original PDF | Source files and supporting material

188-01 • Result family 188

The sharp terminal leave in random triangle removal

Starting from the complete graph on n vertices, repeatedly remove the three edges of a uniformly chosen remaining triangle. We prove that the number of edges left at termination, divided by n3/2, converges in L2 to $1/(2\sqrt{2})$. This proves the triangle case of the sharp-constant conjecture of Joos and Kühn. In particular, the same limit holds in probability and for the normalized expectation.

Original PDF | Source files and supporting material

189-01 • Result family 189

Cycle–clique Ramsey numbers

We prove that $R(C_m, K_n) = (m-1)(n-1)+1$ for every pair of integers $m \ge n \ge 3$ other than $(m,n) = (3,3)$, for which $R(C_3, K_3) = 6$. This establishes the cycle-clique conjecture of Erdős, Faudree, Rousseau and Schelp. The proof combines expansion in a minimal counterexample with a large-clique lemma and an optimization of paths joining clique vertices. These arguments reduce the remaining cases to $3,099$ finite parameter-pattern instances, which are excluded by two exact implementations of proved inference rules. Complete programs and deduction traces accompany the paper.

Original PDF | Source files and supporting material

190-01 • Result family 190

Polynomial removal fails for ordered binary matrices

We construct a fixed 66×66 binary matrix for which ordered matrix removal has no polynomial bound. This disproves the polynomial ordered binary matrix-removal conjecture. Ordered copies preserve the separate row and column orders and match both zeros and ones; removal permits changing entries in either direction.

Original PDF | Source files and supporting material

191-01 • Result family 191

A power improvement in the Heilbronn triangle lower bound

There are absolute constants $\eta, c_1 > 0$ such that, for every sufficiently large integer n, one can choose n points in the unit square so that every triangle they determine has area at least $c_1 n^{-2+\eta}$. Thus the almost n−2 upper-bound formulation of Heilbronn's triangle problem is false. The exponent η is fixed but extremely small.

Original PDF | Source files and supporting material

192-01 • Result family 192

Unbounded Violations of the Square-Root Degree Bound

We disprove the Gopalan-Servedio square-root conjecture, even up to an arbitrary constant factor. For every real C > 0, there is a nonconstant Boolean function $f : \{-1,1\}^n \to \{-1,1\}$ on a finite sign cube such that $\sum_{i=1}^{n} \hat{f}(\{i\}) > C\sqrt{\deg(f)}$. Here $\hat{f}(\{i\})$ is the linear Fourier coefficient associated with the ith input, and $\deg(f)$ is the degree of the real multilinear polynomial representing f.

Original PDF | Source files and supporting material

Algebra

29 manuscripts • Official subject grouping

193-01 • Result family 193

Positivity of Serre's Intersection Multiplicity

We prove Serre's positivity conjecture, including ramified mixed characteristic. If two nonzero finitely generated modules over a regular local ring have tensor product of finite length and complementary dimensions, then their intersection multiplicity is strictly positive.

Original PDF | Source files and supporting material

194-01 • Result family 194

Lech's multiplicity conjecture

We prove Lech's multiplicity conjecture: Hilbert-Samuel multiplicity cannot decrease under a flat local homomorphism of nonzero Noetherian local rings. The result holds in arbitrary dimension, with no restrictions on the residue fields or characteristics.

Original PDF | Source files and supporting material

195-01 • Result family 195

A Complete Local Domain without a Small Cohen-Macaulay Module

We construct a three-dimensional complete Noetherian normal local domain containing $\mathbb{C}$, with residue field $\mathbb{C}$, that has no nonzero finitely generated maximal Cohen-Macaulay module. This disproves the domain form of the small Cohen-Macaulay module conjecture.

Original PDF | Source files and supporting material

196-01 • Result family 196

A Torsion-Free Group Algebra with Zero Divisors

We disprove Kaplansky's zero-divisor conjecture by constructing a finitely presented torsion-free group G for which $\mathbb{F}_2[G]$ has nonzero zero divisors. The group admits a finite two-dimensional classifying space.

Original PDF | Source files and supporting material

197-01 • Result family 197

A Torsion-Free Group Algebra That Is Not Directly Finite

We construct a finitely presented torsion-free counterexample to Kaplansky's direct finiteness conjecture over the field of two elements. The group admits a finite two-dimensional classifying complex.

Original PDF | Source files and supporting material

197-02 • Result family 197

A Counterexample to Kaplansky's Direct-Finiteness Conjecture in Characteristic Two

We disprove Kaplansky's direct-finiteness conjecture by constructing a finite field K of characteristic two, a finitely presented group G, and finite sums $a, b \in K[G]$ with $ab = 1$ but $ba \neq 1$. The group G is nonsofic. The same elements define a cellular automaton on KG that is injective but not surjective, disproving Gottschalk's surjunctivity conjecture.

Original PDF | Source files and supporting material

197-03 • Result family 197

A Counterexample to the Group-Ring Determinant Conjecture

We disprove the unrestricted group-ring Determinant Conjecture. We construct a finitely generated group G and a square matrix over $\mathbb{Z}[G]$ that is invertible over $\mathbb{Q}[G]$ and has Fuglede-Kadison determinant strictly between zero and one. The logarithmic integral defining the determinant is finite.

Original PDF | Source files and supporting material

197-04 • Result family 197

A Counterexample to Kaplansky's Direct-Finiteness Conjecture in Odd Characteristic

We construct a counterexample to Kaplansky's direct-finiteness conjecture in odd characteristic. For one specified odd prime p, we obtain a field K of order p4, a finitely generated group G containing torsion, and finite sums $a, b \in K[G]$ with $ab = 1$ but $ba \neq 1$. The same elements define a cellular automaton on KG that is injective but not surjective.

Original PDF | Source files and supporting material

198-01 • Result family 198

An algebra of infinite little finitistic dimension

We construct a finite-dimensional complex algebra whose finite-dimensional modules have unbounded finite projective dimensions. This disproves the little finitistic-dimension conjecture.

Original PDF | Source files and supporting material

199-01 • Result family 199

An explicit counterexample to the Auslander-Reiten conjecture

We disprove the Auslander-Reiten conjecture for Artin algebras. We construct a finite-dimensional algebra Λ over $k = \mathbb{F}_2(q, H_1, H_2)$ and a finite-dimensional nonprojective left module Z such that $\mathrm{Ext}^i_\Lambda(Z, Z) = \mathrm{Ext}^i_\Lambda(Z, \Lambda) = 0$ for every i > 0. The module is Gorenstein-projective, so the same example also disproves the Gorenstein-projective conjecture. Both counterexamples persist after every extension of k.

Original PDF | Source files and supporting material

199-02 • Result family 199

A counterexample to Tachikawa's second conjecture

We disprove Tachikawa's second conjecture by constructing a finite-dimensional symmetric algebra over $k = \mathbb{F}_2(q, H_1, H_2)$ with a finite-dimensional nonprojective module whose self-extension groups vanish in every positive degree. The associated endomorphism algebra also gives counterexamples to the classical, generalized, and strong Nakayama conjectures, the Auslander-Gorenstein conjecture, and the Wakamatsu tilting conjecture. These conclusions persist after every extension of k.

Original PDF | Source files and supporting material

200-01 • Result family 200

The Artinian Lex-Plus-Powers Betti Theorem

We prove the Eisenbud-Green-Harris and lex-plus-powers conjectures over every characteristic-zero field for homogeneous regular sequences of any positive length, with degrees at least two. For every homogeneous ideal containing such a sequence, the corresponding lex-plus-powers ideal has the same Hilbert function and at least as large a graded Betti number in every homological and internal degree.

Original PDF | Source files and supporting material

200-02 • Result family 200

Commuting Division-Coefficient Forms and the Artinian Eisenbud–Green–Harris Conjecture

We prove the Eisenbud-Green-Harris conjecture over every characteristic-zero field for homogeneous regular sequences of any positive length, with degrees at least two. Thus every homogeneous ideal containing such a sequence has the Hilbert function of a monomial ideal containing the corresponding pure powers.

Original PDF | Source files and supporting material

201-01 • Result family 201

A Counterexample to Kurosh’s Division-Ring Problem

We construct a countable division ring of characteristic zero that is algebraic over its center, generated by two elements as an algebra over that center, and infinite-dimensional over it. This gives a negative answer to the Kurosh problem for division rings.

Original PDF | Source files and supporting material

202-01 • Result family 202

The Blockwise Alperin Weight Conjecture

We prove the numerical blockwise Alperin weight conjecture for every finite group and every prime p. For each p-block B, the number of irreducible Brauer characters in B equals the number of conjugacy classes of B-weights.

Original PDF | Source files and supporting material

203-01 • Result family 203

Donovan's Conjecture over Algebraically Closed Fields

We prove Donovan's conjecture over every algebraically closed field K of characteristic p > 0. For each fixed K and bound on defect-group order, blocks of finite groups represent only finitely many K-linear Morita equivalence classes. The defect groups need not be abelian, and the result includes p = 2.

Original PDF | Source files and supporting material

203-02 • Result family 203

Integral Donovan Finiteness over Witt Vectors

We prove integral Donovan finiteness: for every prime p and positive integer M, the blocks of all finite groups with defect groups of order at most M have only finitely many Morita equivalence classes over $W(\overline{\mathbb{F}}_p)$. The defect groups need not be abelian, and the result includes p = 2. The same bounded-defect finiteness holds over each fixed complete discrete valuation ring of characteristic zero with algebraically closed residue field of characteristic p, including ramified rings.

Original PDF | Source files and supporting material

204-01 • Result family 204

Tensor saturation for even spin groups

We prove the saturation conjecture for Spin$(2n)$, n $\geq$ 2. If three dominant integral weights sum to an element of the root lattice, then the existence of a nonzero tensor invariant after a positive integral dilation implies the existence of one at the original weights.

Original PDF | Source files and supporting material

205-01 • Result family 205

Universal Tensor Squares for Symmetric Groups

For every positive integer n other than 2, 4, and 9, we prove that some irreducible complex representation of Sn has a tensor square containing every irreducible representation. This resolves the tensor square conjecture for symmetric groups affirmatively.

Original PDF | Source files and supporting material

205-02 • Result family 205

A Cyclic Polytabloid Proof of Saxl's Conjecture

For every staircase partition, we prove that the tensor square of the corresponding irreducible complex representation of the symmetric group contains every irreducible representation of that group. This proves Saxl's conjecture.

Original PDF | Source files and supporting material

206-01 • Result family 206

Finite congruence lattices: characterization and undecidability

We give an explicit colored-graph characterization of the finite nonempty lattices that occur as full congruence lattices of finite algebras, and prove that deciding this representation property is undecidable. In particular, the finite lattice representation problem has a negative answer. We also prove that recognition of full subgroup intervals in finite groups is undecidable.

Original PDF | Source files and supporting material

206-02 • Result family 206

A negative solution to the finite lattice representation problem

We give a negative solution to the finite lattice representation problem. We prove that there is a finite nonempty lattice that is not the full congruence lattice of any finite nonempty algebra of any finite signature.

Original PDF | Source files and supporting material

207-01 • Result family 207

The ℓ^1-Bass Conjecture for Discrete Groups

We prove the ℓ1-Bass conjecture for every discrete group. The Hattori-Stallings trace of every idempotent matrix over the complex ℓ1 group algebra is supported on finitely many conjugacy classes of finite-order elements.

Original PDF | Source files and supporting material

207-02 • Result family 207

The Bass trace conjecture and the characteristic-zero Kaplansky idempotent conjecture

We prove the complex group-ring Bass trace conjecture for every discrete group: the Hattori-Stallings trace of a finitely generated projective module over its complex group ring is supported on conjugacy classes of finite-order elements. As a consequence, for every torsion-free group G and every commutative unital domain R of characteristic zero, the only idempotents in RG are 0 and 1. This proves Kaplansky's idempotent conjecture in characteristic zero.

Original PDF | Source files and supporting material

208-01 • Result family 208

Fiber functors for finite symmetric tensor categories in positive characteristic

We prove the finite case of the Benson-Etingof-Ostrik conjecture: every finite symmetric tensor category over an algebraically closed field k of characteristic p > 0 admits a k-linear exact faithful strong symmetric monoidal functor to a finite higher Verlinde category $\mathrm{Ver}_{p^n}(k)$. The result includes characteristic two, and the level n may depend on the category.

Original PDF | Source files and supporting material

209-01 • Result family 209

An integral counterexample to Gersten's conjecture

We construct a two-dimensional ramified regular local ring A of mixed characteristic $(0, 5)$ for which $K_5(A) \to K_5(\mathrm{Frac}\, A)$ has a nonzero kernel. This disproves Gersten's conjecture in its unrestricted integral form.

Original PDF | Source files and supporting material

209-02 • Result family 209

An integral degree-three Gersten counterexample

We construct a two-dimensional ramified regular local ring A in mixed characteristic $(0, 5)$ for which the integral map $K_3(A) \to K_3(\mathrm{Frac}\, A)$ has nonzero kernel. This gives a negative answer to the unrestricted integral Gersten conjecture.

Original PDF | Source files and supporting material

210-01 • Result family 210

Foulkes' conjecture for the sixth symmetric power

We prove the sixth-symmetric-power case of Foulkes' conjecture. For every integer b $\geq$ 6 and every finite-dimensional complex vector space V, there is a $\mathrm{GL}(V)$-equivariant injection $\mathrm{Sym}^6(\mathrm{Sym}^b V) \hookrightarrow \mathrm{Sym}^b(\mathrm{Sym}^6 V)$.

Original PDF | Source files and supporting material

210-02 • Result family 210

Quadratic stabilization of the canonical Foulkes–Howe map

For every finite-dimensional complex vector space V, we prove that the canonical Foulkes-Howe map $\mathrm{Sym}^b(\mathrm{Sym}^a V) \longrightarrow \mathrm{Sym}^a(\mathrm{Sym}^b V)$ is surjective whenever a $\geq$ 2 and $b \geq a(a-1)$. This gives a quadratic stabilization bound independent of $\dim V$.

Original PDF | Source files and supporting material

Probability and statistical mechanics

105 manuscripts • Official subject grouping

211-01 • Result family 211

Random Walks on Critical FK-Ising Maps and Liouville Brownian Motion

We prove that stationary random walks on critical spherical FK-Ising planar maps converge to Liouville Brownian motion on the ordinary unit-area $\sqrt{3}$-quantum sphere, using the geometric and electrical results of the spectral companion. The walk chooses uniformly among all incident half-edges, retaining loops and multiple edges. With the corner measure as the stationary law, the deterministic time acceleration is exactly the number of map edges. For any deterministic metric scale giving the metric-measure limit, convergence retains that same surface and any fixed finite number of conditionally independent walks with their time parameters.

Original PDF | Source files and supporting material

211-02 • Result family 211

Spectral convergence for critical FK-Ising planar maps

Using the conformal and metric-measure companion results and the stated Brownian/Liouville quantum gravity inputs, we prove spectral convergence for critical spherical FK-Ising maps to Liouville Brownian motion on the ordinary unit-area $\sqrt{3}$-quantum sphere. The discrete walk has total attempt rate one, uses every map edge including loops and multiplicities, and has the corner measure as its stationary law. Accelerating time by the number of map edges gives joint convergence of the metric-measure space, all ordered eigenvalues with multiplicities and padding, and the heat trace locally uniformly at strictly positive times. The conductivity and clock constant are one in these conventions.

Original PDF | Source files and supporting material

211-03 • Result family 211

A Linear Clock for Random Walk on Tree-Weighted Planar Maps

We prove that stationary random walk on a planar map sampled with weight equal to its number of spanning trees converges to Liouville Brownian motion on the unit-area $\sqrt{2}$-Liouville quantum sphere. The convergence retains the conditional path law jointly with the measured metric space. The walk chooses uniformly among all incident half-edges and starts from the stationary degree measure. For total attempt rate one and the continuum Dirichlet form with factor 1/2, the time acceleration is exactly the number of map edges. The result uses the companion contour and metric limits for this same ensemble.

Original PDF | Source files and supporting material

211-04 • Result family 211

Canonical conformal limits of subcritical FK planar maps

For every fixed $0 < q < 4$, we prove joint convergence of spherical Fortuin-Kasteleyn planar maps in their flag-triangle uniformization to the corresponding unit-area Liouville quantum gravity sphere decorated by an independent conformal loop ensemble. The convergence includes the area mea-

sure, deterministically rescaled graph distances between all vertex pairs, and the full nested interface collection, with interfaces converging uniformly up to reparameterization.

Original PDF | Source files and supporting material

211-05 • Result family 211

The critical Liouville quantum sphere and geometric limits of FK maps at q= 4

We construct the field and area law of the unit-area critical Liouville quantum sphere as a limit of ordinary subcritical quantum spheres, and equip it with its critical intrinsic metric. We then prove that spherical Fortuin-Kasteleyn planar maps at q = 4, embedded by their equilateral flag uniformizations, converge jointly to this sphere decorated by an independent nested conformal loop ensemble CLE4. With deterministic distance normalization, the convergence includes the area measure, the full embedded distance function, and every macroscopic interface through all positive integer edge counts.

Original PDF | Source files and supporting material

211-06 • Result family 211

Metric-measure limits of subcritical FK and spanning-tree planar maps

We resolve the finite spherical cases of Gwynne and Miller's graph-metric conjecture for critical Fortuin-Kasteleyn maps at each fixed $q \in (0,4)$ and for uniform spanning-tree-decorated maps. After deterministic rescaling of graph distances, these maps, equipped with the vertex probability measure proportional to degree, converge in Gromov-Hausdorff-Prokhorov law to their ordinary unit-area Liouville quantum gravity spheres. Distances use every primal edge, and convergence holds through all positive integer edge counts.

Original PDF | Source files and supporting material

211-07 • Result family 211

Brownian continuum random tree limits of finite Fortuin-Kasteleyn maps above four

We prove the finite-volume continuum-random-tree prediction for critical Fortuin-Kasteleyn planar maps at every fixed q > 4. After rescaling graph distances by a constant times n−1/2, an n-edge map with normalized degree measure converges to the Brownian continuum random tree in the Gromov-Hausdorff-Prokhorov topology. The convergence holds through all positive integer sizes.

Original PDF | Source files and supporting material

212-01 • Result family 212

No bigeodesics in planar first-passage percolation

We prove that planar first-passage percolation with independent identically distributed nonnegative nonatomic edge weights has almost surely no doubly infinite geodesic, provided the minimum of four independent weights has finite second moment. This resolves the planar no-bigeodesics conjecture under that moment assumption. The conclusion rules out all bigeodesics simultaneously, without any regularity assumption on the limit shape.

Original PDF | Source files and supporting material

212-02 • Result family 212

Strict convexity and differentiability of the planar exponential first-passage limit shape

We prove that the limit shape of undirected nearest-neighbor first-passage percolation on $\mathbb{Z}2$ with independent exponential edge weights is strictly convex and has a C1 boundary. This resolves the strict convexity and differentiability conjectures for the planar exponential model. More generally, we prove differentiability of the time-constant norm for every Gamma edge-weight law with positive shape and rate.

Original PDF | Source files and supporting material

213-01 • Result family 213

Critical bond and site percolation on the cubic lattice

We prove that nearest-neighbor Bernoulli bond and site percolation on $\mathbb{Z}3$ have no infinite cluster at their respective critical parameters. The proof combines a finite connection inequality for independent hyperedges with a finite-scale extension estimate and an adaptive exploration.

Original PDF | Source files and supporting material

213-02 • Result family 213

No percolation at criticality on quasi-transitive graphs

We prove that critical Bernoulli bond percolation has no infinite cluster on any infinite connected locally finite quasi-transitive graph with critical probability less than one. This resolves the bond form of the criticality conjecture of Benjamini and Schramm.

Original PDF | Source files and supporting material

214-01 • Result family 214

Nonuniqueness of percolation on nonamenable quasi-transitive graphs

We prove the bond-percolation nonuniqueness conjecture of Benjamini and Schramm: every infinite connected, locally finite, nonamenable quasi-transitive graph has a nonempty interval of parameters for which Bernoulli bond percolation almost surely has infinitely many infinite clusters. We also prove Hutchcroft's stronger operator-threshold conjecture, establishing $p_c < p_{2\to 2} \le p_u$.

Original PDF | Source files and supporting material

215-01 • Result family 215

The canonical massive continuum limit of the two-dimensional O(3) model

We construct a canonical interacting massive continuum limit of the two-dimensional nearest-neighbor $O(3)$ model with unit-length spins, no external field, and no topological term. Normalized by susceptibility and second-moment correlation length, the limit exists as the bare coupling tends to infinity through all positive real values, without selecting subsequences. The limiting fields satisfy the Osterwalder-Schrader axioms and have a nonzero connected four-point correlation on sep-

arated time supports. Their reconstructed theory has a unique vacuum, a nonzero vacuum complement, and a positive Hamiltonian gap on that entire complement.

Original PDF | Source files and supporting material

215-02 • Result family 215

An Isolated Particle Pole for the Two-Dimensional O(3) Spin Field

We consider the continuum spin field constructed from the nearest-neighbor two-dimensional $O(3)$ model by the fixed prescription of the companion paper. We prove that its vector two-point spectral measure has a positive atom corresponding to the lowest mass, separated by a positive gap from all remaining mass support.

Original PDF | Source files and supporting material

215-03 • Result family 215

Exact mass asymptotics for the two-dimensional O(4) lattice model

For the nearest-neighbor $O(4)$ model on the square lattice at inverse temperature β, we prove the exact low-temperature asymptotic $m_{\mathrm{lat}}(\beta) \sim 32 \exp(\pi/4 - 1/2)\sqrt{\beta}\, \exp(-\pi\beta) \qquad (\beta \to \infty)$. Here the mass is the gap of the full Osterwalder-Schrader transfer operator, including rotation-invariant local-observable sectors, in units of one original lattice time step.

Original PDF | Source files and supporting material

215-04 • Result family 215

Sharp mass bounds for the two-dimensional O(4) model

We prove sharp mass bounds for the two-dimensional nearest-neighbor $O(4)$ model. For all sufficiently large inverse couplings β, the full transfer gap, including rotation-invariant local-observable sectors, is bounded above and below by positive multiples of $\sqrt{\beta}e^{-\pi\beta}$. We also prove that the model has a unique periodic local limit and a positive full gap at every finite β > 0, resolving the all-temperature lattice mass-generation conjecture for this periodic state. Under the stated cutoff scaling, the mass bounds are uniform in independently fixed physical units. A continuum spectral interpretation requires separate convergence hypotheses for the transfer semigroup and the block observables.

Original PDF | Source files and supporting material

215-05 • Result family 215

Exponential decay in two-dimensional classical O(n) models

We prove exponential decay of two-point correlations for the classical nearest-neighbor $O(n)$ model on the square lattice, for every n ≥ 3 and every finite positive inverse temperature. The estimate is uniform over finite free-boundary subgraphs and bounded nonnegative edge strengths. This resolves positively the all-temperature exponential spin-decay conjecture for these models.

Original PDF | Source files and supporting material

216-01 • Result family 216

The critical logarithmic correction for the planar XY model

We prove the critical logarithmic correction for the nearest-neighbor cosine XY model on the square lattice. Let $C_{b_c}(r)$ be the two-point correlation at critical inverse temperature, obtained by taking the free-box thermodynamic limit while the two sites remain r lattice steps apart along a coordinate axis. Then $C_{b_c}(r) = B_{\mathrm{XY}} r^{-1/4} (\log r)^{1/8} (1 + o(1)), \qquad B_{\mathrm{XY}} \in (0, \infty),$ as $r \to \infty$.

Original PDF | Source files and supporting material

216-02 • Result family 216

Critical Center Magnetization in the Planar XY Model

Assuming the stated critical-height, local-renormalization, and spin-field inputs from the companion papers, we determine the center magnetization of the planar XY model in a square with aligned boundary spins at its mass-defined critical threshold. As $n \to \infty$, the magnetization is $A_{\mathrm{XY}} n^{-1/8} (\log n)^{1/16} (1 + o(1))$, where A_{XY} is a finite, strictly positive model-specific constant.

Original PDF | Source files and supporting material

216-03 • Result family 216

The Critical Spin Field of the Planar XY Model

Using the companion critical Bessel-height and pin-limit theorems, we prove that at the mass-defined critical inverse temperature of the nearest-neighbor XY model on the square lattice, the spin field on a square with boundary angles fixed to zero converges along the full sequence to the full-variance imaginary exponential of a zero-Dirichlet Gaussian free field with stiffness $2/\pi$. The normalization uses the exact center magnetization and the lattice Green-function factor. Convergence holds in law in $H^{-3}_{\mathrm{loc}}((-1,1)^2)$; all mixed moments and joint laws of fields smeared against smooth compactly supported test functions in $(-1,1)^2$ converge as well.

Original PDF | Source files and supporting material

216-04 • Result family 216

Essential Singularity of the Correlation Length in the Planar XY Model

We prove the Berezinskii-Kosterlitz-Thouless essential singularity for the correlation length of the nearest-neighbor cosine XY model on the square lattice. Let $m(b)$ be the mass obtained by taking first the free-box thermodynamic limit and then the separation limit along a coordinate axis. As the inverse temperature b approaches bc from the massive side, $\sqrt{b_c - b} \log \frac{1}{m(b)} \longrightarrow A_{\mathrm{XY}},$ where A_{XY} is a finite, strictly positive model-specific constant.

Original PDF | Source files and supporting material

216-05 • Result family 216

The critical correlation exponent of the planar XY model

For the ordinary nearest-neighbor XY model on the square lattice, we prove the Berezinskii-Kosterlitz-Thouless prediction that the critical spin-correlation exponent is 1/4. More precisely, at the mass-defined critical inverse temperature, the infinite-volume correlation is $n^{-1/4+o(1)}$. The infinite-volume limit through free square boxes is taken before the separation limit.

Original PDF | Source files and supporting material

216-06 • Result family 216

BKT universality for height and planar spin fields

We prove Gaussian scaling limits for the two-dimensional discrete Gaussian height model throughout its rough phase, including the physical roughening threshold, for every finite square-symmetric interaction set containing the nearest neighbors. After the natural lattice normalization, the critical effective temperature has the universal value 8π. For the ordinary nearest-neighbor Villain and XY models at sufficiently low fixed temperatures, we also prove that their Green-function-normalized spin fields converge to the imaginary exponential of a Dirichlet Gaussian free field.

Original PDF | Source files and supporting material

217-01 • Result family 217

The low-temperature Sherrington-Kirkpatrick free-energy limiting law

For every fixed inverse temperature β > 1, we prove that the zero-field Gaussian Ising Sherrington-Kirkpatrick free energy, centered by its expectation and divided by its standard deviation, converges in distribution to a nondegenerate law as the system size tends to infinity through all integers. We also prove that its variance divided by n1/3 converges to a finite positive constant.

Original PDF | Source files and supporting material

217-02 • Result family 217

The low-temperature Sherrington-Kirkpatrick fluctuation scale

For the zero-field Gaussian Sherrington-Kirkpatrick model at every fixed inverse temperature β > 1, we prove that the standard deviation of the log partition function is $n^{1/6+o(1)}$. The same exponent describes its typical centered absolute fluctuations, establishing the predicted one-sixth exponent in this regime.

Original PDF | Source files and supporting material

218-01 • Result family 218

Quenched SLE_3 limits for general weak random-bond Ising models

For the planar Ising model with bonds $J_e = 1 + \varepsilon\xi_e$, where the ξe have any fixed bounded, nondegenerate, mean-zero iid law, we prove quenched chordal SLE3 convergence for sufficiently small ε > 0. The temperature is the spontaneous-magnetization threshold. Convergence holds in probability over environments for the full oriented curve law in deterministic Jordan-domain approximations.

Original PDF | Source files and supporting material

218-02 • Result family 218

Quenched SLE_3 Universality for the Weak Random-Bond Ising Model

We prove quenched chordal SLE3 convergence for the planar Ising model with sufficiently weak independent symmetric two-valued ferromagnetic bonds. The temperature is the critical point

defined by spontaneous magnetization. Convergence holds in probability over environments for the full oriented curve law in deterministic Jordan-domain approximations.

Original PDF | Source files and supporting material

218-03 • Result family 218

Logarithmic Relative Fluctuations in the Weakly Disordered Planar Ising Model

Assuming the stated deterministic critical-reference estimates, we prove that the relative second moment of the quenched critical spin correlation in the square-lattice Ising model with independent fair bonds $1 \pm \varepsilon$ grows as $(\log r)^{1/4+o(1)}$ for each sufficiently small fixed ε > 0. The correlation is evaluated at the physical critical temperature, with the thermodynamic limit taken before the disorder moments and the large-distance limit.

Original PDF | Source files and supporting material

218-04 • Result family 218

Conformal universality of bulk Ising correlations under weak interactions

We prove conformal universality of mixed bulk spin and energy correlations for the square-lattice Ising model with sufficiently small, square-symmetric, finite-range even multispin perturbations of either sign. One critical-temperature branch and two field normalizations apply in every bounded simply connected C2 Jordan domain with free, plus, or minus boundary conditions, and in the periodic thermodynamic plane state. Energies are centered in their actual states.

Original PDF | Source files and supporting material

218-05 • Result family 218

SLE3 universality for weak finite-range Ising interactions

We prove that every sufficiently small, square-symmetric finite-range perturbation of the planar Ising contour energy has a domain-independent inverse temperature at which its spin interface converges to chordal SLE3. The result allows interactions of either sign, arbitrary admissible exterior contours, and uniformly approximated Jordan domains. Convergence holds for the full oriented curve law in uniform distance modulo increasing reparametrization.

Original PDF | Source files and supporting material

218-06 • Result family 218

Buffered comparison and stopping-band resolution in critical Ising

We prove pointwise comparison and finite stopping-band approximation theorems for the critical nearest-neighbor Ising model on the square lattice. Arbitrary common pinned spins are allowed in the comparison: the likelihood ratio is controlled by a zero-field FK connection probability on the graph with those vertices deleted. For an interface exploration across a band of fixed positive width, finitely many regular signed barriers approximate every good conditional target law in total variation, uniformly over bounded observables and sufficiently fine meshes.

Original PDF | Source files and supporting material

219-01 • Result family 219

Fixed-energy universality for weak Anderson disorder on random regular graphs

For every fixed degree $d \geq 3$, we prove fixed-energy GOE universality for the adjacency matrix of a uniformly random simple labelled d-regular graph with independent uniform diagonal disorder. The disorder strength is positive, sufficiently small, and fixed as the graph grows. At each fixed energy in a compact subinterval of the clean spectral band, the full microscopic eigenvalue point process converges to the GOE bulk process after rescaling by the positive density of states of the corresponding infinite-tree operator. The permitted disorder strength depends on the degree and the distance from the clean spectral edges.

Original PDF | Source files and supporting material

219-02 • Result family 219

GOE bulk universality for fixed-degree random regular graphs

We prove the fixed-degree bulk-universality conjecture for random regular graphs. For every fixed integer $d \geq 3$ and every fixed energy in the open Kesten-McKay bulk, the unfolded eigenvalue point process of the adjacency matrix of a uniform simple labelled d-regular graph converges to the GOE bulk process. The convergence holds along all admissible graph sizes, without additional conditioning.

Original PDF | Source files and supporting material

220-01 • Result family 220

A directional zero-one law for finite-range-dependent random environments

We prove a directional zero-one law for uniformly elliptic nearest-neighbor random walks in stationary, ergodic, finite-range-dependent environments on $\mathbb{Z}d$, $d \geq 3$. For every fixed nonzero real direction, the probability of escape in that direction, averaged over the environment, is either zero or one. Finite-range dependence is imposed on the full transition rows: collections of rows at distance greater than a fixed range are independent, including collections indexed by infinite deterministic sets.

Original PDF | Source files and supporting material

220-02 • Result family 220

A directional zero-one law under strict ellipticity

We prove the directional zero-one conjecture for nearest-neighbor random walks in independent and identically distributed strictly elliptic environments on $\mathbb{Z}d$, $d \geq 3$: the probability of escape in each fixed nonzero real direction is zero or one. Only strict positivity of the transition probabilities is required; no uniform lower bound or moment assumption is imposed.

Original PDF | Source files and supporting material

220-03 • Result family 220

Directional transience implies ballisticity

We prove that almost-sure transience in a fixed direction implies a deterministic limiting velocity with positive projection in that direction for nearest-neighbor random walks in independent and identically distributed uniformly elliptic environments on $\mathbb{Z}$d, d $\geq$ 2. This resolves the ballisticity conjecture positively.

Original PDF | Source files and supporting material

221-01 • Result family 221

The Mézard-Parisi formula for diluted spin glasses

We prove the Mézard-Parisi hierarchical cavity formula for diluted even-arity Ising models in the Panchenko-Talagrand class. This resolves the variational equality conjecture for that class: the limiting pressure equals the infimum of the trial functional over all finite hierarchy depths and trial laws. Only first moments of the interaction and external field are required.

Original PDF | Source files and supporting material

222-01 • Result family 222

The free energy of the Ising random perceptron

We determine the limiting free energy of the Ising perceptron with independent Gaussian patterns for every bounded Borel log-potential, at every fixed positive temperature and pattern density. We give an explicit variational formula for the limit and prove convergence in expectation and probability.

Original PDF | Source files and supporting material

222-02 • Result family 222

Microscopic jamming in the negative spherical perceptron

We prove a sharp feasibility threshold and limiting gap and force laws for the spherical perceptron with margin −1 and quadratic penalty. The limits are taken successively in system size, inverse temperature, and density approaching the threshold from above. They agree for Gaussian coordinates and for the equal mixture of centered Gaussian coordinates with variances $1-\varepsilon$ and $1+\varepsilon$, for every sufficiently small fixed ε. The contact-removed gap cumulative law and the mean-one force cumulative law satisfy $G_J(u) = u^{1-\gamma+o(1)}, \qquad F_J(s) = s^{1+\theta+o(1)},$ as $u \downarrow 0$ and $s \downarrow 0$, with $\gamma = (2+\theta)^{-1}$, $0.4126930 < \gamma < 0.4126934$, and $0.4231063 < \theta < 0.4231088$. A finite numerical certificate for these exponent intervals, together with its mathematical error bounds, is included.

Original PDF | Source files and supporting material

222-03 • Result family 222

The spherical perceptron with bi-orthogonally invariant disorder

We determine the limiting free energy of a spherical perceptron with a bounded continuous activation and a bi-orthogonally invariant disorder matrix. The singular values may have any compact limiting distribution, provided there are no outliers. We give an explicit variational formula for this limit.

Original PDF | Source files and supporting material

222-04 • Result family 222

The free energy of the spherical random perceptron

We prove an exact variational formula for the limiting pressure of the spherical random perceptron with an arbitrary bounded continuous single-pattern potential. The formula holds at every fixed positive density and inverse temperature, with convergence in expectation and in probability.

Original PDF | Source files and supporting material

223-01 • Result family 223

Square-lattice FK interfaces and nested loops for 1 <= q < 4

For every fixed $1 \leq q < 4$, critical square-lattice random-cluster Dobrushin interfaces converge as ordered curves to chordal $\mathrm{SLE}_{\kappa(q)}$, where $\kappa(q) = 4\pi/\arccos(-\sqrt{q}/2)$, confirming the Rohde-Schramm prediction in this parameter range. This holds in every bounded Jordan domain under uniform marked boundary approximation. The complete nested plane loop collections converge to whole-plane $\mathrm{CLE}_{\kappa(q)}$ in a spherical matching topology retaining multiplicities and traversals. At q = 1 we obtain Cardy's formula for square-lattice bond percolation with free boundary edges.

Original PDF | Source files and supporting material

223-02 • Result family 223

Self-dual random-cluster interfaces below one

For every fixed $0 < q < 1$, we prove that the Dobrushin interface of the square-lattice random-cluster model at its self-dual parameter converges to chordal SLEκ, where $\kappa = 4\pi/\arccos(-\sqrt{q}/2) \in (6, 8)$. The approximating domains are simple closed nearest-neighbor lattice polygons with distinct marked vertices, whose marked boundary parametrizations converge uniformly to those of a bounded smooth Jordan domain. Convergence holds along the full mesh sequence in the uniform metric on oriented curves modulo increasing reparametrization. The proof combines finite connection comparisons below one, localization in irregular tiled disks, and a boundary observable that determines the limiting Loewner driver.

Original PDF | Source files and supporting material

223-03 • Result family 223

Quenched SLE Universality for Weakly Disordered FK-Ising Interfaces

We prove that critical FK-Ising interfaces with sufficiently weak, symmetric, independent two-valued bond disorder converge to chordal SLE16/3. The disorder strength is fixed as the mesh tends to zero, and convergence of the conditional curve laws holds in probability over the environment.

Original PDF | Source files and supporting material

223-04 • Result family 223

Thermal FK-Ising interfaces and massive SLE

We prove convergence of thermal FK-Ising interfaces, for every fixed nonzero mass of either sign, on uniformly angle-bounded isoradial lattices in bounded simply connected domains. The limit is independent of the lattice and the admissible domain approximation. For positive mass it is the unique massive $\mathrm{SLE}_{16/3}$ law with locally finite-energy drift prescribed by a massive boundary value problem; negative mass follows by duality and reversal. The theorem also allows Carathéodory approximations whose diameters diverge.

Original PDF | Source files and supporting material

223-05 • Result family 223

Natural Occupation Measures for Critical Square-Lattice FK Interfaces

We prove that the rescaled counting measure of a critical square-lattice Fortuin-Kasteleyn Dobrushin interface in the unit square converges to the Minkowski-content measure of its Schramm-Loewner limit, for every fixed cluster weight $1 \le q < 4$. A single deterministic constant times the predicted power of the mesh gives the normalization. Convergence is joint with the ordered curve and includes the total mass, giving the scaling limit of the interface's total number of steps.

Original PDF | Source files and supporting material

223-06 • Result family 223

Conformal Limits of Critical Square-Lattice Random-Cluster Interfaces

For every fixed $1 \le q \le 4$, critical square-lattice random-cluster Dobrushin interfaces converge to chordal $\mathrm{SLE}_{\kappa(q)}$, where $\kappa(q) = 4\pi/\arccos(-\sqrt{q}/2)$, and the complete nested plane loop collections converge to whole-plane $\mathrm{CLE}_{\kappa(q)}$. The results hold under uniform marked Jordan boundary approximation and retain loop multiplicities and traversals. At q = 1 we obtain Cardy's formula for square-lattice bond percolation with free boundary edges.

Original PDF | Source files and supporting material

224-01 • Result family 224

From critical crossings to quenched near-critical universality in Voronoi percolation

Taking Cardy's crossing formula for critical Poisson-Voronoi percolation as an input, we prove a universal joint limit for the near-critical crossing thresholds of rational polygonal quadrilaterals under the monotone coupling. Conditional on the Poisson tessellation, the threshold law converges in probability over tessellations to the same law as on the triangular lattice. Each model is normalized by its own expected number of color-pivotal sites for a unit-square crossing.

Original PDF | Source files and supporting material

224-02 • Result family 224

A Pivotal Amplitude for Voronoi Percolation from Cardy's Formula

Taking Cardy's conformal crossing formula for critical planar Poisson-Voronoi percolation in every bounded Jordan quadrilateral as an input, we prove that the expected number of color-pivotal

cells for a unit-square crossing is asymptotic to a positive constant times $\varepsilon^{-3/4}$, where the point intensity is ε^{-2}. No rate of convergence in Cardy's formula is required.

Original PDF | Source files and supporting material

224-03 • Result family 224

Cardy’s formula for critical Poisson-Voronoi percolation

We prove Cardy's formula for annealed crossing probabilities in critical planar Poisson-Voronoi percolation in every bounded Jordan quadrilateral. This proves the annealed crossing-probability form of the conformal-invariance conjecture for this model.

Original PDF | Source files and supporting material

225-01 • Result family 225

The Gaussian free field limit of the balanced six-vertex model with variance multiplier 1/arcsin(c/2)

We prove that the height function of the square-lattice six-vertex model with weights $a = b = 1$ and $0 < c \leq 2$, in the plane state obtained from balanced tori, converges to a multiple of the Gaussian free field. For unit height jumps and Green kernel $-(2\pi)^{-1} \log|x - y|$, the squared multiplier is $1/\arcsin(c/2)$.

Original PDF | Source files and supporting material

226-01 • Result family 226

The curve scaling limit of half-plane double dimers

We prove that the complete double-dimer loop ensemble for the Temperleyan square lattice in the upper half-plane converges to nested CLE4. The convergence holds along the full mesh limit and matches every macroscopic loop as an unparametrized curve. This resolves the half-plane Temperleyan form of the double-dimer CLE4 scaling-limit conjecture.

Original PDF | Source files and supporting material

227-01 • Result family 227

Universality of critical quench autocorrelations in the Sherrington-Kirkpatrick model

We prove joint functional convergence of the stationary and quench autocorrelations of zero-field Sherrington-Kirkpatrick heat-bath dynamics at inverse temperature β = 1, with the same random limit for Gaussian and Rademacher couplings. Each site has a rate-one clock, mean spin autocorrelations are multiplied by n1/3, and waiting times and lags are measured in units n2/3. The quench starts from independent fair spins, and convergence is uniform on compact sets of positive waiting times and lags. The quench limit is selected by these initial states and relaxes to the stationary limiting autocorrelation as the waiting time tends to infinity.

Original PDF | Source files and supporting material

227-02 • Result family 227

Functional universality of critical SK autocorrelations

We prove that the stationary spin autocorrelation of the zero-field Sherrington-Kirkpatrick model at inverse temperature β = 1 has a common functional scaling limit for Gaussian and Rademacher couplings. With rate-one heat-bath clocks at every site, time is scaled by n2/3 and the mean spin autocorrelation is multiplied by n1/3. The functions converge in law uniformly on compact positive-time intervals. Their limiting law is not a point mass: it retains sample-to-sample randomness, and its functions decay to zero at large times.

Original PDF | Source files and supporting material

227-03 • Result family 227

A spectral gap throughout the high-temperature Sherrington-Kirkpatrick phase

For every fixed inverse temperature $0 < \beta < 1$, we prove that the unscaled spectral gap of single-site heat-bath dynamics for the zero-field Gaussian Sherrington-Kirkpatrick model is bounded away from zero with probability tending to one over the disorder. Equivalently, the Gibbs law satisfies a dimension-free Poincaré inequality for all functions.

Original PDF | Source files and supporting material

227-04 • Result family 227

Cutoff throughout the high-temperature Sherrington-Kirkpatrick phase

We prove worst-case total-variation cutoff for the zero-field Gaussian Sherrington-Kirkpatrick heat-bath dynamics at every fixed inverse temperature $0 \leq \beta < 1$. With rate-one refresh at each spin, the cutoff location is $\log n/(2\lambda(\beta))$ for a positive deterministic rate $\lambda(\beta)$. The location for uniformly chosen single-site update attempts is n times as large. Convergence is in probability over the disorder.

Original PDF | Source files and supporting material

227-05 • Result family 227

Critical slowing down in the Sherrington-Kirkpatrick model

At the critical inverse temperature β = 1, we prove slow mixing for zero-field Gaussian Sherrington-Kirkpatrick heat-bath dynamics from typical equilibrium configurations held fixed as initial states. For every deterministic sequence $t_n = o(n^{2/3})$ in rate-one-per-site time, the Gibbs mass of initial states whose time-tn total-variation distance from equilibrium exceeds 1/4 tends to one in probability over the disorder. The same statement holds for every deterministic integer sequence $k_n = o(n^{5/3})$ of uniform-site update attempts.

Original PDF | Source files and supporting material

227-06 • Result family 227

Stretched-exponential barriers for typical SK initial states

At every fixed inverse temperature β > 1, we prove a stretched-exponential obstruction to mixing for the zero-field Sherrington-Kirkpatrick model from typical equilibrium configurations held fixed as initial states. The Gibbs mass of states whose total-variation distance from equilibrium at time

$\exp(n^{1/10000})$ exceeds 1/4 tends to one in probability over the disorder. This holds for rate-one-per-site heat-bath dynamics and after the same stated number of discrete update attempts.

Original PDF | Source files and supporting material

227-07 • Result family 227

A typical-start upper bound for low-temperature SK Glauber dynamics

For every fixed inverse temperature β > 1, we prove subexponential mixing for rate-one-per-site heat-bath dynamics in the zero-field Gaussian Sherrington-Kirkpatrick model from a typical equilibrium configuration. If one Gibbs-sampled configuration is held fixed as the initial state, its total-variation distance from equilibrium is at most 1/4 by time $\exp(n^{1-1/40000000})$, with joint probability tending to one over the disorder and the sampled state.

Original PDF | Source files and supporting material

227-08 • Result family 227

Critical mixing in the Sherrington-Kirkpatrick model

At the critical inverse temperature β = 1, we determine the worst-start total-variation mixing exponent for single-site heat-bath dynamics in the zero-field Gaussian Sherrington-Kirkpatrick model. The mixing time is $n^{2/3+o(1)}$ when every site has rate one, or $n^{5/3+o(1)}$ attempted uniform-site updates, in probability over the disorder.

Original PDF | Source files and supporting material

228-01 • Result family 228

A continuum temperature singularity for a radial pair potential

We construct a stable radial pair potential in three dimensions whose canonical free energy has a strict downward derivative jump at one common finite positive inverse temperature throughout an open interval of positive densities. The potential has a divergent repulsive core, a nontrivial attractive interval, and an integrable tail satisfying $\phi(r) = o(r^{-3})$. Its free-cube canonical free energy is finite at every positive inverse temperature and density.

Original PDF | Source files and supporting material

228-02 • Result family 228

A radial continuum phase transition with algebraic decay

We construct a bounded, continuous, stable radial pair potential in three dimensions with $|\phi(r)| \leq Cr^{-3-1/32}$ for r ≥ 1. Throughout an open interval of positive densities, its canonical thermodynamic free energy is finite at every positive inverse temperature and has a strict downward derivative jump at one common finite positive inverse temperature.

Original PDF | Source files and supporting material

229-01 • Result family 229

The Reconstruction Threshold for the Ferromagnetic Four-State Potts Model

We establish the exact Kesten-Stigum reconstruction threshold for the ferromagnetic four-state Potts broadcast model on every regular d-ary tree with d $\geq$ 2 and every Poisson Galton-Watson tree of mean d > 0. Reconstruction occurs exactly when $d\lambda^2 > 1$; we prove nonreconstruction at and below the threshold, including equality. In the Poisson model the whole tree is observed and the reconstruction advantage is averaged without conditioning on survival. The proof uses reproducible exact-arithmetic verification of polynomial inequalities.

Original PDF | Source files and supporting material

229-02 • Result family 229

A Capacity Criterion for Four-State Potts Reconstruction on Trees

For the ferromagnetic four-state broadcast model with $0 < \lambda < 1$, we prove that reconstruction on a bounded-degree deterministic rooted tree occurs exactly when its L3 capacity with edge resistances $\lambda^{-2|e|}$ is positive. This gives an exact criterion without regularity or growth-rate assumptions on the tree, including at the exponential critical boundary.

Original PDF | Source files and supporting material

229-03 • Result family 229

The exact reconstruction threshold for the three-state symmetric channel

We determine the exact reconstruction threshold for the symmetric three-state broadcast process on every regular b-ary tree, b $\geq$ 2, and every observed Poisson Galton-Watson tree of mean d > 1. Reconstruction occurs exactly when $d\lambda^2 > 1$, with d = b in the regular model; there is nonreconstruction at equality for either sign of the channel parameter. The Poisson advantage is averaged over trees and spins without conditioning on survival. This resolves the all-degree three-state regular-tree prediction. Combining the Poisson theorem with known tree-to-graph and algorithmic results gives the exact weak-recovery threshold for the symmetric three-community sparse stochastic block model with independent uniform labels, fixed within- and between-community rates $a, b > 0$, and mean degree $(a+2b)/3 > 1$: recovery is possible exactly when $(a-b)^2 > 3(a+2b)$. Above this threshold it is achievable in $O(n \log n)$ time; at or below it, weak recovery is information-theoretically impossible.

Original PDF | Source files and supporting material

230-01 • Result family 230

An exact Hausdorff gauge for SLE

For each $0 < \kappa < 8$, we construct a deterministic Hausdorff gauge that almost surely assigns positive finite measure to every nontrivial compact positive-time segment of chordal Schramm-Loewner evolution. This answers Schramm's Hausdorff-measure existence problem in this parameter range. The entire trace has finite expected gauge measure in each bounded box.

Original PDF | Source files and supporting material

230-02 • Result family 230

An explicit exact Hausdorff gauge for SLE

For each fixed $0 < \kappa < 8$, let $d = 1 + \kappa/8$. The gauge $h(r) = r^d(\log\log(1/r))^{(2-d)/2}$ at sufficiently small radii almost surely gives positive finite Hausdorff measure to every nontrivial positive-time compact segment of chordal SLEκ. This gives an explicit solution to Schramm's Hausdorff-measure problem in this parameter range. The entire trace has finite expected measure in every bounded disk. The exponent-one iterated-logarithm gauge suggested by Schramm is not sigma-finite on any such segment.

Original PDF | Source files and supporting material

231-01 • Result family 231

The free uniform spanning forest is a factor of IID

We prove that the free uniform spanning forest is a factor of IID on every infinite connected locally finite simple unweighted graph. One Borel rule works for all such graphs and uses no root, answering affirmatively the general factor question for unimodular random graphs. We also show that every translation-invariant strongly Rayleigh process indexed by a countable group is a factor of IID, including invariant determinantal processes with Hermitian positive-contraction kernels. This group-action conclusion requires no amenability.

Original PDF | Source files and supporting material

232-01 • Result family 232

Gaussian free-field limits of weighted integer Lipschitz heights

We prove a Gaussian free field scaling limit for weighted integer Lipschitz heights on the triangular lattice. With zero boundary values and a factor x for each edge on which the height changes, the field converges after division by a positive constant depending only on x to the zero-Dirichlet Gaussian free field, for every fixed $x \in [1/\sqrt{2}, 1]$. The convergence holds as a random distribution on every bounded C2 Jordan domain under inside lattice approximations with uniformly convergent boundary parametrizations. This includes the uniform height model and the predicted critical endpoint.

Original PDF | Source files and supporting material

232-02 • Result family 232

The Gaussian free field limit of integer Lipschitz heights with two-arc boundary data

We prove that the centered uniform odd integer height function on triangular-lattice approximations of a smooth simply connected domain, with neighboring differences zero or two and boundary values +1 and −1 on two arcs, converges to a universal multiple of the Dirichlet Gaussian free field. This resolves the field part of Schramm's Problem 2.2. We give an absolutely convergent finite-volume formula for the normalization. The proof combines reflection positivity, a spectral sum rule, and boundary comparison with Gaussian moment identities.

Original PDF | Source files and supporting material

232-03 • Result family 232

Uniform real Lipschitz surfaces on the triangular lattice

We prove the Gaussian free field and SLE_4 scaling limits for uniformly sampled real nearest-neighbor Lipschitz heights on the triangular lattice, resolving Schramm's Problem 2.3. On approximations of smooth simply connected domains, the centered height field converges as a random distribution to a multiple of the Dirichlet Gaussian free field. At one tuned two-arc boundary amplitude, the zero-height interface converges in uniform curve distance to chordal SLE_4. We identify the relation between the field variance and the boundary height in terms of an implicit stationary tangent-flux coefficient.

Original PDF | Source files and supporting material

233-01 • Result family 233

The joint scaling limit of critical Ashkin-Teller currents

For each fixed point on the critical Ashkin-Teller line, including the four-state Potts endpoint, we prove the conjectured joint scaling limit of the height and both current-cluster collections in every bounded Jordan domain, for every admissible polygonal approximation. The height converges to a Gaussian free field with the predicted coupling constant, and the clusters converge to canonical recursive two-valued local sets of that same field. The joint limit includes all nesting depths and the distinguished wired boundary cluster.

Original PDF | Source files and supporting material

234-01 • Result family 234

All-temperature pressure for orthogonally invariant Ising spin glasses

We determine the limiting pressure of an Ising spin glass with Haar orthogonal eigenvectors, a compact limiting spectral law, and no asymptotic outliers, at every fixed temperature. The pressure converges in expectation and almost surely to a variational formula. We also give the limiting pressure with deterministic external fields whose empirical laws converge in first-moment transport distance, and derive a formula for the zero-field ground-state energy by taking temperature to zero.

Original PDF | Source files and supporting material

235-01 • Result family 235

A Limiting Satisfiability Threshold for Every Fixed Clause Size

For every fixed integer $k \geq 3$, random k-SAT with independent uniformly signed clauses on distinct variables, sampled with replacement, has a finite positive limiting satisfiability threshold. We credit Gaia Carenini [[5]](https://eccc.weizmann.ac.il/report/2026/229/) with priority for resolving the satisfiability conjecture. This paper gives an alternative proof, using concentration of a capped last satisfiable index and a comparison between different system sizes.

Original PDF | Source files and supporting material

235-02 • Result family 235

Linear Variance of the Random 3-SAT Hitting Time

For random 3-SAT on n Boolean variables, with independent uniformly signed clauses on three distinct variables sampled with replacement, we prove that the first unsatisfiable prefix has variance $\Theta(n)$. The upper bound removes the logarithmic loss in the earlier variance estimate; the matching lower bound follows from Wilson's transition-width theorem.

Original PDF | Source files and supporting material

235-03 • Result family 235

Variance of the Random k-SAT Hitting Time

Let Hn be the index of the first unsatisfiable prefix in random k-SAT on n variables, with independent uniformly signed clauses using k distinct variables and sampled with replacement. For every fixed k $\geq$ 4, we prove Var$(H_n) = \Theta_k(n)$. For k = 3, the variance is bounded below by a positive multiple of n and above by a constant multiple of $n \log n$; the companion paper on random 3-SAT sharpens this to $\Theta(n)$. The same upper bounds proved here hold after clipping at any fixed positive multiple of n.

Original PDF | Source files and supporting material

235-04 • Result family 235

Computing the Random 3-SAT Threshold

The limiting satisfiability threshold of uniform random 3-SAT is a computable real. We credit Gaia Carenini [[4]](https://eccc.weizmann.ac.il/report/2026/229/) with priority for resolving the satisfiability conjecture, which establishes the threshold's existence. We prove that one finite deterministic machine can approximate it to any prescribed accuracy. A deletion estimate gives explicit lower certificates, while a finite hierarchical approximation of the soft pressure gives upper certificates at every larger rational density. A fair search through these certificates halts without requiring a computable rate of finite-size convergence.

Original PDF | Source files and supporting material

236-01 • Result family 236

The sharp factor-of-IID threshold for the free Ising model on regular trees

For every integer d $\geq$ 3 and inverse temperature β $\geq$ 0, the free zero-field ferromagnetic Ising measure on the infinite d-regular tree is a factor of IID exactly when tanh $\beta \leq (d-1)^{-1/2}$. We prove the positive implication, including equality, resolving the conjecture of Nam, Sly and Zhang; the strict converse is known. The spin factor can be chosen to commute with every tree automorphism on every label input.

Original PDF | Source files and supporting material

237-01 • Result family 237

Radial transfer estimates and polygon length laws for honeycomb walks

We prove critical diameter-tail exponents −2 for unrooted honeycomb polygons and −2/3 for length-weighted polygons, together with a truncated second-length-moment bound of exponent

2/3. Consequently, polygons conditioned to have length at least n have diameter $n^{3/4+o(1)}$ in probability under either weight.

Original PDF | Source files and supporting material

237-02 • Result family 237

Critical honeycomb chords with prescribed boundary endpoints

Critical self-avoiding walks between prescribed, macroscopically separated boundary ports of a regular honeycomb hexagon have length $R^{4/3+o(1)}$ in probability, where R is the scale of the hexagon. We prove the corresponding statements for half-plane arches, parallel cuts and non-parallel pure cuts. The half-plane law also has mean length $R^{4/3+o(1)}$. A separate strip argument gives endpoint mean laws on one density-one set of heights and in an aligned, critically weighted mixture of all even heights in a macroscopic interval.

Original PDF | Source files and supporting material

237-03 • Result family 237

Cylinder loop weights and planar nesting

We determine the growth exponent, at every fixed positive loop fugacity, of the critical honeycomb partition function for disjoint polygons separating two prescribed markers on a balanced cylinder. At fugacity two the cylinder exponent is 1/6; the corresponding planar nesting exponent and middle-strip nesting exponent are 1/12.

Original PDF | Source files and supporting material

237-04 • Result family 237

Mass and covering exponents for fixed-length honeycomb walks

For uniform self-avoiding walks of every sufficiently large integer length on the honeycomb lattice, we prove diameter exponent 3/4 and simultaneous local-mass and covering exponent 4/3, with arbitrary positive exponent slack and arbitrary polynomial failure probability.

Original PDF | Source files and supporting material

237-05 • Result family 237

Signed cylinder propagation and marked polygons on the honeycomb lattice

At the critical activity and loop fugacity two, we prove a 1/6 partition exponent for disjoint honeycomb polygons separating opposite marks on a balanced infinite cylinder, and a 1/12 planar nesting exponent. For ordered first-exit chords in a regular hexagon of side R, with both boundary ports summed and diameter at least $R/10$, the finite critical partition is $R^{3/4+o(1)}$ and the normalized mean length is $R^{4/3+o(1)}$. A separate two-bond estimate on tilted cylinders with controlled site proportions bounds the length-square mass of planar polygons of diameter at most H, modulo translations, by $H^{2/3+o(1)}$.

Original PDF | Source files and supporting material

237-06 • Result family 237

Cylinder amplitudes and logarithmic bridge-length windows on the honeycomb lattice

We prove a logarithmic window of critical honeycomb bridge lengths: summed through height $2h(\log h)^{1/64}$, bridges of lengths between $h^{4/3}(\log h)^{-1/8}$ and $2h^{4/3}(\log h)^{1/2}$ have mass at least $h^{3/4}(\log h)^{-C}$.

Original PDF | Source files and supporting material

237-07 • Result family 237

Marked polygon correlations and one-arc bounds

At the critical honeycomb vertex activity, the squared-length mass of simple polygons of diameter at most H, counted modulo translations, is at most $H^{2/3+o(1)}$. We also prove a quantitative two-mark cylinder estimate and a polynomial one-arc bound uniform even for arbitrarily unequal marked intervals.

Original PDF | Source files and supporting material

237-08 • Result family 237

Disk transfer representations and confined bridge mass

We prove that critical honeycomb bridges crossing a strip of width R have mean length $R^{4/3+o(1)}$, with the same exponent after confinement to a fixed multiple of the strip width. The initial boundary port is fixed, the terminal port is summed, and all finite lengths receive their critical weights. The bridge mass is comparable to R−1/4; the half-plane arch kernel at endpoint separation m is comparable to m−5/4.

Original PDF | Source files and supporting material

237-09 • Result family 237

Polynomial vacuum representations and bridge mass for honeycomb walks

We prove that, under the all-length critical measure, a honeycomb bridge crossing a strip of h layers, with its initial port fixed and its terminal port free, has mean length $h^{4/3+o(1)}$. We also obtain central visit probability $R^{-2/3+o(1)}$ and mean length $R^{4/3+o(1)}$ for macroscopic free-boundary chords in a regular hexagon.

Original PDF | Source files and supporting material

237-10 • Result family 237

Renewal and changes of law for critical honeycomb walks

For critical honeycomb self-avoiding walk, we prove spatial exponent 3/4 for the infinite irreducible-bridge law and for bridges of every sufficiently large compatible even length. We establish the corresponding thermal laws at every large discount scale, including all positive length and spatial moments. For unrestricted uniform walks, the endpoint lower law holds on a common set of lengths of natural density one. We also determine the near-critical exponential correlation scale and small-force free-energy exponent, and prove spatial local lower bounds that permit conditioning on a prescribed terminal vertex.

Original PDF | Source files and supporting material

237-11 • Result family 237

Critical strip-crossing mass on the honeycomb lattice

At the critical weight of the regular honeycomb lattice, the total weight of self-avoiding paths crossing a strip of height N is comparable to N−1/4. The first horizontal-displacement moment of return paths is comparable to N3/4.

Original PDF | Source files and supporting material

237-12 • Result family 237

Uniform marked-polygon estimates and sharp finite bridge moments

We prove a uniform bound for critical honeycomb polygons through two axial marks on a periodic staircase. The bound retains an explicit power of the ratio between the period and the marked separation. We also prove sharp finite strip estimates: bridge mass of order h−1/4, first-length mass at most $Ch^{13/12}$, and mass at least $ch^{-1/4}$ on bridges with length at least $ch^{4/3}$.

Original PDF | Source files and supporting material

237-13 • Result family 237

Cap-selected amplitudes and triangle chords for honeycomb walks

At the critical honeycomb fugacity, self-avoiding port-to-port chords in an equilateral lattice triangle of side R, summed over both boundary endpoints and restricted to diameter at least $R/100$, have partition sum comparable to R3/4 and mean length $R^{4/3+o(1)}$. We also determine the amplitude selected by two vacuum caps on a cylinder of circumference N and prove that it grows as $N^{1/6+o(1)}$.

Original PDF | Source files and supporting material

238-01 • Result family 238

Optimal-order mixing of the Thorp shuffle

We prove that the Thorp shuffle on 2^d cards mixes in $\Theta(d)$ complete shuffles. The full permutation law after $1600d$ shuffles converges to uniform in total variation as $d \to \infty$, uniformly over the initial deck, while a support count gives a lower bound of $2d - O(1)$.

Original PDF | Source files and supporting material

238-02 • Result family 238

Random coordinate frames and partial permutation laws

For the Thorp shuffle on 2^d cards, we prove that the worst-start total-variation distance after $32800d$ physical shuffles tends to zero as $d \to \infty$. Combining our fixed-list estimate with the companion Fourier transfer improves this bound to $512d$ shuffles. These results follow from bounds on partially observed permutation laws in random coordinate frames.

Original PDF | Source files and supporting material

238-03 • Result family 238

From partial permutation information to Fourier bounds

We show how information about partial permutations controls full permutation laws. Let n tend to infinity through multiples of eight. If the images of a uniformly chosen $7n/8$ labels approach the uniform injection law in average total variation, and the sign mean tends to zero, then the product of two independent permutations with the given law converges to uniform on Sn.

Original PDF | Source files and supporting material

238-04 • Result family 238

Conditional information under deterministic coordinate sweeps

We bound the information remaining after paths of specified cards in the Thorp shuffle on $n = 2^d$ cards have been observed. After a fixed number of deterministic coordinate sweeps, the joint endpoint law of further cards is close to uniform on the available positions, on average over the observed paths, provided a fixed positive fraction of labels lies outside both lists. Combined with a Fourier transfer, these bounds give full-deck mixing after $2048d$ physical shuffles.

Original PDF | Source files and supporting material

238-05 • Result family 238

Conditional permutations in a revealed switching environment

Fix half the labels in a Thorp shuffle on 2^d positions and reveal their complete trajectories. We prove that, after an explicit absolute number of coordinate sweeps, the conditional permutation of the remaining labels approaches uniform in expected total variation as $d \to \infty$, uniformly in the initial layout. The resulting constructions give full-deck mixing in a constant multiple of d physical shuffles.

Original PDF | Source files and supporting material

238-06 • Result family 238

Routing densities and representation contraction for Thorp sweeps

For $N = 2^d$ cards, we prove that an absolute number of coordinate sweeps of the Thorp shuffle brings the full permutation law to total-variation distance tending to zero from uniform. The mixing time therefore has optimal order $\Theta(\log N)$ in physical shuffles.

Original PDF | Source files and supporting material

238-07 • Result family 238

Row-column symmetry and contraction of coordinate sweeps

We prove that the Thorp shuffle on $N = 2^d$ labeled cards has full-permutation total-variation mixing time $\Theta(\log N)$ in physical shuffles. An absolute number of coordinate sweeps suffices for the upper bound.

Original PDF | Source files and supporting material

238-08 • Result family 238

Random-subspace tests and trace smoothing for coordinate sweeps

We prove that an absolute number of coordinate sweeps of the Thorp shuffle on $n = 2^d$ positions brings the full permutation to total-variation distance at most $\frac{1}{2}n^{-5}$ from uniform, uniformly over the initial deck. Thus $O(d)$ physical shuffles suffice, which is optimal in order.

Original PDF | Source files and supporting material

238-09 • Result family 238

Compatibility entropy and the spectrum of a Thorp sweep

We prove that the Thorp shuffle on $n = 2^d$ cards mixes in $\Theta(d)$ physical shuffles. After a sufficiently large fixed number of coordinate sweeps, the total-variation distance of the full permutation from uniform tends to zero as $d \to \infty$.

Original PDF | Source files and supporting material

238-10 • Result family 238

Signed tensor densities and diagram budgets for the Thorp shuffle

We prove that a fixed number of coordinate sweeps of the Thorp shuffle on 2^d cards makes the squared L2 distance of the full permutation density from uniform tend to zero as $d \to \infty$. In particular, the total-variation mixing time is $\Theta(d)$ physical shuffles.

Original PDF | Source files and supporting material

238-11 • Result family 238

Conditional coordinate sweeps and analytic transfer

For coordinate sweeps with near-uniform line laws and line sizes among powers of two in a suitable fixed interval, we prove a Schatten-moment bound after conditioning on any feasible collection of prescribed card trajectories. An analytic transfer to binary sweeps then shows that a fixed number of sweeps mixes the Thorp shuffle on 2^d cards in $O(d)$ physical steps, matching the order of the support lower bound.

Original PDF | Source files and supporting material

238-12 • Result family 238

A strict four-row permanent inequality and permutation moments

We prove a permanent inequality with exponent strictly below two for laws on S4 sufficiently close to uniform and with exactly uniform coordinate marginals. Applied to a four-row recursion, it shows that an absolute number of coordinate sweeps brings the full permutation of the Thorp shuffle on $N = 2^d$ cards to total-variation distance tending to zero from uniform as $N \to \infty$. Thus the mixing time has the optimal order $O(\log N)$ in physical shuffles.

Original PDF | Source files and supporting material

239-01 • Result family 239

The sharp exponential rate of singularity for symmetric Bernoulli matrices

Let An be a symmetric $n \times n$ matrix whose entries on and above the diagonal are independent uniform signs. We prove Pr(det $A_n = 0) = \left(\frac{1}{2} + o(1)\right)^n$.

Original PDF | Source files and supporting material

239-02 • Result family 239

The sharp singularity rate for biased symmetric sign matrices

Let An be a symmetric random matrix whose entries on and above the diagonal are independent signs, equal to 1 with fixed probability $p \in (0,1) \setminus \{1/2\}$. We prove that $\mathbb{P}(\det A_n = 0) = (p^2 + (1-p)^2 + o(1))^n$. The rate is attained by the event that two rows agree.

Original PDF | Source files and supporting material

Mathematical logic

8 manuscripts • Official subject grouping

240-01 • Result family 240

A CH obstruction to a prescribed categoricity threshold

Assuming the continuum hypothesis, we construct an abstract elementary class with Löwenheim-Skolem number $\aleph_0$ that is categorical in every sufficiently large cardinal but has at least two nonisomorphic models of cardinality $\beth_{\omega_2}$. Thus categoricity does not transfer down to the proposed bound $\beth_{(2^{\aleph_0})^+}$, which equals $\beth_{\omega_2}$ under CH. Consequently, if ZFC is consistent, the prescribed-threshold form of Shelah's categoricity conjecture is not provable in ZFC.

Original PDF | Source files and supporting material

240-02 • Result family 240

Eventual categoricity for abstract elementary classes

We prove Shelah's eventual categoricity conjecture for abstract elementary classes in ZFC. For each infinite bound on the Löwenheim-Skolem number there is a uniform threshold such that categoricity in any one cardinal at or above that threshold implies categoricity in every cardinal at or above the same threshold.

Original PDF | Source files and supporting material

241-01 • Result family 241

Rigidity of the Turing degrees

We prove that every order automorphism of the full partial order of Turing degrees is the identity, resolving the rigidity conjecture for the Turing degrees positively.

Original PDF | Source files and supporting material

242-01 • Result family 242

Single-fold Diophantine representations

Every recursively enumerable set of natural-number tuples has a polynomial Diophantine representation with exactly one complete auxiliary tuple for each member. This proves the single-fold conjecture and, consequently, the finite-fold conjecture.

Original PDF | Source files and supporting material

243-01 • Result family 243

Choiceless polynomial time with counting does not capture polynomial time

We prove that choiceless polynomial time with counting does not capture polynomial time on unordered finite structures, confirming the noncapture conjecture of Blass, Gurevich and Shelah. A linear-consistency query over $\mathbb{F}3$ in a fixed binary vocabulary is decidable in polynomial time but not in the full counting formalism.

Original PDF | Source files and supporting material

243-02 • Result family 243

Witnessed symmetric choice is strictly stronger than choiceless polynomial time with counting

We prove that witnessed symmetric choice strictly increases the expressive power of choiceless polynomial time with counting. A fixed sentence with one witnessed-choice occurrence defines a Boolean query on every finite input that is not definable in the original counting formalism.

Original PDF | Source files and supporting material

244-01 • Result family 244

The Partition Principle does not imply Choice

We prove that the Partition Principle does not imply the Axiom of Choice: if ZF is consistent, then so is ZF with the Partition Principle, Choice for ordinal-indexed families, and the negation of the Axiom of Choice. Separately, over every countable transitive model of ZFC, we construct a transitive symmetric model of this theory with the same ordinals and no new countable sequences of ground elements.

Original PDF | Source files and supporting material

245-01 • Result family 245

Weak and strong normalization in pure type systems

We prove that every weakly β-normalizing pure type system is strongly β-normalizing. Both properties quantify over all legal expressions in all valid contexts, and reduction acts inside type annotations. No functionality hypothesis is required. This resolves the β-Barendregt-Geuvers-Klop conjecture.

Original PDF | Source files and supporting material

Group theory

22 manuscripts • Official subject grouping

246-01 • Result family 246

A Modulus Proof of Cannon’s Conjecture

We prove that every hyperbolic group whose boundary is homeomorphic to the two-sphere admits a proper cocompact isometric action on hyperbolic three-space with finite kernel. This resolves Cannon's conjecture positively.

Original PDF | Source files and supporting material

247-01 • Result family 247

An infinite finitely presented residually finite 2-group

We prove that the infinite, ordinarily finitely presented periodic Steinberg group $\Gamma = \mathrm{St}_{12}(R)$ of a companion paper is residually finite. Its finite-index subgroup $G = \ker(\Gamma \to \mathrm{St}_{12}(\mathbb{F}_2))$ is infinite, ordinarily finitely presented, and residually finite, and every element of G has finite 2-power order. The orders of its elements are unbounded. Phases of long words in the companion's graded algebra allow finite degree truncations to detect all elements of Γ, including the central kernel.

Original PDF | Source files and supporting material

247-02 • Result family 247

An infinite finitely presented periodic group

We construct an infinite group with an ordinary finite presentation in which every element has finite order, answering the finitely presented Burnside question negatively. We also construct an infinite-dimensional finitely presented nonunital nil associative algebra over $\mathbb{F}2$ that is Jacobson radical but not nilpotent. Its unitization is finitely presented, algebraic, and infinite-dimensional. These algebras answer the finitely presented nil- and radical-algebra nilpotence questions and the finite-presentation version of Kurosh's algebraic finiteness question negatively.

Original PDF | Source files and supporting material

248-01 • Result family 248

Thompson's group F is nonamenable

We prove that Thompson's group F is nonamenable. This confirms Geoghegan's conjecture and resolves the amenability problem for F.

Original PDF | Source files and supporting material

249-01 • Result family 249

A finitely generated counterexample to the Eilenberg-Ganea conjecture

We construct a finitely generated residually finite group of integral cohomological dimension two and geometric dimension three, disproving the Eilenberg-Ganea conjecture.

Original PDF | Source files and supporting material

250-01 • Result family 250

Finite algebraic envelopes and the Boone-Higman conjecture

We prove the Boone-Higman conjecture. A finitely generated group has decidable word problem if and only if it embeds in a finitely presented simple group.

Original PDF | Source files and supporting material

250-02 • Result family 250

Simple F∞ overgroups of groups with decidable word problem

Every finitely generated group with decidable word problem embeds in a simple group of type F∞. This resolves the higher-finiteness strengthening of the Boone-Higman conjecture.

Original PDF | Source files and supporting material

250-03 • Result family 250

A universal group of type F∞

We construct a single group of type F∞ containing every finitely presented group. Its finitely generated subgroups, up to isomorphism, are exactly the finitely generated recursively presented groups. This answers the F∞ form of the higher-dimensional Higman embedding question.

Original PDF | Source files and supporting material

251-01 • Result family 251

Unitarizability implies amenability for discrete groups

We prove that a discrete group is amenable if and only if every uniformly bounded representation on a complex Hilbert space is similar to a unitary representation. This resolves Dixmier's unitarizability problem affirmatively for discrete groups.

Original PDF | Source files and supporting material

251-02 • Result family 251

Strong Ulam Stability Characterizes Amenability

A countable discrete group is amenable if and only if it is strongly Ulam stable: every sufficiently accurate unitary almost representation, on any complex Hilbert space, is uniformly close in operator norm to a genuine representation on the same space. We prove the converse to Kazhdan's amenable stability theorem, answering the question of Burger, Ozawa, and Thom. The inclusion of infinite-dimensional Hilbert spaces is essential.

Original PDF | Source files and supporting material

252-01 • Result family 252

A torsion-free hyperbolic group that is not residually finite

We construct a torsion-free word-hyperbolic group that is not residually finite, answering the residual-finiteness question for hyperbolic groups negatively.

Original PDF | Source files and supporting material

253-01 • Result family 253

An infinite finitely presented simple amenable group

We construct an infinite finitely presented simple amenable group, giving a positive answer to the finite-presentation existence question for simple amenable groups.

Original PDF | Source files and supporting material

254-01 • Result family 254

Harmonic heights and the Artin K(pi,1) conjecture

We prove that the standard Salvetti complex of every Artin group with finitely many standard generators is aspherical. This resolves the Artin $K(\pi,1)$ conjecture in finite rank.

Original PDF | Source files and supporting material

254-02 • Result family 254

An Artin group with no geometric CAT(0) action

We construct an Artin group on 116 generators that admits no proper, cocompact isometric action on a nonempty proper CAT(0) space. This refutes the CAT(0) conjecture for Artin groups.

Original PDF | Source files and supporting material

254-03 • Result family 254

Parabolic intersections in Artin groups

We prove that every intersection of parabolic subgroups of any finite-rank Artin group, with arbitrary finite or infinite Coxeter labels, is parabolic. This resolves the Parabolic Intersection Conjecture affirmatively.

Original PDF | Source files and supporting material

255-01 • Result family 255

Quasi-isometric recognition of virtually polycyclic groups

We prove that every finitely generated group quasi-isometric to a finitely generated virtually polycyclic group is virtually polycyclic. This resolves the lattice-recognition conjecture of Eskin, Fisher and Whyte, which allows the ambient solvable Lie group in the conclusion to differ from the one in the hypothesis.

Original PDF | Source files and supporting material

256-01 • Result family 256

Nonsingular systems of equations over arbitrary groups

Every finite nonsingular system of equations over an arbitrary group has a simultaneous solution in an overgroup. This proves Howie's conjecture on nonsingular systems.

Original PDF | Source files and supporting material

256-02 • Result family 256

The Kervaire theorem for groups

We prove that no free product of a nontrivial group with an infinite cyclic group is normally generated by one element. This resolves the Kervaire conjecture positively.

Original PDF | Source files and supporting material

257-01 • Result family 257

A hyperbolic group with no geometric CAT(0) action

We construct a hyperbolic group with a finite classifying space that admits no geometric action on a proper complete CAT(0) space. Consequently, a finite aspherical simplicial complex with a linear combinatorial disk-filling inequality need not have a finite locally CAT(0) homotopy model, and hence need not have a finite locally CAT(−1) model. Here metric models carry geodesic length metrics inducing the given complex topology.

Original PDF | Source files and supporting material

258-01 • Result family 258

Baumslag-Solitar-free one-relator groups are hyperbolic

We prove Gersten's conjecture: every finitely generated one-relator group containing no subgroup isomorphic to a Baumslag-Solitar group BS(m, n), for nonzero integers m, n, is word-hyperbolic.

Original PDF | Source files and supporting material

258-02 • Result family 258

Virtual compact specialness of hyperbolic one-relator groups

We prove that every word-hyperbolic one-relator group is virtually compact special. Combined with the work of Kielak-Linton, this resolves Wise's virtual free-by-cyclic conjecture for hyperbolic one-relator groups: every such group is virtually free-by-cyclic, with the free kernel allowed to have infinite rank.

Original PDF | Source files and supporting material

259-01 • Result family 259

A group without fixed price

We construct a finitely generated group with two essentially free probability-measure-preserving actions of different costs. This gives a negative answer to Gaboriau's general fixed-price problem. The group is an amalgam of a free group of rank 100 with a direct product. Its Bernoulli action has cost bounded away from one, whereas finite height extensions have costs tending to one.

Original PDF | Source files and supporting material

Mathematical physics

59 manuscripts • Official subject grouping

260-01 • Result family 260

The spacetime Penrose inequality with charge and original-data rigidity

Under the stated energy, decay, and trapping hypotheses, we prove the sharp charged spacetime Penrose upper-area inequality for one-ended three-dimensional initial data with source-free electric and magnetic fields. The theorem allows arbitrary second fundamental form, nonzero ADM momentum, and disconnected boundary. Writing m for invariant ADM mass, Q for total charge magnitude, and rA for the minimum-enclosing-area radius, the bound is m ≥ Q and $r_A \leq m + \sqrt{m^2 - Q^2}$. The polynomial mass bound $m \geq (r_A + Q^2/r_A)/2$ is asserted only when $r_A > Q$. When m > Q, equality under the stated connected, outermost, outer-area-minimizing future-horizon hypotheses identifies the original data as a smooth spacelike slice of dyonic Reissner-Nordström, including smooth attachment at the future horizon.

Original PDF | Source files and supporting material

260-02 • Result family 260

A Charged Reduction of the Spacetime Penrose Inequality in Spatial Dimensions at Least Four

Using the companion neutral spacetime Penrose theorem exactly at its stated strong-decay, future-trapped, positive-area, and future-timelike scope, we prove the sharp purely electric upper-area inequality in every spatial dimension n ≥ 4 for one-ended charged data satisfying the charged dominant energy condition and $\mathrm{div}_g E = 0$, with the specified decay, integrability, and finite-flux assumptions. The data may have arbitrary interior topology and ADM momentum, with a future or past trapping sign chosen independently on each boundary component. Positive enclosing area and strict ADM timelikeness are conclusions. The polynomial mass rearrangement is asserted only when the $(n-2)$nd power of the enclosing-area radius exceeds $|Q|$. When $m > |Q|$, equality under the stated connected, outermost, outer-area-minimizing future-horizon hypotheses recovers the original metric, second fundamental form, and electric field as a global spacelike slice of a Reissner-Nordström-Tangherlini exterior. The equality classification includes only slices whose induced magnetic two-form vanishes.

Original PDF | Source files and supporting material

260-03 • Result family 260

Spacetime Penrose inequalities: enclosing area, charge, and rigidity

We prove sharp spacetime Penrose inequalities for invariant ADM mass and minimum enclosing area in three and four spatial dimensions. Under their respective designated-end conventions, the neutral numerical results allow arbitrary second fundamental form, nonzero momentum, disconnected weakly future trapped boundary, and finitely many ends. Under the stated horizon hypotheses, equality for the neutral inequalities reconstructs the original exterior data as spacelike slices of Schwarzschild spacetime in three dimensions and Schwarzschild-Tangherlini spacetime in four; the four-dimensional equality theorem concerns a connected, one-ended exterior. We also prove a three-dimensional electric-magnetic charged upper-area inequality for one-ended data with nonzero momentum, with separate purely electric rest-frame corollaries allowing finitely

many ends. Finally, for each fixed transverse-traceless seed and prescribed decaying solution branch, we prove a local Schwarzschild-anti-de Sitter inequality.

Original PDF | Source files and supporting material

260-04 • Result family 260

The Kerr-Newman Penrose Inequality for Axisymmetric Electrovacuum Exteriors

We prove the sharp Kerr-Newman Penrose inequality $m^2 \geq \frac{A}{16\pi} + \frac{Q^2}{2} + \frac{\pi(Q^4+4J^2)}{A}$. Here $Q^2 = Q_e^2 + Q_b^2$, and J is the conserved total angular momentum, including its electromagnetic contribution. The result applies to smooth axisymmetric electrovacuum exteriors in the stated one-ended topological and decay class, with a connected outermost, outer-area-minimizing future marginally outer trapped boundary, Coulomb electromagnetic asymptotics, zero ADM momentum, and the explicitly assumed physical-area condition $A \geq 4\pi\sqrt{Q^4+4J^2}$. No maximality assumption is made. On the strict area branch, equality within this class characterizes the original data as an admissible spacelike exterior slice of a subextremal dyonic Kerr-Newman spacetime, with the boundary mapped smoothly to a future-horizon cross-section or the bifurcation sphere.

Original PDF | Source files and supporting material

260-05 • Result family 260

Electromagnetic tails and the Kerr-Newman Penrose inequality

We construct smooth axisymmetric electrovacuum exteriors that violate the Kerr-Newman Penrose inequality when J is the bare gravitational ADM angular momentum and the electromagnetic fields have only $O(r^{-2})$ decay, allowing angularly varying leading tails. The examples have zero total electric and magnetic charge even though both electromagnetic fields are nonzero. They have a connected outermost, outer-area-minimizing future marginally outer trapped boundary and lie strictly on the physical area branch. We also obtain equality examples whose original data admit no Kerr-Newman spacelike realization with the corresponding mass, angular momentum, and charges. These counterexamples do not refute formulations using conserved total angular momentum with its electromagnetic correction or stronger Coulomb asymptotics.

Original PDF | Source files and supporting material

260-06 • Result family 260

The nonmaximal anti-de Sitter Penrose Inequality and original-data rigidity

We prove the spacetime Penrose Inequality for smooth three-dimensional initial-data exteriors with one spherical asymptotically anti-de Sitter end, satisfying the stated decay and integrability assumptions, the dominant energy condition, and a compact weakly future outer trapped boundary. We assume that the hyperbolic metric four-flux is future timelike; its Lorentz norm is the mass, and the area is the infimum over full enclosing cuts. On the connected outermost, outer area-minimizing horizon subclass, equality characterizes the original data as a spacelike hypersurface in Schwarzschild-anti-de Sitter spacetime with matching metric mass. No maximality, evolution, or auxiliary solvability assumption is used.

Original PDF | Source files and supporting material

260-07 • Result family 260

The Penrose inequality for maximal asymptotically hyperbolic initial data

We prove the sharp Penrose inequality for three-dimensional maximal asymptotically hyperbolic initial data with spherical conformal infinity. Under the dominant energy condition, the stated decay and integrability assumptions, and a future-timelike mass covector, the invariant mass is bounded below by the Schwarzschild-anti-de Sitter mass associated with the minimum enclosing area of a weakly future outer-trapped boundary. The boundary may be disconnected, and no restriction is imposed on the compact topology.

Original PDF | Source files and supporting material

260-08 • Result family 260

A local Penrose inequality for conformal perturbations of Schwarzschild-anti-de Sitter data

We prove the asymptotically hyperbolic Penrose inequality for sufficiently small maximal vacuum conformal perturbations of a positive-mass Schwarzschild-anti-de Sitter exterior, for every fixed decaying transverse-traceless seed and every solution branch satisfying the stated decay and mass assumptions. The bound uses the area of the marginally outer trapped boundary itself; it requires neither outermostness nor outer area-minimization, and no bulk-versus-boundary domination condition. For sufficiently small positive parameters, equality holds for radial seeds and the inequality is strict otherwise.

Original PDF | Source files and supporting material

260-09 • Result family 260

The spacetime Penrose inequality and enclosing area

We prove the sharp spacetime Penrose inequality for smooth one-ended initial-data exteriors in every spatial dimension $n \geq 3$, under the dominant energy condition, weak future trapping, the stated differentiated decay and positive enclosing area. These hypotheses force the ADM energy-momentum to be future timelike. Area is the infimum over full enclosing cuts in the original metric. In dimensions three and four, two metric derivatives and one tensor derivative suffice, and both positive enclosing area and future timelikeness follow from the hypotheses.

Original PDF | Source files and supporting material

260-10 • Result family 260

Conformal flow and the Riemannian Penrose inequality with minimizing frontiers

We give a detailed conformal-flow proof of the numerical Riemannian Penrose inequality in every dimension $n \geq 3$ for complete asymptotically flat exteriors with nonnegative scalar curvature and full compact frontiers that are outer minimizing and locally perimeter minimizing. The metric extends smoothly through the possibly singular frontier; neither spin nor frontier connectedness is assumed. The proof follows the conformal-flow approach of Bray, Bray-Lee, and Bi-Zhu.

Original PDF | Source files and supporting material

260-11 • Result family 260

Equality and rigidity in the spacetime Penrose inequality

Equality in the spacetime Penrose inequality identifies the original initial data as a spacelike slice of a Schwarzschild-Tangherlini exterior, smoothly attached to its horizon, under the stated outermostness and decay hypotheses. We prove this in every spatial dimension $n \geq 3$ for a strong-decay exterior class and for weaker decay in dimensions three and four. In dimension three the finite-component theorem forces a single spherical horizon and gives qualitative strictness for disconnected boundaries.

Original PDF | Source files and supporting material

260-12 • Result family 260

Boundary graph deformations for the spacetime Penrose inequality

We construct graph and conformal deformations of trapped initial-data exteriors in three and four spatial dimensions. The resulting metrics have nonnegative scalar curvature, strictly negative inner mean curvature, a lower bound protecting every enclosing cut, and an arbitrarily small upper error in ADM energy. We also give a direct four-dimensional maximal vacuum construction that retains a noncompact decaying second fundamental form. These constructions give boundary routes to the corresponding numerical Penrose inequalities.

Original PDF | Source files and supporting material

260-13 • Result family 260

Area-controlled end replacement and the Bondi Penrose inequality in the CKS class

We replace a three-dimensional Cha-Khuri-Sakovich hyperboloidal end by an asymptotically flat end while preserving the dominant energy condition and each fixed compact interior. For strictly future-timelike CKS initial-data charge, the replacements lose asymptotically no enclosing area and their ADM masses tend to the invariant Bondi mass. Combining this construction with the companion numerical spacetime Penrose theorem yields the sharp Bondi bound for possibly disconnected weakly future trapped boundaries in this class. Horizon-regular Schwarzschild exteriors attain equality at every positive mass.

Original PDF | Source files and supporting material

261-01 • Result family 261

Absolutely Continuous Spectrum for Weak-Disorder Anderson Models in Dimensions at Least Three

For each fixed dimension $d \geq 3$ and each fixed sufficiently small positive disorder strength, we prove that the Anderson operator on $\mathbb{Z}d$ with independent uniform site potentials almost surely has purely absolutely continuous spectrum with nonzero weight on an open energy interval. The interval may depend on d but is independent of the disorder strength. This settles the purely absolutely continuous energy-range question in Simon's Problem 1 for this model.

Original PDF | Source files and supporting material

261-02 • Result family 261

Pure-Point Spectrum for the Two-Dimensional Anderson Model at Every Positive Disorder

For each fixed positive disorder strength, we prove that the nearest-neighbor Anderson operator on the square lattice with independent uniform site potentials almost surely has pure-point spectral type throughout its spectrum. This resolves the pure-point assertion of the two-dimensional Anderson localization conjecture for the uniform single-site law.

Original PDF | Source files and supporting material

262-01 • Result family 262

Equality cases in the sharp one-dimensional matrix Lieb-Thirring inequality

We classify all equality cases in the sharp one-dimensional Lieb-Thirring inequality for every finite matrix size and $1/2 < \gamma < 3/2$. For measurable Hermitian positive semidefinite potentials W with $\int_{\mathbb{R}} \operatorname{tr}(W^{\gamma+1/2}) < \infty$, equality holds precisely for direct sums, in one constant unitary basis, of scalar one-bound-state solitons and zero channels. The nonzero solitons may have independent scales and centers.

Original PDF | Source files and supporting material

262-02 • Result family 262

Sharp one-dimensional Lieb-Thirring inequalities for matrix potentials

We prove the sharp one-dimensional Lieb-Thirring inequality for every finite matrix size and every exponent $1/2 < \gamma < 3/2$. The optimal constant is the scalar one-bound-state constant, independently of the matrix size. The inequality bounds the full sum of negative eigenvalue moments for every measurable Hermitian positive semidefinite potential W satisfying $\int_{\mathbb{R}} \operatorname{tr}(W^{\gamma+1/2}) < \infty$. The matrices may have arbitrary rank and need not commute at different points.

Original PDF | Source files and supporting material

262-03 • Result family 262

Sharp one-dimensional Lieb-Thirring constants

We resolve affirmatively the remaining cases of the scalar one-dimensional Lieb-Thirring conjecture: for every $\frac{1}{2} < \gamma < \frac{3}{2}$, the optimal constant is the one-bound-state constant. The estimate holds for every nonnegative potential in $L^{\gamma+1/2}(\mathbb{R})$, with all negative eigenvalues included.

Original PDF | Source files and supporting material

263-01 • Result family 263

Uniform excess charge for Coulomb molecules and the outer radius of neutral atoms

We prove that a molecule with M fixed nuclei of real charges at least one and total nuclear charge Z strictly binds at most $Z + CM$ electrons, where C is universal. The result holds for the full nonrelativistic Coulomb Hamiltonian with two spin states and arbitrary distinct nuclear positions. For neutral atoms with integer nuclear charge $Z \geq 1$, the first ionization energy and, for every ground state, the radius outside which one half of an electron remains are each bounded above and below by positive universal constants.

Original PDF | Source files and supporting material

263-02 • Result family 263

Generalized ionization energies for full Coulomb atoms

We prove the energy part of the generalized ionization conjecture for full nonrelativistic Coulomb atoms with two electron spin states. The energy needed to remove m electrons is asymptotic to $a_{\mathrm{TF}} m^{7/3}$ whenever $m \to \infty$ and $Z/m \to \infty$, with the Thomas-Fermi constant in atomic units. We also obtain both conjectured iterated limits.

Original PDF | Source files and supporting material

263-03 • Result family 263

Generalized outer-electron radii of neutral Coulomb atoms

We prove the radius part of the generalized ionization conjecture for neutral nonrelativistic Coulomb atoms with two spin states. For every choice of ground states, the upper and lower large-nuclear-charge limits of the radius defined by an expected exterior electron mass m are both asymptotic to $(81\pi^2/2)^{1/3} m^{-1/3}$ as m tends to infinity.

Original PDF | Source files and supporting material

264-01 • Result family 264

Generic Future Inextendibility with Square-Integrable Connection Near a Fixed Kerr Spacetime

For each fixed rotating subextremal Kerr background with mass M > 0 and rotation $0 < |\mathfrak{a}| < M$, we prove that the smooth vacuum data near its complete two-ended bridge whose full maximal globally hyperbolic development admits a future $C^0 \cap W^{1,2}_{\mathrm{loc}}$ extension form a meagre set in the weighted smooth topology. The extension metric is continuous and nondegenerate, and its weak connection is locally square-integrable. The neighborhood requires smallness of only the tenth weighted seminorm, while the topology tests every finite order.

Original PDF | Source files and supporting material

264-02 • Result family 264

Generic C1 Future Inextendibility Near Rotating Subextremal Kerr Spacetimes

For each fixed Kerr spacetime with mass M > 0 and rotation $0 < \mathfrak{a} < M$, we prove that a dense Gδ set of nearby smooth, complete two-ended vacuum data has a maximal globally hyperbolic development with no future C1 extension. The neighborhood and genericity are defined in a weighted smooth topology, and ambient extensions may be nonvacuum.

Original PDF | Source files and supporting material

264-03 • Result family 264

Quantitative Near-Kerr Evolution and Generic C2 Future Inextendibility

For every fixed rotating subextremal Kerr bridge, we prove that smooth vacuum data admitting a C2 future extension of their full maximal globally hyperbolic development form a meagre set in a neighborhood defined by one finite-order seminorm. The topology allows arbitrary symbol-bounded asymptotically flat tails and imposes no symmetry.

Original PDF | Source files and supporting material

265-01 • Result family 265

A two-dimensional area law from a global spectral gap

We prove an entropy area law for the unique ground state of a finite-range Hamiltonian on any finite induced subgraph of the square lattice. A lower bound on the spectral gap of the full Hamiltonian and fixed bounds on the local dimension, interaction range, and interaction strength suffice. For every set of sites, its entanglement entropy is bounded by a constant times the number of edges crossing its boundary, independently of the size and shape of the domain.

Original PDF | Source files and supporting material

265-02 • Result family 265

Polynomial PEPS approximation of gapped square-grid ground states

We prove that the unique ground state of a uniformly gapped nearest-neighbor Hamiltonian on an $L \times L$ square lattice admits a projected entangled-pair state approximation with bond dimension polynomial in L and global vector error at most L−1 after normalization. Only the gap of the full Hamiltonian is assumed. The result is an existence theorem, with constants uniform over Hamiltonians of fixed local dimension, interaction strength, and gap.

Original PDF | Source files and supporting material

266-01 • Result family 266

The maximum number of mutually unbiased bases in dimension six

We prove that the maximum number of mutually unbiased orthonormal bases in $\mathbb{C}6$ is three, resolving Zauner's dimension-six MUB conjecture. The upper bound is computer-assisted: under the stated binary64 arithmetic and compiler conditions, a complete execution of the documented verification pipeline excludes four arbitrary complex bases.

Original PDF | Source files and supporting material

266-02 • Result family 266

Exact Fourier certificates for complex Hadamard matrices of order six

We prove the Fourier-vanishing conjecture of Matolcsi, Ruzsa, and Weiner: every complex Hadamard matrix of order six outside Tao's cubic equivalence class has a vanishing character sum at every permutation of $(1, 1, 1, -1, -1, -1)$. We also give an exact certificate excluding seven mutually unbiased bases in $\mathbb{C}6$; by Weiner's completion theorem, this yields an upper bound of five. Both certificates use only integer and rational arithmetic, and the complete verifier is included.

Original PDF | Source files and supporting material

267-01 • Result family 267

Bose-Einstein condensation at positive temperature in the dilute hard-sphere gas

We prove Bose-Einstein condensation at positive temperature in the three-dimensional dilute hard-sphere gas. For each fixed exclusion distance and every sufficiently small fixed density, there is a strictly positive temperature, independent of the volume, at which the exact canonical Gibbs state has a positive condensate fraction in the thermodynamic limit. The condensate occupies the constant orbital.

Original PDF | Source files and supporting material

267-02 • Result family 267

Quantum Depletion and Momentum Distribution in the Dilute Hard-Sphere Bose Gas

We prove the full scaled Bogoliubov momentum distribution for the depleted particles in the three-dimensional hard-sphere Bose gas at zero temperature. The thermodynamic limit is taken at each fixed density before the dilute limit, uniformly over all pure and mixed ground states. All thermodynamic accumulation values have the same dilute asymptotic: the limiting nonzero-momentum occupation measure has total mass $8/(3\sqrt{\pi})$, giving the depletion fraction $\frac{8}{3\sqrt{\pi}}\sqrt{\rho a^3} + o(\sqrt{\rho a^3})$ for density ρ and hard-sphere exclusion distance a.

Original PDF | Source files and supporting material

267-03 • Result family 267

Quantum Depletion for Fixed Bounded Repulsive Potentials

We prove the Bogoliubov quantum-depletion asymptotic for the ground state of a three-dimensional Bose gas with a fixed bounded, nonnegative, radial interaction of finite range and positive scattering length a. The thermodynamic limit is taken at each fixed density ρ before the dilute limit. Every thermodynamic accumulation value of the fraction outside the constant mode is $\frac{8}{3\sqrt{\pi}}\sqrt{\rho a^3} + o(\sqrt{\rho a^3})$ as $\rho \downarrow 0$. The assertion is uniform over ground-state density matrices and does not require the occupation to have a unique thermodynamic limit.

Original PDF | Source files and supporting material

267-04 • Result family 267

A density-uniform condensate bound for dilute Bose gases

For each fixed bounded measurable, nonnegative, radial interaction of finite range in three dimensions that is not zero almost everywhere, we prove a positive lower bound on the constant-orbital condensate fraction that is uniform over all sufficiently small densities and all temperatures between zero and the square of the density. The interaction and density remain fixed in the thermodynamic limit.

Original PDF | Source files and supporting material

267-05 • Result family 267

Ground-state condensation in the dilute hard-sphere gas

We prove Bose-Einstein condensation in every ground state of a dilute three-dimensional hard-sphere Bose gas. At every sufficiently small fixed gas parameter, the constant orbital contains a positive fraction of the particles in the thermodynamic limit. The fraction can be chosen independently of the gas parameter, and the conclusion holds for arbitrary complex ground states.

Original PDF | Source files and supporting material

268-01 • Result family 268

The periodic spin-one Haldane gap

We prove a uniform positive spectral gap for the pure antiferromagnetic spin-one Heisenberg chain on even periodic rings, establishing the positive even-periodic formulation of the spin-one Haldane conjecture.

Original PDF | Source files and supporting material

268-02 • Result family 268

A boundary-field gap for the spin-one Heisenberg chain

We prove a uniform spectral gap for odd open spin-one antiferromagnetic Heisenberg chains with the same fixed magnetic field at both endpoints. The field $h = 3/5$ gives a gap greater than $\log(10)/392$ on every chain of $2L+1$ sites with L ≥ 960. Tasaki's index theorem then gives index −1 for every subsequential local limit of these boundary-selected ground states.

Original PDF | Source files and supporting material

269-01 • Result family 269

Uniform Stability of the Spherical Laughlin Gap

We prove that the fermionic Laughlin V1 Hamiltonian at filling 1/3 on expanding round spheres retains a unique ground state and a uniform spectral gap under sufficiently weak bounded real scalar one-body potentials projected to the lowest Landau level. With coefficient one for each pair projector and Laughlin flux $q = 3(N-1)$, the gap and perturbation threshold are uniform for all sufficiently large particle numbers and all potential profiles of supremum norm at most one. The result uses the uniform unperturbed Fock-space gap and gives existential constants.

Original PDF | Source files and supporting material

269-02 • Result family 269

A Fock-space inequality and the Laughlin spectral gap

We prove the spherical fermionic Laughlin spectral-gap conjecture for the full V1 interaction. With coefficient one for each pair projector, the gap above the Laughlin state at filling 1/3 on the round sphere is at least 1/25 for all sufficiently large systems. More generally, $H_Q^2 \geq \gamma H_Q$ for some fixed $\gamma > 1/25$ on the entire lowest-Landau-level Fock space for all sufficiently large flux Q, independently of particle number.

Original PDF | Source files and supporting material

270-01 • Result family 270

The unique threshold bound state of the SU(N) BFSS model

For every finite N $\geq$ 2, we prove that undeformed SU(N) BFSS matrix quantum mechanics has exactly one normalizable zero-energy state after removing the center of mass. This establishes the threshold-bound-state conjecture.

Original PDF | Source files and supporting material

270-02 • Result family 270

Positive eigenvalues of the relative SU(2) BFSS Hamiltonian

We prove that the relative SU(2) BFSS Hamiltonian has infinitely many positive eigenvalues tending to infinity, with square-integrable eigenvectors. The operator is defined by closing the gauge-invariant supercharge form. This refutes, at N = 2, the exclusion of normalizable positive-energy states stated in the original BFSS paper.

Original PDF | Source files and supporting material

271-01 • Result family 271

Bloch's Law for Finite-Range Heisenberg Ferromagnets in Three Dimensions

We prove Bloch's T3/2 law for the spontaneous magnetization of three-dimensional quantum Heisenberg ferromagnets at every fixed positive quantum spin. The result holds for every non-negative symmetric finite-range interaction whose support generates $\mathbb{Z}3$, including spatially anisotropic couplings. The leading magnetization deficit has the exact coefficient determined by the determinant of the quadratic one-magnon dispersion. The thermodynamic limit is taken before the right field derivative at zero, and the low-temperature limit is taken last.

Original PDF | Source files and supporting material

271-02 • Result family 271

The first lattice correction to Bloch's law

We prove the first lattice correction to Bloch's law for the three-dimensional nearest-neighbor quantum Heisenberg ferromagnet at every fixed spin $S = \frac{1}{2}, 1, \frac{3}{2}, \ldots$. The spontaneous-magnetization deficit agrees with the full ideal-magnon density up to $o(\beta^{-5/2})$. In addition to the leading Bloch term, this gives the correction $3\zeta(5/2)(\beta S)^{-5/2}/(128\pi^{3/2})$. The magnetization is the right derivative at zero field of the thermodynamic pressure; the volume limit precedes the field derivative, and the low-temperature limit is taken last.

Original PDF | Source files and supporting material

271-03 • Result family 271

The spherical magnetization law for the three-dimensional quantum Heisenberg ferromagnet

We prove the spherical magnetization law for the three-dimensional nearest-neighbor isotropic quantum Heisenberg ferromagnet at every fixed positive quantum spin and every sufficiently low fixed positive temperature. As the even periodic cubes grow, the symmetric zero-field magnetization converges in moments to a uniform direction with a deterministic positive magnitude. This

magnitude equals the right derivative at zero field of the infinite-volume pressure. The law is expressed through self-adjoint linear combinations of the spin components and requires no joint measurement of noncommuting observables.

Original PDF | Source files and supporting material

271-04 • Result family 271

Spontaneous magnetization in the quantum Heisenberg ferromagnet

For every dimension d $\geq$ 3 and every spin $S \in \{\frac{1}{2}, 1, \frac{3}{2}, ...\}$, we prove that the nearest-neighbor isotropic quantum Heisenberg ferromagnet has a translation-invariant, spontaneously magnetized equilibrium state at every sufficiently low positive temperature. The same state satisfies the KMS condition for the zero-field dynamics and has magnetization at least $S/4$. This resolves the low-temperature ordering problem in the spontaneous-magnetization formulation.

Original PDF | Source files and supporting material

272-01 • Result family 272

Entanglement with zero distillable secret key in local dimension ten

We construct an entangled state on $\mathbb{C}^{10} \otimes \mathbb{C}^{10}$ with zero distillable secret key for the local-instrument protocols specified here. They allow joint processing of all copies and unlimited two-way public communication, with the input as the only shared private resource and an eavesdropper holding a purification and the complete public record. The construction also disproves the unrestricted two-map PPT-composition conjecture. A separate trace-preserving PPT channel on $M_{21}(\mathbb{C})$ has a square that is not entanglement breaking.

Original PDF | Source files and supporting material

273-01 • Result family 273

The entropy photon-number inequality

We prove the entropy photon-number inequality for two independent bosonic inputs with finite mean energy, for every finite number of modes. This resolves the entropy photon-number conjecture in this finite-energy setting while allowing arbitrary entanglement among the modes within either input. As a consequence, we determine the exact minimum output entropy at fixed input entropy for every finite tensor power of an identical thermal attenuator, over finite-energy inputs that may be entangled across modes. We also obtain the exact classical capacity region of the degraded two-receiver pure-loss bosonic broadcast channel under a mean photon constraint.

Original PDF | Source files and supporting material

274-01 • Result family 274

Product-projection localization and the QAC0 parity lower bound

We prove that constant-depth quantum circuits with arbitrary one-qubit and unbounded-arity Toffoli gates cannot compute parity with any fixed positive worst-case advantage using polynomially many qubits. Ancillas start in zero, only one output qubit is measured, and all final garbage is unrestricted. This resolves Moore's parity conjecture in the measured-output model.

Original PDF | Source files and supporting material

274-02 • Result family 274

Regular trajectories, pruning and quantum parity

We prove that constant-depth quantum circuits with arbitrary one-qubit gates and unbounded-arity Toffoli gates cannot compute parity with any fixed positive worst-case advantage using polynomially many total qubits. Ancillary qubits are initialized to $|0\rangle$, one output qubit is measured, and all other final registers may be discarded without restriction. This resolves Moore's parity conjecture in the measured-output model.

Original PDF | Source files and supporting material

275-01 • Result family 275

Continuum Coulomb hardness with binary nuclear charges

We prove that approximating the electronic Coulomb spectral infimum in three-dimensional space is QMA-hard when positive integer nuclear charges are encoded in binary. The nuclei have distinct rational positions, the electron number is unary, and the energy is minimized over all antisymmetric continuum states and spin sectors. A deterministic classical polynomial-time reduction produces instances with threshold separation at least one. The nuclear charges may be exponentially large, but every output has polynomial bit length.

Original PDF | Source files and supporting material

275-02 • Result family 275

QMA-hardness of continuum Coulomb energy with unit nuclear charges

We prove that approximating the electronic ground-energy infimum for clamped unit-charge nuclei is QMA-hard on the full spinful fermionic continuum space. A deterministic classical polynomial-time reduction produces polynomially many nuclei at distinct rational positions and polynomially many electrons, with polynomial rational bit lengths and threshold separation at least one. No orbital basis, magnetic field, or additional external potential is supplied, and no binding assumption is imposed.

Original PDF | Source files and supporting material

276-01 • Result family 276

Classical capacity and entropy inequalities for generalized amplitude damping

We determine the unassisted classical capacity of every qubit generalized amplitude-damping channel, for all damping strengths and thermal occupations. It equals the one-shot Holevo capacity and is given by a one-variable maximum. A binary pure-state ensemble attains the one-use optimum, and its independent products attain the optimum at every block length. We also prove that one-shot Holevo capacity, minimum output entropy, and regularized unassisted classical capacity are additive under tensor product with every finite-dimensional completely positive trace-preserving partner channel.

Original PDF | Source files and supporting material

277-01 • Result family 277

Threshold parallel repetition for finite-dimensional entangled games

For every finite two-player game with entangled value v < 1, we prove exponential decay for the probability of winning at least a $v + \delta$ fraction of k independent repetitions, uniformly over all finite-dimensional joint strategies. The result allows arbitrary correlated question distributions and holds for every $0 < \delta < 1 - v$ and k ≥ 1. Its universal rate is proportional to $\delta^5/(1 + \log(|\mathcal{A}||\mathcal{B}|))$, where $\mathcal{A}, \mathcal{B}$ are the answer alphabets. For each fixed question distribution, we also obtain an explicit cubic rate in δ.

Original PDF | Source files and supporting material

278-01 • Result family 278

A Coulomb ground-state density without Kohn-Sham ensemble representation

We construct a finite three-electron Coulomb molecule whose spin-summed ground-state density cannot be reproduced by any ground-state ensemble of noninteracting electrons in a single real, spin-independent local potential in $L^{3/2}(\mathbf{R}^3) + L^{\infty}(\mathbf{R}^3)$. The molecule has two equal positive integer nuclear charges, specified by an exact finite nonnumerical formula.

Original PDF | Source files and supporting material

279-01 • Result family 279

Exact quantum factoring over a fixed finite gate set

We give a polynomial-time uniform quantum circuit family that outputs the complete prime factorization of every integer N ≥ 2 with probability one. A fixed finite set of bounded-arity gates suffices, and both the gate count and the number of qubits have polynomial worst-case bounds in the input length.

Original PDF | Source files and supporting material

280-01 • Result family 280

Strongly rational unitary vertex operator algebras and conformal nets

Every simple unitary strongly rational vertex operator algebra generates a completely rational conformal net. All its simple grading-restricted modules are unitarizable, its canonical fusion forms are positive, and the Carpi-Weiner-Xu functor gives a braided unitary tensor equivalence from its grading-restricted finite-length module category onto the finite-index sectors of the net. Gui's extension theorems then identify normalized irreducible finite-index local extensions of these nets with simple CFT-type conformal extensions of the vertex operator algebras.

Original PDF | Source files and supporting material

281-01 • Result family 281

QAOA attains the SK ground-state energy in the thermodynamic-first limit

We prove that the Quantum Approximate Optimization Algorithm (QAOA) approaches the ground-state energy per spin of the Gaussian zero-field Sherrington-Kirkpatrick model when system size tends to infinity first and circuit depth then increases. For every accuracy, some finite depth and deterministic angles, independent of system size and disorder, achieve that accuracy in the limiting

expected energy per spin using the standard cost Hamiltonian and transverse-field mixer. This proves the eventual Parisi-optimality conjecture of Basso, Farhi, Marwaha, Villalonga, and Zhou in its fixed-parameter thermodynamic formulation. We give no quantitative bound on the required depth or efficient angle-selection procedure.

Original PDF | Source files and supporting material

281-02 • Result family 281

Full support of the zero-temperature Sherrington-Kirkpatrick order parameter

We prove that every admissible integrable minimizer of the zero-temperature Parisi functional for the pure, zero-field Sherrington-Kirkpatrick model has full relative Stieltjes support on $[0,1)$. Thus its support has no gaps at any overlap scale below one. We use the covariance normalization $\xi(t) = t^2/2$.

Original PDF | Source files and supporting material

282-01 • Result family 282

Scale and conformal symmetry in four-dimensional operational quantum field theory

We prove that scale symmetry implies local conformal symmetry for a class of four-dimensional unitary, positive-energy quantum field theories with a Poincaré- and scale-invariant vacuum. The class has a discrete, bounded-below spectrum of scaling dimensions with finite multiplicities, and each original field has finite scaling support. The original fields and their adjoints have compatible affiliated realizations in a causally commuting bounded local net, and a physical local dilatation current generates scale transformations through its local Ward identities. The operational framework admits sharply localized reconstructed fields after their joint products and compatible affiliated realizations in the same net have been established. Under these hypotheses, the stress tensor admits a symmetric, conserved, traceless improvement with unchanged translation and Lorentz charges. The corresponding conformal currents satisfy unbroken local Ward identities on the physical field algebra. The conclusion is local; integration to a global conformal action on the whole bounded net remains a separate question.

Original PDF | Source files and supporting material

283-01 • Result family 283

Polynomial-Time Unitary Synthesis from a Boolean Oracle

We give a positive answer to the constant-error formulation of the Aaronson-Kuperberg unitary synthesis problem. For every n, a quantum oracle circuit generated in polynomial time from n alone can approximate the channel of every n-qubit unitary to full diamond-norm error at most 1/2, after a suitable Boolean oracle is chosen. Using the fixed gates $H, T, T^\dagger$, CNOT, the circuit has polynomially many qubits, elementary gates, and oracle calls, and its oracle queries have polynomial length. The oracle may depend on the target unitary; the theorem does not give an efficient classical procedure for constructing it.

Original PDF | Source files and supporting material

284-01 • Result family 284

A Nearly Quartic Separation Between Randomized and Quantum Query Complexity

We construct total Boolean functions with a nearly quartic separation between bounded-error randomized and quantum query complexity. Writing these complexities as $R(f)$ and $Q(f)$, the examples rule out every universal bound $R(f) = O((1 + Q(f))^{\alpha})$ with $\alpha < 4$. Thus the known quartic upper bound has the optimal exponent, disproving the conjectured cubic bound. Both complexities count worst-case bit queries with error at most 1/3 on every input.

Original PDF | Source files and supporting material

Operator algebras

31 manuscripts • Official subject grouping

285-01 • Result family 285

A torsion-free counterexample to reduced Baum-Connes injectivity

We disprove rational injectivity of the coefficient-free reduced Baum-Connes assembly map for torsion-free groups. Specifically, we construct a finitely generated torsion-free discrete group whose degree-zero assembly map has a kernel class of infinite order.

Original PDF | Source files and supporting material

285-02 • Result family 285

A torsion-free counterexample to the Kadison-Kaplansky projection conjecture

We construct a finitely generated torsion-free discrete group whose reduced group C$*$-algebra contains a projection other than zero and the identity. This disproves the Kadison-Kaplansky projection conjecture.

Original PDF | Source files and supporting material

285-03 • Result family 285

An irrational-trace counterexample to reduced Baum-Connes

We construct a finitely generated discrete group whose reduced group C$*$-algebra contains a projection of irrational canonical trace. Its K0-class lies outside the image of the coefficient-free reduced Baum-Connes assembly map. This disproves the coefficient-free reduced Baum-Connes conjecture for countable discrete groups.

Original PDF | Source files and supporting material

286-01 • Result family 286

Arithmeticity of twisted finite correspondences for lattices over local fields

We prove arithmetic exhaustion for bifinite correspondences between arbitrarily scalar-twisted group factors of ICC groups commensurable with property-(T) lattices over characteristic-zero local fields and twisted group factors of arbitrary countable ICC groups. Every such correspondence is a closed summand of a finite direct sum of models obtained from actual finite-index subgroup isomorphisms and finite-dimensional projective representations of the matched cocycle ratio. The proof includes reducible products, mixed real and finite places, and quaternionic and Cayley rank-one factors.

Original PDF | Source files and supporting material

286-02 • Result family 286

The arithmetic category and stable recovery of lattice factors

We prove stable canonical recovery for scalar-twisted group factors of ICC groups commensurable with property-(T) lattices over characteristic-zero local fields. Every specified stable isomorphism to a scalar-twisted factor of any countably infinite ICC group recovers the group, the cocycle up to a normalized cochain, and equal amplification scales, with an implementing partial isometry in the given matrix corners. We compute the arithmetic correspondence subcategory for arbitrary countable ICC groups, including all bounded maps, summands, conjugates, and Connes fusion, and combine it with the companion's arithmetic exhaustion theorem. Multiplication also recovers arbitrary finite-index factor neighbors, with the exact projective condition on the matched cocycle ratio.

Original PDF | Source files and supporting material

287-01 • Result family 287

An isomorphism of the free group factors

We solve the free group factor isomorphism problem affirmatively by proving that $L(\mathbb{F}_2)$ and $L(\mathbb{F}_3)$ are isomorphic as tracial von Neumann algebras. The classical free group factor alternative then implies that all interpolated free group factors, including $L(\mathbb{F}_\infty)$, are isomorphic and have fundamental group $\mathbb{R}_{>0}$.

Original PDF | Source files and supporting material

288-01 • Result family 288

Kadison's similarity theorem through uniform derivation estimates

We prove that every bounded complex-linear unital algebra homomorphism from a unital complex C$*$-algebra into the bounded operators on an arbitrary Hilbert space is similar to a $*$-homomorphism. This resolves Kadison's similarity conjecture positively. We also obtain one universal hyperreflexivity constant for all unital von Neumann algebras on arbitrary complex Hilbert spaces.

Original PDF | Source files and supporting material

289-01 • Result family 289

Universal strong Kadison-Kastler stability

We prove that sufficiently close unital von Neumann algebras are conjugate by a unitary arbitrarily close to the identity. The tolerance depends only on the prescribed distance of that unitary from the identity, uniformly over all algebras, representations, and Hilbert spaces. This resolves the strong Kadison-Kastler conjecture.

Original PDF | Source files and supporting material

289-02 • Result family 289

Near Inclusions of von Neumann Algebras Without Small Spatial Embeddings

We give a negative answer to the unrestricted small-spatial-embedding problem for one-sided near inclusions of von Neumann algebras. On separable complex Hilbert spaces, we construct pairs of unital von Neumann algebras with common identity whose one-sided gaps tend to zero. Spatial

embeddings of the source into the target exist, but every implementing unitary stays a fixed positive distance from the identity. The obstruction therefore concerns small implementing unitaries, rather than the existence of spatial embeddings.

Original PDF | Source files and supporting material

289-03 • Result family 289

Close Separable C*-Algebras Without Spatial Conjugacy

We disprove the separable C∗-algebraic spatial form of the Kadison-Kastler conjecture. For every ε > 0, we construct unital, norm-separable C∗-algebras on a common separable complex Hilbert space, with the same identity and Kadison-Kastler distance less than ε, that are not conjugate by any unitary. The two algebras have the same von Neumann closure.

Original PDF | Source files and supporting material

290-01 • Result family 290

Expected amenable subalgebras preserving core commutants

We prove that every inclusion $N \subset M$ of von Neumann algebras with separable preduals and a faithful normal conditional expectation contains an expected amenable subalgebra $P \subset N$ such that $P' \cap c(M) = N' \cap c(M)$, where $c(M)$ denotes the continuous core of M. This resolves the relative bicentralizer conjecture in this setting.

Original PDF | Source files and supporting material

290-02 • Result family 290

Bounded recovery for modular spectral averages

We prove a bounded spectral recovery theorem for a von Neumann algebra with a faithful normal state whose centralizer consists only of scalars. A positive averaged squared norm for an arbitrary bounded Hilbert-space operator on shrinking modular spectral bands can be recovered on uniformly bounded algebra elements with shrinking spectral support. We apply this theorem to spectral intertwining rigidity and obtain an alternative proof of bicentralizer triviality for type III1 factors with separable predual.

Original PDF | Source files and supporting material

291-01 • Result family 291

Equivariant Jiang-Su Stability for Amenable Actions in the Unital Stably Finite Case

Every action of a countable discrete amenable group on a simple, separable, unital, infinite-dimensional, nuclear, stably finite complex C∗-algebra that is already Jiang-Su stable absorbs the trivial action on the Jiang-Su algebra up to cocycle conjugacy. No restriction is imposed on the action on the tracial-state simplex. This proves the unital, stably finite case of Szabó's Conjecture A on automatic equivariant Jiang-Su stability.

Original PDF | Source files and supporting material

291-02 • Result family 291

Cuntz comparison and Jiang-Su absorption

We prove that strict comparison in the extended-functional sense implies Jiang-Su absorption for separable simple nuclear non-elementary C∗-algebras. This resolves the corresponding implication of the Toms-Winter regularity problem, including nonunital algebras and allowing unbounded traces. More generally, every separable nuclear C∗-algebra whose Cuntz semigroup is almost unperforated and fully almost divisible absorbs the Jiang-Su algebra.

Original PDF | Source files and supporting material

291-03 • Result family 291

Nuclear dimension and Jiang-Su stability without elementary subquotients

For separable nuclear C∗-algebras with no nonzero elementary ideal subquotients, finite nuclear dimension is equivalent to Jiang-Su stability. More generally, $\dim_{\mathrm{nuc}}(A_0 \otimes \mathcal{Z}) \leq 1$ for every separable nuclear A0. This proves Robert and Tikuisis's Conjecture (C1) and the nuclear-dimension equivalence in the nonsimple Toms-Winter regularity question.

Original PDF | Source files and supporting material

291-04 • Result family 291

Tracial projection methods and uniform property Gamma

For a simple, separable, unital, infinite-dimensional, nuclear, stably finite C∗-algebra with traces, real rank zero of the uniform tracial ultrapower of its uniform tracial completion implies uniform property Γ. This answers Problem XXI of Schafhauser, Tikuisis and White affirmatively. Independently, strict comparison implies Jiang-Su absorption for simple, separable, unital, infinite-dimensional nuclear algebras, resolving the unital Toms-Winter conjecture. Under comparison tested on traces of finite target rank, we also obtain uniform property Γ and absorption for simple, separable, nuclear, stably projectionless algebras whose densely finite traces are all bounded and have a nonempty compact normalized base.

Original PDF | Source files and supporting material

292-01 • Result family 292

An explicit obstruction to nuclear norm-ultrapower embeddings

We give a negative answer to Kirchberg's norm-ultrapower embedding problem. We construct an explicit separable unital full group C∗-algebra that admits no unital embedding into Bω, for any nonzero unital nuclear C∗-algebra B and any free ultrafilter ω on $\mathbb{N}$. In particular, it does not embed unitally into $\mathcal{O}_2^\omega$.

Original PDF | Source files and supporting material

293-01 • Result family 293

Invariant-projection counterexamples for every irrational rotation

For every irrational angle, we construct a continuous nonnegative circle weight with exactly one zero and logarithmic integral $-\infty$ whose weighted rotation has no nontrivial invariant projection in the associated hyperfinite type II1 factor. This answers negatively the question of Zhu,

Fang, and Shi, with the angle prescribed in advance. The resulting operator is nonzero and norm-quasinilpotent.

Original PDF | Source files and supporting material

293-02 • Result family 293

Backward intertwiners and a transitive commutant

We give a negative answer to the hyperinvariant-subspace problem by constructing, on every infinite-dimensional separable complex Hilbert space, a nonzero bounded norm-quasinilpotent operator with no nonzero proper closed hyperinvariant subspace. Its commutant is a proper strongly closed unital transitive complex operator algebra.

Original PDF | Source files and supporting material

294-01 • Result family 294

A counterexample to Kaplansky's quasitrace conjecture and failure of tensor-product stable finiteness

We refute Kaplansky's quasitrace conjecture by constructing a separable unital complex C$*$-algebra that admits normalized 2-quasitraces but has no tracial state. As a consequence, we show that the minimal tensor product of two unital simple stably finite complex C$*$-algebras can be properly infinite, even when one factor is the reduced free-group algebra $C_r^*(\mathbb{F}_2)$.

Original PDF | Source files and supporting material

295-01 • Result family 295

Vanishing of higher bounded Hochschild cohomology

We prove that every bounded Hochschild cocycle of degree at least two on a complex von Neumann algebra, with values in the algebra itself, has a bounded primitive. Together with the established degree-one inner-derivation theorem, this resolves the Kadison-Ringrose cohomology conjecture positively.

Original PDF | Source files and supporting material

296-01 • Result family 296

Relative generation and the generator problem for finite factors

We prove that every type II1 factor with separable predual is generated by one operator, equivalently by two self-adjoint operators. Combined with the established direct-integral reduction to this case, this gives an affirmative solution of the generator problem for von Neumann algebras with separable predual.

Original PDF | Source files and supporting material

297-01 • Result family 297

A counterexample to Naimark's problem in ZFC

We construct in ZFC a unital infinite-dimensional simple complex C$*$-algebra with a faithful tracial state whose nonzero irreducible representations are all unitarily equivalent. This gives a negative answer to Naimark's problem without additional set-theoretic assumptions.

Original PDF | Source files and supporting material

298-01 • Result family 298

A finite-entropy separation of microstates and nonmicrostates free entropy

We answer the finite-entropy equality question for microstates and nonmicrostates free entropy negatively. We construct a bounded self-adjoint tuple X in a von Neumann algebra with faithful normal tracial state such that $-\infty < \chi(X) \leq \chi^*(X) - \frac{1}{2} < \infty$. Here χ is the original microstates entropy with an operator-norm cutoff and a limsup over matrix sizes. The counterexample uses a large but fixed number of variables.

Original PDF | Source files and supporting material

299-01 • Result family 299

The Kirchberg-Rørdam character criterion

We prove the Kirchberg-Rørdam character criterion: for every free ultrafilter on $\mathbb{N}$, a nonzero unital separable complex C∗-algebra absorbs the Jiang-Su algebra if and only if its norm central-sequence algebra has no characters. We also prove that the infinite minimal tensor power of every nonzero unital separable complex C∗-algebra without characters is $\mathcal{Z}$-stable and contains a unital copy of $\mathcal{Z}$.

Original PDF | Source files and supporting material

300-01 • Result family 300

Approximation Paving over Arbitrary Maximal Abelian Subalgebras

We prove the approximation-paving conjecture of Popa and Vaes for every maximal abelian subalgebra of a complex von Neumann algebra. For each $0 < \varepsilon < 1$, every self-adjoint operator is a strong limit of self-adjoint operators of norm at most three times its norm, each admitting a norm paving with error at most ε times the approximant's own norm. The number of projections is at most $C\varepsilon^{-6}$ for a universal constant C. No separability or conditional-expectation hypothesis is required.

Original PDF | Source files and supporting material

300-02 • Result family 300

Quadratic Strong-Operator Paving over Arbitrary Maximal Abelian Subalgebras

We prove the quadratic strong-operator paving conjecture of Popa and Vaes. For every $0 < \varepsilon < 1$, every self-adjoint element of a von Neumann algebra admits strong-operator paving over each maximal abelian subalgebra with at most $5 \times 10^8 \varepsilon^{-2}$ projections. The bound is uniform over representations and requires no separability or conditional-expectation assumption.

Original PDF | Source files and supporting material

301-01 • Result family 301

The trace cone classifies Razak-Jacelon stabilizations

We prove that the canonical topological cone of all extended lower-semicontinuous tracial weights determines a separable nuclear C$*$-algebra after tensoring with the Razak-Jacelon algebra and the compact operators, answering Robert's trace-cone classification question positively. The isomorphism realizes the prescribed cone map, with arbitrary ideal structure and without a density assumption on the finite domains of the weights.

Original PDF | Source files and supporting material

302-01 • Result family 302

Filtered products and boundary-preserving compression in complex cobordism

We prove that, in sufficiently large smash powers, an MU-null restriction to a finite pointed subcomplex becomes stably null on the entire union of products with a fixed positive proportion of restricted factors. This coherent vanishing theorem yields boundary-preserving compression of cube-valued maps on every compact metrizable input space. When the torus slot bundle embeds continuously into a trivial bundle of rank less than twice the source rank, the compression places a positive proportion of slots on their boundaries in arbitrarily large powers and fixes every original boundary slot exactly. An example at equality shows that the strict rank inequality cannot be removed.

Original PDF | Source files and supporting material

302-02 • Result family 302

Radius of comparison equals half the mean dimension

We prove the integer-action case of the Phillips-Toms conjecture: for every minimal homeomorphism of an infinite compact metrizable space, the radius of comparison of its crossed product equals one half of its mean dimension, including equality at infinity. For these systems, zero mean dimension and the small boundary property are equivalent to Jiang-Su stability and to finite nuclear dimension of the crossed product; in this case its nuclear dimension is at most one.

Original PDF | Source files and supporting material

303-01 • Result family 303

Weak pure infiniteness and O-infinity absorption

We prove that every complex C$*$-algebra in which each positive element is properly infinite is strongly purely infinite. This answers the ordinary-to-strong part of Kirchberg-Rørdam's comparison question. For exact algebras, we also prove that proper infiniteness of one fixed finite amplification of every positive element implies proper infiniteness of each positive element. Consequently, every separable nuclear algebra with this fixed-amplification property absorbs $\mathcal{O}_\infty$, without assumptions of unitality or simplicity.

Original PDF | Source files and supporting material

Topology

26 manuscripts • Official subject grouping

304-01 • Result family 304

The Hilbert-Smith conjecture in every finite dimension

We prove the Hilbert-Smith conjecture in every finite dimension: every locally compact second-countable Hausdorff group acting faithfully and jointly continuously on a connected Hausdorff second-countable finite-dimensional topological manifold without boundary is a Lie group.

Original PDF | Source files and supporting material

305-01 • Result family 305

A boundary-only obstruction to four-dimensional disk embedding

We disprove the four-dimensional disc embedding conjecture without a fundamental-group hypothesis, even when no homotopy classes or output framings are prescribed. We construct a compact oriented smooth four-manifold containing finitely many disc maps with framed algebraic dual spheres whose boundary circles bound no disjoint locally flat discs. Consequently, the free group on two generators is not good in the sense of Freedman-Quinn, and neither is any group containing it as a subgroup.

Original PDF | Source files and supporting material

305-02 • Result family 305

A marked tensor obstruction to four-dimensional disk embedding

We disprove the unrestricted four-dimensional disk-embedding conjecture. We construct immersed disks in a compact oriented smooth four-manifold with framed algebraic dual spheres satisfying the usual equivariant intersection and reduced self-intersection conditions, but with no pairwise disjoint locally flat replacements that preserve the boundary maps and induced normal framings. The obstruction holds even when the replacement disks' relative homotopy classes are not prescribed.

Original PDF | Source files and supporting material

305-03 • Result family 305

A PD4 group without an aspherical manifold model

We construct a finitely presented integral Poincaré duality group of dimension four that has a finite classifying space but is not the fundamental group of any closed aspherical topological four-manifold. This gives a negative answer to Wall's manifold-realization question in dimension four.

Original PDF | Source files and supporting material

306-01 • Result family 306

Purely cosmetic surgery on knots in the three-sphere

We prove the purely cosmetic surgery conjecture for knots in the three-sphere. Distinct Dehn surgery slopes on a nontrivial smooth knot never give orientation-preservingly homeomorphic manifolds.

Original PDF | Source files and supporting material

307-01 • Result family 307

Failure of rational injectivity for maximal coarse assembly

We construct a uniformly discrete bounded-geometry space whose maximal coarse assembly map has an infinite-order element in its kernel. The space is a coarse disjoint union of finite connected graphs of uniformly bounded degree, so maximal coarse assembly need not be rationally injective even for such graph unions.

Original PDF | Source files and supporting material

307-02 • Result family 307

A counterexample to the coarse Novikov conjecture

We disprove the coarse Novikov conjecture: ordinary coarse assembly need not be rationally injective for uniformly discrete spaces of bounded geometry. We construct a coarse disjoint union of finite graphs of uniformly bounded degree and an infinite-order class in its degree-one coarse K-homology whose image under ordinary coarse assembly in the K-theory of the reduced, locally compact Roe algebra vanishes.

Original PDF | Source files and supporting material

308-01 • Result family 308

Finite Smith-Toda Complexes at Varying Primes

For every nonnegative integer n, we construct a Smith-Toda complex $V(n)$ at some prime p depending on n. It is a finite p-local spectrum whose Brown-Peterson homology is $\mathrm{BP}_*/(p, v_1, \ldots, v_n)$ with its canonical comodule structure and generator in degree zero. Each listed generator is killed to its first power.

Original PDF | Source files and supporting material

308-02 • Result family 308

A finite Smith-Toda complex V(4) at the prime 1009

We construct a Smith-Toda complex $V(4)$ at the prime 1009. It is an ordinary finite 1009-local spectrum whose Brown-Peterson homology is $BP_*/(1009, v_1, v_2, v_3, v_4)$, with its canonical comodule structure and generator in degree zero.

Original PDF | Source files and supporting material

309-01 • Result family 309

The Kervaire invariant problem at the prime three

We solve the Kervaire invariant problem at the prime three for the standard Kervaire classes in the mod-three Adams spectral sequence. These classes survive precisely at indices 0, 2, and 3, and each surviving detection coset contains an element of additive order three. In particular, there is an order-three Kervaire element in stem 322.

Original PDF | Source files and supporting material

310-01 • Result family 310

Rational homology and Quillen's conjecture

We prove Quillen's conjecture for all finite groups and all primes. More precisely, if a finite group G has trivial largest normal p-subgroup $O_p(G)$, then the poset of nontrivial elementary abelian p-subgroups of G has nonzero augmented reduced rational homology. This establishes the stronger rational-homology form of the conjecture.

Original PDF | Source files and supporting material

311-01 • Result family 311

Stabilizer orbits and thick tensor ideals of dualizable K(n)-local spectra

We classify the prime and radical ideals of the Lubin-Tate deformation ring invariant under an open Morava stabilizer subgroup, at every prime and positive height. This proves Chai's invariant-ideal conjecture in the standard finite-residue-field formulation. It also proves the Hovey-Strickland conjecture: the dualizable $K(n)$-local category has exactly $n+2$ thick tensor ideals, and its Balmer spectrum is a chain of $n+1$ points.

Original PDF | Source files and supporting material

312-01 • Result family 312

The Grothendieck homotopy hypothesis via elementary expansions

We prove the Grothendieck homotopy hypothesis for every Grothendieck coherator in the Ara-Henry convention: its weak globular infinity-groupoids recover the homotopy theory of spaces. We also resolve Henry's pushout conjecture, showing that elementary expansions preserve components and all homotopy groups of cellular infinity-groupoids.

Original PDF | Source files and supporting material

313-01 • Result family 313

Finite generation for the K(n)-local sphere

We prove that the homotopy groups of the $K(n)$-local sphere are finitely generated $\mathbb{Z}$p-modules in every integer degree, for every prime p and positive height n. Equivalently, the same conclusion holds after $K(n)$-localizing any finite p-local spectrum. This answers the degreewise finiteness question of Hovey and Hovey-Strickland.

Original PDF | Source files and supporting material

314-01 • Result family 314

Cyclic length and chromatic fixed-point loss

For a finite p-group G and a subgroup H, we prove that the optimal chromatic fixed-point loss equals the shortest length of a subnormal chain from H to G with cyclic quotients. The equality holds at every prime and every nonnegative height, resolving positively the equality proposed by Kuhn and Lloyd.

Original PDF | Source files and supporting material

315-01 • Result family 315

The Singer conjecture in dimension four

We prove the four-dimensional Singer conjecture: the L2-Betti numbers of the universal cover of a closed connected aspherical topological four-manifold vanish outside degree two. More generally, the same vanishing holds for finite connected aspherical integral Poincaré complexes of formal dimension four, including nonorientable ones.

Original PDF | Source files and supporting material

316-01 • Result family 316

The Stable Hurewicz Image of the Sphere at Two

We prove Curtis's conjecture: in every positive degree, the mod-two stable Hurewicz image of the sphere is spanned by the images of η, ν, σ and the Kervaire-invariant-one classes that exist. Consequently, Eccles's conjecture holds for every sphere Sn with n > 0.

Original PDF | Source files and supporting material

317-01 • Result family 317

Thomason Model Structures in Every Strict Higher Dimension

We prove the higher-dimensional Thomason model-structure conjecture of Ara and Maltsiniotis. For every $1 \leq n \leq \infty$, the category of small strict globular n-categories admits a proper combinatorial model structure that is Quillen equivalent to simplicial sets. Its weak equivalences and fibrations are detected by the twice-extended Street nerve $\mathrm{Ex}^2 N_n$, and the Quillen equivalence is given by $c_n \mathrm{Sd}^2 \dashv \mathrm{Ex}^2 N_n$. Thus strict higher categories model the homotopy theory of spaces in every positive finite dimension and in dimension ω.

Original PDF | Source files and supporting material

318-01 • Result family 318

Filtered chromatic splitting at generic primes

For every n ≥ 1 and prime $p > n+1$, we construct a 2^n-stage ordered filtration of $L_{n-1} L_{K(n)} S^{\wedge}_p$ with the classical chromatic-splitting cofibers. The map from the first stage to the target is the canonical localization unit.

Original PDF | Source files and supporting material

318-02 • Result family 318

The height-three chromatic overlap: an explicit filtration and its attachments

For every prime p $\geq$ 5, we construct an explicit eight-stage filtration of $L_2 L_{K(3)} \mathbb{S}^{\wedge}_p$ by the local-sphere layers in the height-three chromatic-splitting pattern. The map from its first stage to the overlap is the canonical unit, and two signed fracture formulas identify all attachments for the chosen local maps and compatibility homotopy. A companion canonical-map theorem further implies that the first height-one attachment is nonzero.

Original PDF | Source files and supporting material

318-03 • Result family 318

A rational obstruction to strong chromatic splitting at height three

For every prime p $\geq$ 5, the canonical map $L_0 L_{K(3)} S \to L_0 L_{K(2)} L_{K(3)} S$ for the sphere spectrum S is nonzero on π−3. Consequently, the height-three strong chromatic splitting formula is false in this range, even as an equivalence of underlying $E(2)$-local spectra without specified summand maps.

Original PDF | Source files and supporting material

318-04 • Result family 318

Failure of finite assembly for a chromatic overlap at the prime three

At the prime three and height three, the chromatic overlap cannot be constructed from the rational, height-one, and height-two local spheres by finitely many sums, shifts, cofibers, and retracts in the category of $E(2)$-local modules over the derived 3-complete sphere. This gives a negative answer to the ordinary finite-assembly question for chromatic overlaps.

Original PDF | Source files and supporting material

318-05 • Result family 318

Counterexamples to weak chromatic splitting: sphere kernels and descent exponents

For the derived p-completion of the sphere, the canonical weak chromatic splitting map has no homotopy retraction at height p for every prime p $\geq$ 5, and at height $p+1$ for every prime p $\geq$ 7. The classical sphere product $\beta_1^{(p-1)^2}$ gives a nonzero kernel class in both ranges.

Original PDF | Source files and supporting material

319-01 • Result family 319

Counterexamples to the Hahn-Wilson conjecture at height two

We disprove the Hahn-Wilson conjecture at height two. For every sufficiently large prime p, we construct a connective p-complete spectrum X of exact fp-type two outside the ordinary thick subcategory generated by the specified standard form of $\mathrm{BP}\langle 2\rangle^{\wedge}_p$. The same spectrum satisfies both localization comparisons $L^f_2 X \simeq L_2 X$ and $L_{T(2)} X \simeq L_{K(2)} X$.

Original PDF | Source files and supporting material

320-01 • Result family 320

Nonhomeomorphic closed aspherical four-manifolds with the same homotopy type

We construct closed connected aspherical topological four-manifolds that are homotopy equivalent but not homeomorphic, disproving the homeomorphism-existence formulation of the Borel conjecture in dimension four. Their common fundamental group is word-hyperbolic. One of the manifolds also has a self-homotopy equivalence not homotopic to a homeomorphism.

Original PDF | Source files and supporting material

321-01 • Result family 321

A Counterexample to Wall's D(2) Problem

We give a negative answer to Wall's finite $D(2)$ problem. We construct a finite connected three-dimensional CW complex satisfying the $D(2)$ finiteness condition but not homotopy equivalent to any finite CW complex of dimension at most two. The example has infinite fundamental group.

Original PDF | Source files and supporting material

Functional analysis

19 manuscripts • Official subject grouping

322-01 • Result family 322

A positive solution to Tingley’s problem

Every surjective isometry between the unit spheres of real Banach spaces extends uniquely to a surjective real-linear isometry, giving an affirmative solution to Tingley's problem. If distances between different radii are not preserved, we realize the positive maximal defect in a possibly enlarged pair of Banach spaces. We then align extremal chords using common supports and Darbo's fixed-point theorem and obtain a contradiction from support and convexity estimates.

Original PDF | Source files and supporting material

323-01 • Result family 323

Relative independence of the separable quotient problem

The separable quotient problem asks whether every infinite-dimensional Banach space has a separable infinite-dimensional quotient. We prove that this statement is independent of ZFC, relative to the consistency of ZFC with a measurable cardinal. This holds over both the real and complex fields. For spaces of norm density $\aleph 1$, we obtain independence relative to the consistency of ZFC alone.

Original PDF | Source files and supporting material

324-01 • Result family 324

Lipschitz Equivalent Separable Banach Spaces Need Not Be Linearly Isomorphic

There are separable real Banach spaces that are globally bi-Lipschitz equivalent but not linearly isomorphic. This gives a negative answer to the separable Banach-space Lipschitz-isomorphism problem.

Original PDF | Source files and supporting material

324-02 • Result family 324

Bi-Lipschitz Absorption of c0 Without a Linear Copy of c0

We construct a separable real Banach space Z that contains no linear copy of c0, yet is bi-Lipschitz equivalent to $Z \oplus_\infty c_0$. The same space contains a bi-Lipschitz image of every separable metric space.

Original PDF | Source files and supporting material

325-01 • Result family 325

A direct proof of the complete Crouzeix inequality

We give a direct proof of the sharp constant-two numerical-range inequality for matrix-valued polynomials in all finite base and coefficient dimensions. This resolves the complete Crouzeix conjec-

ture in its matrix formulation, including matrices whose numerical ranges are points or line segments.

Original PDF | Source files and supporting material

325-02 • Result family 325

The complete Crouzeix theorem: optimal similarity and a common positive boundary representation

We resolve the complete Crouzeix conjecture by proving the sharp constant-two numerical-range inequality for every bounded operator on a complex Hilbert space and every matrix-valued polynomial. No separability assumption is needed. The closure of the numerical range is a complete 2-spectral set, and the bound extends to finite matrix-valued functions holomorphic near that closure. For a finite matrix and a bounded convex domain containing its numerical range with regular real-analytic Jordan boundary, the optimal similarity making its conformal disk image contractive is attained with condition number at most two. For the similar matrix, one continuous positive boundary density of mass the identity represents the evaluation of every matrix-valued function holomorphic near the closed domain.

Original PDF | Source files and supporting material

326-01 • Result family 326

The cotype-cotype conjecture under the approximation property

We prove the cotype-cotype conjecture under the ordinary approximation property. A nonzero real Banach space with this property is K-convex if and only if both the space and its dual have finite Rademacher cotype, possibly with different exponents.

Original PDF | Source files and supporting material

327-01 • Result family 327

Nontrivial Markov Type Forces Superreflexivity

We prove that every real Banach space with Markov type $p > 1$ is superreflexive. Together with the known converse, this characterizes superreflexivity by nontrivial Markov type. This answers Naor's question: every real Banach space with nontrivial Markov type admits an equivalent uniformly smooth norm.

Original PDF | Source files and supporting material

328-01 • Result family 328

Fixed Points of Nonexpansive Maps in Reflexive Banach Spaces

Every nonexpansive selfmap of a nonempty closed bounded convex subset of a real reflexive Banach space has a fixed point. This resolves the reflexive-space fixed point problem for the given norm.

Original PDF | Source files and supporting material

329-01 • Result family 329

Counterexamples to the duality conjecture for metric entropy

We disprove Pietsch's dimension-free duality conjecture for metric entropy. For every proposed pair of universal constants, we construct origin-symmetric convex bodies that violate the corresponding covering-entropy inequality, already when the covering body is a cube.

Original PDF | Source files and supporting material

330-01 • Result family 330

A uniformly discrete counterexample to bounded approximation in Lipschitz-free spaces

We construct a countable uniformly discrete metric space whose real Lipschitz-free space has the approximation property but fails the bounded approximation property, answering Kalton's question negatively. The identity can be approximated on every compact set by finite-rank operators, but no uniform bound on their norms is possible.

Original PDF | Source files and supporting material

331-01 • Result family 331

Asymptotic midpoint uniform convexity and unbounded diamond distortion in a reflexive tree space

We construct a separable reflexive real Banach space whose given norm is asymptotically midpoint uniformly convex but which admits no asymptotically uniformly convex equivalent norm. Its averaged midpoint modulus is at least $\sqrt{1+t^2/12}-1$, and the countably branching diamond of depth k has distortion at least $\sqrt{1+k/12}$ in this space. This gives a negative answer to the reflexive diamond converse for asymptotic uniform convexifiability.

Original PDF | Source files and supporting material

331-02 • Result family 331

Midpoint lenses in segment spaces

For a real segment-forest dual with unbounded finite component heights and the real infinite-height coordinate predual, we bound the tail of an arbitrary displacement in a symmetric lens by $2\sqrt{R^2-\|x\|^2}$, where R is the lens radius and x is its finitely supported center. Both given norms are asymptotically midpoint uniformly convex, although neither space admits an equivalent asymptotically uniformly convex norm. The finite-height forest dual is reflexive.

Original PDF | Source files and supporting material

331-03 • Result family 331

Distortion of countably branching diamonds from midpoint and tree energies

We derive quantitative distortion bounds for countably branching diamond graphs from midpoint estimates and direct tree energies. In the dual of a finite-height segment forest, every distortion-D embedding of the depth-k diamond satisfies $D^2 \geq 1+k/4$. The same bound holds in the infinite-height coordinate predual. We also obtain power-type distortion bounds for path-cost and recursive tree norms.

Original PDF | Source files and supporting material

331-04 • Result family 331

Exact asymptotic moduli in a Daugavet subspace of L1

For an infinite-dimensional real subspace of L1 whose unit ball is totally bounded in measure and whose norm has the Daugavet property, we compute the averaged midpoint and one-sided asymptotic moduli at every unit center. They are $\max\{t/2, t-1\}$ and $\max\{0, t-2\}$, respectively. The same weak-neighborhood geometry excludes every equivalent asymptotically uniformly convex norm. A quantitative realization of the Kadets-Werner construction supplies a space with both hypotheses.

Original PDF | Source files and supporting material

331-05 • Result family 331

Midpoint convexity from bounded tree potentials and path costs

Four real Banach spaces defined by bounded tree potentials satisfy the averaged asymptotic midpoint bound $\hat{\delta}(t) \geq \sqrt{1+t^2/4} - 1$ for $0 < t < 1$, while none admits an asymptotically uniformly convex (AUC) renorming. On finite-height trees, the globally constrained norm equals the least additive cost of a Hilbert vector and root paths. The quadratic path and segment-start outer Hilbert sums are reflexive and asymptotically midpoint uniformly convex, and admit no equivalent AUC norm.

Original PDF | Source files and supporting material

331-06 • Result family 331

Independent products in real L1: asymptotic midpoint convexity without AUC renormings

For a countably branching tree, the closed real L1 spans of products of independent exponential or Gaussian-square multipliers along its paths have positive averaged asymptotic midpoint moduli and admit no equivalent asymptotically uniformly convex norm. The renorming obstruction holds for every positive nonconstant mean-one multiplier with finite second moment.

Original PDF | Source files and supporting material

331-07 • Result family 331

Midpoint convexity from two recursive potentials

We study tree norms computed by two least nonnegative fields whose difference is the vector. For Euclidean child aggregation, two root-sum spaces have an averaged asymptotic midpoint modulus of at least $t^3/128$ for $0 < t < 1$, including a reflexive joining-root space. A reflexive construction with a fixed zero root also satisfies a homogeneous cubic estimate. These spaces admit no equivalent asymptotically uniformly convex norm. We also obtain sixth-power and cubic estimates when the aggregation exponent depends on the height of a finite component.

Original PDF | Source files and supporting material

332-01 • Result family 332

Metric Markov Cotype Two of ℓ1

We prove that the real Banach space ℓ1 has metric Markov cotype two, answering a question of Mendel and Naor. As a consequence, every Lipschitz map from an arbitrary subset of a real Hilbert space into ℓ1 extends to the whole Hilbert space with a universal multiplicative loss in its Lipschitz constant, resolving Ball's extension problem for this target.

Original PDF | Source files and supporting material

Differential geometry

49 manuscripts • Official subject grouping

333-01 • Result family 333

Smooth isometric immersions of closed surfaces into Euclidean four-space

Every closed smooth Riemannian surface admits a smooth isometric immersion into Euclidean four-space, without an orientability assumption. This resolves the closed-surface form of the classical four-dimensional isometric-immersion problem.

Original PDF | Source files and supporting material

334-01 • Result family 334

A Smooth Metric with No Local Isometric Immersion into Three-Space

We construct a smooth positive-definite Riemannian metric on $(-1,1)^2$ that agrees with the Euclidean metric to every order at the origin, yet no neighborhood of the origin admits a smooth isometric immersion into Euclidean three-space. This gives a negative answer to the unrestricted smooth local isometric realization problem for surfaces.

Original PDF | Source files and supporting material

335-01 • Result family 335

An integral scalar curvature bound for real simplicial volume

We prove the integral scalar-curvature inequality proposed by Gromov. For every dimension n $\geq$ 3, there is a constant $a_n > 0$, depending only on n, such that $\int_M (\mathrm{Scal}_g^-)^{n/2}\, dV_g \geq a_n \|M\|$ for every closed connected oriented smooth n-manifold M and every smooth Riemannian metric g. Here $\mathrm{Scal}_g^- := \max\{0, -\mathrm{Scal}_g\}$, and $\|M\|$ is real simplicial volume. The proof uses the nonnegative-scalar-curvature vanishing theorem of the companion paper on rational inessentiality.

Original PDF | Source files and supporting material

335-02 • Result family 335

Positive scalar curvature forces rational inessentiality

We prove that every closed connected oriented smooth manifold admitting strictly positive scalar curvature is rationally inessential: its rational fundamental class maps to zero under the classifying map. No spin or fundamental-group hypothesis is needed, and there is no upper dimension bound. This proves the Gromov-Lawson aspherical conjecture; moreover, every nonnegative-scalar-curvature metric on a closed aspherical manifold is flat. We also show that every closed oriented smooth manifold of positive dimension with nonnegative scalar curvature has zero real simplicial volume, proving the qualitative vanishing consequence of Gromov's conjectural comparison.

Original PDF | Source files and supporting material

336-01 • Result family 336

Spectral scalar curvature and uniform Urysohn width

For every n $\geq$ 4, a complete connected smooth Riemannian n-manifold without boundary satisfying $-4\Delta +$ Scal ≥ 1 as a quadratic-form inequality admits a continuous map to a simplicial complex of dimension at most $n-2$ whose entire fibers have diameter bounded only in terms of n. The bound is measured in the original metric. This extends the uniform Urysohn width theorem from a pointwise scalar-curvature lower bound to a spectral lower bound.

Original PDF | Source files and supporting material

336-02 • Result family 336

Spectral scalar curvature and Urysohn width in dimension three

Every connected complete smooth Riemannian three-manifold without boundary satisfying $-4\Delta +$ Scal $\geq \lambda > 0$ as a quadratic-form inequality admits a continuous map to a graph whose entire fibers have diameter at most $500/\sqrt{\lambda}$ in the original metric. No orientability, spin, compactness, or bounded-geometry assumption is required.

Original PDF | Source files and supporting material

336-03 • Result family 336

Positive scalar curvature and uniform codimension-two width

We prove the quantitative continuous form of Gromov's scalar-curvature conjecture in every dimension n $\geq$ 4. Every complete connected smooth boundaryless n-manifold with scalar curvature at least one admits a continuous map to a simplicial complex of dimension at most $n-2$ whose entire fibers have diameter bounded only in terms of n. We also obtain the continuous macroscopic-dimension conclusion for universal covers of closed manifolds with positive scalar curvature in every dimension n $\geq$ 2.

Original PDF | Source files and supporting material

337-01 • Result family 337

Generalized Cartan-Hadamard isoperimetry and Euclidean equality rigidity

We resolve the generalized Cartan-Hadamard isoperimetric conjecture in every dimension. In a complete simply connected smooth manifold with sectional curvature at most $\kappa \leq 0$, every finite-volume set of finite ambient perimeter has perimeter at least that of the equal-volume ball in curvature κ. For bounded positive-volume sets, equality in the Euclidean comparison holds precisely when the set agrees up to null sets with an open region isometric, with its induced metric, to a round Euclidean ball.

Original PDF | Source files and supporting material

337-02 • Result family 337

Sharp integral fillings in CAT(0) spaces

Every compactly supported integral n-cycle, n $\geq$ 2, in a proper CAT(0) space bounds a compactly supported integral current with the sharp Euclidean mass bound. The theorem allows arbitrary integer multiplicities and unrestricted ambient dimension. In particular, we prove the Euclidean Cartan-Hadamard isoperimetric conjecture in dimensions at least three.

Original PDF | Source files and supporting material

338-01 • Result family 338

Uniformization of complete Kähler manifolds with positive bisectional curvature

We prove that every complete connected noncompact Kähler manifold with strictly positive holomorphic bisectional curvature is biholomorphic to complex Euclidean space. This resolves Yau's uniformization conjecture positively in every complex dimension.

Original PDF | Source files and supporting material

339-01 • Result family 339

Entropy equality and local symmetry in negative curvature

For every closed connected smooth Riemannian manifold of dimension at least three with strictly negative sectional curvature, we prove that normalized Liouville measure has maximal entropy for the unit-speed geodesic flow if and only if the metric is locally symmetric. This resolves Katok's entropy rigidity conjecture positively in these dimensions, including all rank-one symmetric types at arbitrary scale.

Original PDF | Source files and supporting material

340-01 • Result family 340

A counterexample to the nearby Lagrangian conjecture

We disprove the unrestricted nearby Lagrangian conjecture. For some sufficiently large even integer N, we construct a closed exact smoothly embedded Lagrangian in $T^*(S^9 \times S^{N-1})$ that is diffeomorphic to the base but is not Hamiltonian isotopic to the zero section.

Original PDF | Source files and supporting material

341-01 • Result family 341

Deforming hypersymplectic four-manifolds to hyperkähler triples

Every smooth normalized positive triple of closed two-forms on a closed connected oriented four-manifold admits a smooth deformation, with each cohomology class fixed, to a hyperkähler triple. This resolves Donaldson's hypersymplectic deformation conjecture.

Original PDF | Source files and supporting material

342-01 • Result family 342

Taming implies compatibility on four-manifolds

We prove that every smooth almost complex structure on a closed four-manifold which is tamed by a symplectic form is compatible with a symplectic form. This gives a positive solution to Donaldson's tamed-to-compatible conjecture. We also characterize the compatible cone by strict intersection positivity on the closure of the invariant part of the taming cone.

Original PDF | Source files and supporting material

343-01 • Result family 343

Symplectic Ball Packings in Higher Dimensions

We prove that, for all integers n ≥ 3 and k ≥ 1 and all positive real capacities $R_1, \dots, R_k$, the closed standard symplectic $2n$-balls of these capacities embed disjointly into the interior of a ball of capacity R > 0 if and only if $\sum_{i=1}^{k} R_i^n < R^n, \qquad R_i + R_j < R \quad (i \neq j)$. Capacity is π times the squared Euclidean radius, and each embedding is defined on a neighborhood of its closed source ball. This proves Siegel and Yao's Conjecture A.

Original PDF | Source files and supporting material

344-01 • Result family 344

The metric Blaschke theorem

We prove the metric Blaschke conjecture: every connected closed smooth Riemannian manifold of positive dimension whose global injectivity radius equals its diameter is, up to scale, a standard compact rank-one symmetric space.

Original PDF | Source files and supporting material

345-01 • Result family 345

Infinitely many closed geodesic images on every Riemannian sphere

We resolve the sphere case of the closed-geodesic infinitude problem: every smooth Riemannian metric on the standard sphere Sn, n ≥ 2, has infinitely many prime closed geodesics with pairwise distinct images. The two-sphere case is classical; the result in higher dimensions includes degenerate metrics. Using finite covers and theorems of Perelman and Rademacher-Taimanov, we also obtain the same conclusion for every smooth Riemannian metric on a nonempty closed smooth three-manifold.

Original PDF | Source files and supporting material

346-01 • Result family 346

A Codimension-One Bound for the Singular Set of a Stationary Integral Varifold

The singular set of every stationary integral m-varifold in an open Euclidean set has Hausdorff dimension at most $m - 1$, in every positive dimension and codimension. This sharp bound proves the Euclidean singular-set conjecture recorded by Brena, Decio, and De Lellis.

Original PDF | Source files and supporting material

346-02 • Result family 346

Almost-everywhere regularity of stationary integral varifolds

We resolve the almost-everywhere regularity conjecture of Brena, Decio, and De Lellis for stationary integral varifolds of arbitrary positive dimension and codimension in Euclidean open sets. The singular set has zero measure in the dimension of the varifold: near almost every support point, the varifold is a constant positive integer multiple of a smooth embedded minimal submanifold. The corresponding statement also holds on round spheres.

Original PDF | Source files and supporting material

347-01 • Result family 347

Hamiltonian Fixed Points Below the Stable Morse Number in Dimension Twenty-Two

We disprove the stable Morse lower bound for nondegenerate Hamiltonian fixed points with examples in fixed real dimension twenty-two. For every integer m $\geq$ 1, we construct a simply connected closed Kähler manifold with stable Morse number $80 + 1968m$ and a Hamiltonian diffeomorphism with exactly $80 + 1952m$ fixed points, all nondegenerate and with contractible orbit loops. The deficit is therefore unbounded, and the fixed-point count is at most 127/128 of the stable Morse number.

Original PDF | Source files and supporting material

347-02 • Result family 347

Sharpness of the Cyclic Integral Floer Bound Below the Stable Morse Number

We construct a simply connected closed Kähler manifold of real dimension 3332 and minimal Chern number one whose Hamiltonian fixed-point count attains the cyclic integral Floer bound while falling below the stable Morse number. The Hamiltonian diffeomorphism has exactly $1\,872\,232$ fixed points, all nondegenerate and with contractible orbit loops, whereas the stable Morse number is $1\,872\,264$. Thus the stronger stable Morse bound fails by exactly 32, even when the cyclic integral bound is sharp.

Original PDF | Source files and supporting material

347-03 • Result family 347

Hamiltonian Fixed Points Below the Stable Morse Number

We disprove the stable Morse lower bound for nondegenerate Hamiltonian fixed points by constructing a simply connected closed Kähler manifold with sixteen fewer fixed points than its stable Morse number. All the corresponding periodic orbits are contractible. The construction combines a Hamiltonian involution with integral homology torsion at two different primes; its proof uses finite-dimensional Morse theory and complex blowups.

Original PDF | Source files and supporting material

347-04 • Result family 347

A nondegenerate counterexample to the Morse-number Arnold bound

We construct a smooth one-periodic Hamiltonian on a closed symplectic twelve-manifold whose time-one map has fewer fixed points than the manifold's ordinary Morse number. Every fixed point is nondegenerate and has a contractible Hamiltonian trajectory. This disproves the Morse-number form of the Arnold conjecture. The deficit can be arbitrarily large among twelve-dimensional examples.

Original PDF | Source files and supporting material

347-05 • Result family 347

Three fixed points on the symplectic quadric threefold

We construct a smooth Hamiltonian diffeomorphism of the complex quadric threefold with exactly three fixed points, at least one of which is degenerate. The critical number and the unit-inclusive rational cup length of this manifold are both four. Thus the example disproves the unrestricted critical-number and rational cup-length forms of the Arnold conjecture.

Original PDF | Source files and supporting material

348-01 • Result family 348

Zero-Plane Rigidity for Einstein Four-Manifolds

We prove that a closed Einstein four-manifold with positive Einstein constant, nonnegative sectional curvature, and a zero-curvature plane has universal Riemannian cover isometric to a product of two round two-spheres. Under the normalization $\mathrm{Ric} = 3g$, both spheres have radius $1/\sqrt{3}$. The proof extends coupled estimates for the two Weyl curvature blocks to the boundary of the sectional-curvature cone and determines their equality case.

Original PDF | Source files and supporting material

348-02 • Result family 348

An L^2 Einstein Gap for Nonnegatively Curved Four-Manifolds

We prove a universal, scale-invariant L2 gap for the trace-free Ricci tensor on simply connected closed four-manifolds with nonnegative sectional curvature. If the trace-free Ricci energy is sufficiently small, the manifold is diffeomorphic to S4, $\mathbb{CP}^2$, or $S^2 \times S^2$. The proof uses the classification of positive-Einstein, nonnegatively curved four-manifolds supplied by the companion zero-plane rigidity theorem, stated explicitly as the classification premise of our result. No auxiliary curvature, volume, diameter, injectivity-radius, or Sobolev bound is required. The conclusion concerns the smooth manifold, not an isometry of the original metric.

Original PDF | Source files and supporting material

348-03 • Result family 348

Positively curved Einstein four-manifolds

We prove the classification conjecture for connected smooth closed Einstein four-manifolds with strictly positive sectional curvature. Up to positive scaling and isometry, every such manifold is the round four-sphere, the complex projective plane with its Fubini-Study metric, or real projective four-space with its round metric. No orientability assumption is needed.

Original PDF | Source files and supporting material

349-01 • Result family 349

The Solomon-Yau least-volume theorem

We prove the Solomon-Yau least-volume conjecture for minimal hypersurfaces of unit round spheres: in every dimension $m \geq 2$, a closed connected minimal immersion with non-totally-geodesic image has volume at least the smallest m-dimensional minimal Clifford product. Volume is measured on the domain, so covering multiplicities are included.

Original PDF | Source files and supporting material

350-01 • Result family 350

Sharp nodal length on smooth surfaces

We prove that a nonzero real Laplace eigenfunction with eigenvalue λ > 0 on a fixed smooth closed connected Riemannian surface has nodal length at most $C\sqrt{\lambda}$. Together with the known lower bound, this proves Yau's conjecture in this setting.

Original PDF | Source files and supporting material

350-02 • Result family 350

Smooth counterexamples to Yau's nodal upper bound in dimensions three and four

We construct a smooth metric on the three-sphere, arbitrarily close to the round metric in the smooth topology, and a smooth metric on $S^2 \times \mathbb{T}^2$ for which sequences of exact real Laplace eigenfunctions have unbounded nodal measure divided by the square root of the eigenvalue. Each sequence belongs to one fixed metric. Thus the upper-bound part of Yau's nodal conjecture fails for smooth metrics in dimensions three and four.

Original PDF | Source files and supporting material

350-03 • Result family 350

Power-law violations of Yau's nodal upper bound

We construct a smooth Riemannian metric on $S^4 \times S^1$ and a sequence of real Laplace eigenfunctions whose nodal four-volume grows faster than $\lambda^{1/2+\epsilon_0}$ for one fixed $\epsilon_0 > 0$. This disproves the smooth upper-bound assertion in Yau's nodal-set conjecture and the proposed bound with an arbitrarily small positive power loss.

Original PDF | Source files and supporting material

351-01 • Result family 351

A closed Ricci flow with bounded scalar curvature and finite-time curvature blowup

We disprove the scalar-curvature extension conjecture in its unrestricted all-dimensions form. In sufficiently high dimension, we construct a Ricci flow on a closed manifold whose scalar curvature remains uniformly bounded while full curvature diverges at a finite maximal time. In one fixed sufficiently high dimension, the examples have two-sided power-law curvature blowup with arbitrarily large exponents.

Original PDF | Source files and supporting material

351-02 • Result family 351

Bounded scalar curvature and smooth extension of four-dimensional Ricci flow

We prove that a smooth Ricci flow on a closed real four-manifold extends on the same manifold through every finite time at which its scalar curvature remains uniformly bounded. This resolves the scalar-curvature extension problem in dimension four.

Original PDF | Source files and supporting material

351-03 • Result family 351

Path selection and an elliptic inequality on degenerating Ricci-flat trees

We prove a sequential elliptic inequality for the renormalized Einstein-Hilbert functional on a fixed finite tree of four-dimensional Ricci-flat spaces. As the joining lengths diverge and the scale-neutral weighted Ricci error M tends to zero, the functional satisfies $E = o(M)$. The root end has decay exponent greater than one, and every joined quotient group is nontrivial. We also prove the multivariable path-selection theorem for asymptotic expansions used to obtain this estimate.

Original PDF | Source files and supporting material

352-01 • Result family 352

A finite-time singularity of Calabi flow on projective space

We construct a smooth Kähler metric in the Fubini-Study class of $\mathbb{CP}^{10}$ whose Calabi flow develops unbounded scalar curvature in finite time. This disproves Chen's smooth long-time existence conjecture, even in a class containing a constant-scalar-curvature metric.

Original PDF | Source files and supporting material

353-01 • Result family 353

A Smooth Nonquadratic Entire Affine Maximal Graph in Dimension Ten

We construct a smooth nonquadratic entire graph in dimension ten that solves the classical affine maximal equation and has positive-definite Hessian everywhere. This gives a smooth counterexample to the entire-graph affine Bernstein assertion in dimension ten. Hessian positivity is pointwise; no global uniform lower bound or completeness of the Berwald-Blaschke metric is asserted.

Original PDF | Source files and supporting material

353-02 • Result family 353

The affine Bernstein theorem in dimensions three through nine

We prove the Euclidean-complete affine Bernstein conjecture in dimensions three through nine: a smooth locally uniformly convex affine maximal graph is an elliptic paraboloid whenever its induced Euclidean metric is complete. The same conclusion holds, without an initial graph assumption, for connected smooth open (noncompact and without boundary) affine-complete locally uniformly convex immersed hypersurfaces that are classically affine maximal in these dimensions.

Original PDF | Source files and supporting material

354-01 • Result family 354

The Isoperimetric Conjecture for the Cubic Flat Three-Torus

We prove the isoperimetric conjecture for the cubic flat three-torus and classify all minimizers, including the equality cases at the transition volumes $4\pi/81$ and $1/\pi$. The minimizing regions are balls, circular tubes about shortest closed geodesics, coordinate slabs, and their complements.

Original PDF | Source files and supporting material

355-01 • Result family 355

Uniqueness of tangent flows at the first singular time of embedded surface mean-curvature flow

We prove that, at each singular point of the first singular time of the mean-curvature flow of a smooth compact connected embedded surface without boundary in $\mathbb{R}3$, all fixed-center rescalings converge locally smoothly on compact negative-time intervals to one multiplicity-one homothetic self-shrinker flow. The limit is unique in the original ambient coordinates, including its position and axes. No mean-convexity assumption or prescribed tangent model is required.

Original PDF | Source files and supporting material

356-01 • Result family 356

Gigli’s distributional curvature characterization of Alexandrov spaces

For every integer n $\geq$ 2 and κ $\in \mathbb{R}$, we prove that a complete separable metric space is an n-dimensional Alexandrov space of curvature at least κ if and only if, with reference measure $\mathcal{H}^n$, it is a full-support RCD$((n-1)\kappa, n)$ space whose distributional sectional curvature is at least κ in Gigli's original global test classes. This resolves Gigli's characterization conjecture in dimensions at least two.

Original PDF | Source files and supporting material

356-02 • Result family 356

Weak Hessian bounds along every geodesic in RCD spaces

On a full-support RCD(K, N) space with $1 < N < \infty$, we prove that a bounded globally Lipschitz function whose distributional Hessian is bounded above by a bounded continuous function satisfies the corresponding second-derivative inequality along every minimizing geodesic.

Original PDF | Source files and supporting material

357-01 • Result family 357

Bi-Lipschitz Coordinates at Regular Points of Noncollapsed RCD Spaces

We resolve the regular-point bi-Lipschitz conjecture in the noncollapsed setting. For every integer n $\geq$ 2 and every real K, every regular point of a noncollapsed RCD(K, n) space has an open neighborhood bi-Lipschitz homeomorphic to an open subset of $\mathbb{R}$n. The bi-Lipschitz constant depends only on n, and the neighborhood uses the restricted ambient distance.

Original PDF | Source files and supporting material

358-01 • Result family 358

A Three-Manifold Without Conjugate Points and Without a Nonpositively Curved Metric

We construct a closed connected orientable smooth three-manifold that admits a smooth Riemannian metric without conjugate points but admits no smooth Riemannian metric of nonpositive sectional curvature.

Original PDF | Source files and supporting material

359-01 • Result family 359

A negatively pinched Kähler threefold without bounded holomorphic coordinates

We construct a contractible domain in complex dimension three with a complete Kähler metric whose real sectional curvatures lie between two finite negative constants. It admits no bounded holomorphic map to $\mathbb{C}3$ with nowhere-vanishing Jacobian and is therefore not biholomorphic to a bounded domain. This gives a negative answer to the negatively pinched Kähler uniformization question.

Original PDF | Source files and supporting material

359-02 • Result family 359

One-sided negative sectional curvature and the holomorphic Liouville property

We construct, in some sufficiently large fixed finite complex dimension m, a domain in $\mathbb{C}$m diffeomorphic to $\mathbb{R}^{2m}$ that admits a complete Kähler metric with real sectional curvature at most -1 and has only constant bounded holomorphic functions. Its sectional curvatures are unbounded below. The construction gives a negative answer to the one-sided bounded-holomorphic-function question.

Original PDF | Source files and supporting material

360-01 • Result family 360

Global Support and Convex Injectivity Domains under Weak MTW

We prove that weak Ma-Trudinger-Wang curvature on a smooth, connected, compact Riemannian manifold of dimension at least two without boundary implies convexity of every tangent injectivity domain, resolving Villani's conjecture in this setting. Conjugate cut points are allowed. More generally, every ordinary subgradient of a squared-distance cost potential is a minimizing velocity with a global supporting mountain. No density hypothesis is used.

Original PDF | Source files and supporting material

360-02 • Result family 360

Uniform Bi-Holder Transport from Weak MTW

We prove that the weak Ma-Trudinger-Wang condition on a fixed smooth connected compact boundaryless Riemannian manifold of dimension at least two implies a common Hölder estimate for optimal transport maps and their inverses over the entire class of probability densities with

fixed positive upper and lower bounds. The maps have homeomorphic representatives, conjugate cut points are allowed, and no density regularity is assumed.

Original PDF | Source files and supporting material

361-01 • Result family 361

A counterexample to integer-degree harmonic dimension comparison

For some even n $\geq$ 8 and integer k $\geq$ 2, we construct a complete smooth metric on $\mathbb{R}$n with nonnegative Ricci curvature whose space of real harmonic functions of pointwise polynomial growth at most k has dimension larger than the Euclidean harmonic-polynomial dimension. The metric is Euclidean near the origin, has asymptotic volume ratio strictly between zero and one, and has nonunique tangent cones at infinity. This answers the integer-degree form of Yau's dimension comparison question in the negative.

Original PDF | Source files and supporting material

361-02 • Result family 361

A Three-Dimensional Counterexample to Integer-Degree Harmonic Dimension Comparison

For every $4/9 < v < 1$ and $1 < c < 9v/4$, all sufficiently large integers k admit a complete smooth metric on $\mathbb{R}$3 with nonnegative Ricci curvature, asymptotic volume ratio v, and at least $c(k+1)^2$ linearly independent real harmonic functions of pointwise growth at most k. This answers Yau's integer-degree dimension comparison question negatively in dimension three, with a fixed-factor excess over the Euclidean count. For each $1 < c < 9/4$, these metrics can be chosen arbitrarily close to the Euclidean metric in global bi-Lipschitz distance. The metric may depend on k.

Original PDF | Source files and supporting material

Partial differential equations

29 manuscripts • Official subject grouping

362-01 • Result family 362

Global classical solutions of the three-dimensional relativistic Vlasov-Maxwell system

We prove global existence and uniqueness for arbitrary smooth admissible initial data in the three-dimensional, one-species relativistic Vlasov-Maxwell system. The particle density is initially compactly supported, and the electromagnetic fields have finite energy and bounded derivatives of all orders. The solution remains smooth on every finite time interval. This resolves the large-data global classical regularity problem for this model, without size or symmetry restrictions on the data.

Original PDF | Source files and supporting material

363-01 • Result family 363

Nonuniqueness with local conservation for the hard-sphere Boltzmann equation

We prove nonuniqueness for the three-dimensional periodic hard-sphere Boltzmann equation among global entropy solutions satisfying exact local conservation of mass, momentum, and kinetic energy. We construct one nonnegative initial density with bounded velocity support and finite mass, energy, and absolute entropy that gives rise to two distinct such solutions. Both are strongly continuous in L1, and their collision gain and loss terms are integrable with every polynomial velocity weight on every bounded time interval.

Original PDF | Source files and supporting material

363-02 • Result family 363

Nonuniqueness for the periodic hard-sphere Boltzmann equation

We prove nonuniqueness for the periodic hard-sphere Boltzmann equation by constructing two distinct global renormalized solutions with the same nonnegative initial density on $\mathbb{T}^3 \times \mathbb{R}^3$. This density has bounded velocity support and finite mass, energy, and absolute entropy. Both solutions conserve local mass and total momentum and satisfy the global energy and entropy-dissipation inequalities. On a common initial interval, they are strongly continuous in L1, and their collision gains and losses are integrable.

Original PDF | Source files and supporting material

364-01 • Result family 364

The Boltzmann-Grad limit for stable radial potentials on regular kinetic intervals

We derive the nonlinear Boltzmann equation from a grand-canonical Newtonian gas with initial pair exclusion. We treat stable, finite-range radial potentials that are C2 except for an allowed repulsive singularity at the origin, and C1 initial probability densities with spatially summable Gaussian bounds on the density and its spatial gradient. All fixed-order rescaled factorial marginals converge in L1, uniformly throughout every finite interval on which the classical kinetic solution

has a uniform Gaussian bound. The result allows attractive wells and dynamically formed clusters, without restrictions on the differential scattering cross-section.

Original PDF | Source files and supporting material

364-02 • Result family 364

Hard-sphere fluctuations on the regular Boltzmann lifespan

We prove a finite-dimensional central limit theorem away from equilibrium for a deterministic grand-canonical hard-sphere gas in three dimensions. The initial probability density is smooth, with spatially summable Gaussian velocity bounds on the density and its spatial gradient. The limit holds on every finite interval on which the classical Boltzmann solution has uniform Gaussian velocity decay. The empirical measure is centered by its exact microscopic expectation. Its fluctuations converge to the Gaussian solution of the linear fluctuating Boltzmann equation.

Original PDF | Source files and supporting material

365-01 • Result family 365

Determination of a metric and a unitary connection from one boundary patch

We prove that zero-frequency boundary measurements on any nonempty open boundary patch determine both a smooth Riemannian metric and a smooth unitary connection on the trivial Hermitian rank-two bundle over a compact connected smooth manifold of dimension at least three with smooth boundary. Both inputs and observations are restricted to the same patch. The metric and connection are determined up to a diffeomorphism and a unitary gauge that restrict to the identity on the measured patch.

Original PDF | Source files and supporting material

365-02 • Result family 365

Smooth anisotropic uniqueness in the Calderón problem from one boundary patch

We resolve the smooth anisotropic Calderón uniqueness problem with arbitrary same-patch measurements. A smooth Riemannian metric on a compact connected manifold of dimension at least three is determined, up to a diffeomorphism fixing the measured patch, by its zero-frequency Dirichlet-to-Neumann energy form with both input and observation on any nonempty open boundary patch.

Original PDF | Source files and supporting material

365-03 • Result family 365

Nonuniqueness for bounded measurable scalar conductivities in three dimensions

We construct two distinct uniformly positive bounded measurable scalar conductivities on a ball in $\mathbb{R}3$ with the same full Dirichlet-to-Neumann operator. Both conductivities equal one near the boundary. This gives nonuniqueness in the scalar Calderón problem at bounded measurable regularity.

Original PDF | Source files and supporting material

366-01 • Result family 366

Interior regularity of planar Mumford-Shah minimizers

We prove the interior regularity assertion of the planar Mumford-Shah conjecture for reduced absolute minimizers with bounded fidelity data. Every interior point of the closed discontinuity set has a neighborhood consisting of a $C^{1,\alpha}$ arc, an arc ending at that point, or three such arcs meeting at 120 degrees. Only finitely many global connected components meet any relatively compact open set.

Original PDF | Source files and supporting material

367-01 • Result family 367

The critical dimension for one-phase Bernoulli minimizers

We prove that seven is the critical dimension for the one-phase Bernoulli problem: every nonzero one-homogeneous global minimizer in dimensions at most six is flat, while a nonflat one-homogeneous global minimizer exists in dimension seven. It follows that the interior free boundary of a local minimizer is smooth in dimensions at most six. In dimension n $\geq$ 7, its singular set has Hausdorff dimension at most $n-7$, and this bound is sharp. In dimension seven, the singular set is locally finite.

Original PDF | Source files and supporting material

368-01 • Result family 368

Strong diffeomorphic approximation in three dimensions for 1≤p≤2

We resolve the three-dimensional Ball-Evans approximation problem for $1 \leq p \leq 2$. Every $W^{1,p}$ homeomorphism between arbitrary bounded domains in $\mathbb{R}3$ is a strong $W^{1,p}$ limit of smooth diffeomorphisms onto the same target.

Original PDF | Source files and supporting material

368-02 • Result family 368

Strong diffeomorphic approximation in three dimensions for p>2

We resolve the three-dimensional Ball-Evans approximation problem for every finite p > 2. Every $W^{1,p}$ homeomorphism between arbitrary bounded domains in $\mathbb{R}3$ can be approximated strongly in $W^{1,p}$ by smooth diffeomorphisms onto the same target.

Original PDF | Source files and supporting material

369-01 • Result family 369

Strict hot spots and absence of interior critical points on smooth simply connected planar domains

We prove the strict hot spots conjecture for smooth bounded simply connected planar domains. More precisely, every nonzero eigenfunction for the first positive Neumann eigenvalue has nonvanishing gradient in the interior, so all its global maxima and minima lie on the boundary. This holds even when the eigenvalue is multiple.

Original PDF | Source files and supporting material

370-01 • Result family 370

The Subcritical Hénon-Lane-Emden Conjecture

We prove the subcritical Hénon-Lane-Emden conjecture: for every dimension n ≥ 2, positive powers p, q, and real weights A, B, the system has no strictly positive entire solution when $(n+A)/(p+1)+(n+B)/(q+1) > n-2$. Solutions need only be continuous at the origin and classical elsewhere, without a condition at infinity. For n ≥ 3 and $A, B > -2$, combining this result with the radial existence theorem of Bidaut-Véron and Giacomini proves Phan's Conjecture C in full: a positive radial entire solution exists whenever the strict inequality fails.

Original PDF | Source files and supporting material

371-01 • Result family 371

Stable self-similar blowup for a supercritical defocusing Schrödinger equation on the torus

We prove stable self-similar finite-time blowup for a supercritical defocusing nonlinear Schrödinger equation on the twelve-dimensional torus. For a sufficiently large odd power, the blowup initial data contain a nonempty open set in a high Sobolev space. As a consequence, Gaussian Fourier initial data of arbitrarily high Sobolev regularity can have positive probability of finite-time blowup.

Original PDF | Source files and supporting material

372-01 • Result family 372

Global Uniqueness for the Smooth Isotropic Elasticity Inverse Problem

We prove that the full static displacement-to-traction map uniquely determines both real smooth Lamé moduli on every bounded connected smooth domain in ℝ3, provided μ > 0 and $3\lambda + 2\mu > 0$ on the closure. This resolves the smooth three-dimensional isotropic elastic Calderón uniqueness problem under these positivity assumptions.

Original PDF | Source files and supporting material

373-01 • Result family 373

A counterexample to the Monge ansatz for the three-marginal Coulomb cost

We construct a smooth compactly supported probability density on ℝ3, with a smooth compactly supported square root, for which the three-marginal Coulomb transport minimum is not attained by any pair of measure-preserving Borel maps. Nevertheless, the Monge and Kantorovich infima agree: we construct preserving maps whose costs approach the Kantorovich minimum. For every d ≥ 2 and s > 0, we also construct a smooth identical marginal with the same nonattainment and equality-of-infima properties for the inverse-power interaction $\sum_{i<j} |x_i - x_j|^{-s}$ on ℝd.

Original PDF | Source files and supporting material

374-01 • Result family 374

Sharp One-Third Stability of Brenier Maps

For the uniform probability measure ρ on a compact convex body in ℝd, d ≥ 2, quadratic optimal transport maps are one-third Hölder continuous in $L^2(\rho)$ with respect to the target's 2-Wasserstein

distance. The constant is uniform over all targets supported in a fixed compact set, and the exponent is optimal. A three-atom family on a fixed cube disproves Letrouit's conjectured uniform square-root estimate.

Original PDF | Source files and supporting material

375-01 • Result family 375

A positive resolution of De Giorgi's conjecture in dimension eight

We resolve De Giorgi's conjecture positively in dimension eight: every entire C2 solution $u : \mathbb{R}^8 \to (-1, 1)$ of $\Delta u = u^3 - u$ with an everywhere positive directional derivative is a planar heteroclinic. We prove that every stable solution $v : \mathbb{R}^7 \to [-1, 1]$ of this equation is a constant well or a planar heteroclinic, without an energy-growth assumption.

Original PDF | Source files and supporting material

376-01 • Result family 376

Finite Instructions and Solenoidal Shear Flows

We realize finite reciprocal affine instruction maps by smooth incompressible shear flows on a flat three-torus. Applied to a reversible one-head recorder with a finite transition table, this gives a complete machine-to-fluid construction: a fixed particle enters a fixed open strip exactly when a given machine halts, with zero initial velocity, zero pressure, and a solenoidal mean-zero force that is periodic after initialization. The construction acts on full closed rectangles and controls every intermediate trajectory. Separate heights resolve overlap between sources and targets, and an endpoint-size estimate controls all excursions independently of the reciprocal scaling factor.

Original PDF | Source files and supporting material

376-02 • Result family 376

A Fixed Particle Test for Computation in a Forced Viscous Flow

We construct smooth external forces for incompressible Navier-Stokes flow on a fixed flat three-torus at any fixed positive computable viscosity, from zero initial velocity, such that a fixed particle enters a fixed open set exactly when a prescribed Turing machine halts. Every mixed derivative of the force and velocity is bounded and square-integrable in time in spatial supremum norm. The construction uses the machine's ordinary instructions, records their history in a third coordinate, and compensates for finer spatial gates by longer time steps.

Original PDF | Source files and supporting material

376-03 • Result family 376

Universal Computation with Eventually Stationary Navier-Stokes Forcing

At every fixed positive computable viscosity, we construct smooth mean-zero forces on the flat three-torus that become stationary after time one and make a fixed particle, initially in a fluid at rest, enter a fixed open set exactly when a prescribed Turing machine halts. The force has bounded derivatives of every order and a finite effective description. A reversible recording table is realized on whole planar rectangles by Hamiltonian motions, then driven by a mean-zero spatial clock. Separate choices give periodic forcing from time zero or a force whose derivatives are square-integrable in time.

Original PDF | Source files and supporting material

376-04 • Result family 376

Geometric Programs for Solenoidal Forcing

We construct smooth solenoidal mean-zero forces, periodic from time zero, for which a fixed particle on a flat three-torus reaches a fixed open strip exactly when a given machine halts. The initial fluid velocity and the pressure are zero. We also realize positive diagonal maps on coding sheets by incompressible shears, with explicit normal compensation for changes of planar area, and reciprocal maps on whole boxes of positive thickness. Complete local inverses, initialization rules, and intermediate trajectory bounds connect these geometric constructions to finite computations.

Original PDF | Source files and supporting material

376-05 • Result family 376

Prefix Instructions and Incompressible Flows

Finite prefix instructions may change area and erase information. We give explicit history processors and smooth incompressible motions that retain that information and realize every instruction on its full domain. A first application assigns each machine and finite input a smooth mean-zero force on the flat unit three-torus, at any fixed positive computable viscosity. The solution starts from rest; a fixed particle enters a fixed open strip exactly when the machine halts. The force repeats with period one after an initial loading interval. We then prove alternative realizations using full boxes, normal compensation, invariant planes, and a spatial clock. The constructions specify their initialization, comparison class, derivative bounds, and continuous-time observation. Exact formulas and effective cutoffs provide finite descriptions of the fields and all their derivatives.

Original PDF | Source files and supporting material

376-06 • Result family 376

Computation under Rapidly Vanishing Navier-Stokes Forcing

At any fixed positive computable viscosity, smooth forces can make a fixed fluid particle detect the halting of an arbitrary machine, starting from rest, while every mixed derivative of the force and velocity decreases faster than every inverse power of time. We give three complete memory constructions: compact moving curls, alternating fractional coordinates, and a periodic lattice on the flat three-torus. Each machine step takes one unit of physical time. The constructions retain earlier records at separated spatial scales and use an exact open detector with a shrinking signal.

Original PDF | Source files and supporting material

376-07 • Result family 376

Velocity-Field Detection of Computation in Forced Navier-Stokes Flows

We construct smooth forces for three-dimensional incompressible Navier-Stokes flow, from rest at any fixed positive computable viscosity, whose velocity field detects whether a prescribed machine halts. On the unit flat torus, a pointwise test of the third velocity component uses successively shorter stirring intervals in a fixed spatial region. On $\mathbb{R}^2 \times \mathbb{T}$, an integral test uses an expanding array of translation regions and a force with globally bounded mixed derivatives. The vertical velocity solves an advection-diffusion equation: short bursts control its pointwise error in the first construction, and a moving cutoff controls the total escaped mass in the second.

Original PDF | Source files and supporting material

376-08 • Result family 376

Scalar Potentials and Slow Clocks for Forced Fluid Computation

We construct smooth forces of fixed compact spatial support for three-dimensional incompressible Navier-Stokes flow from rest whose particle at the origin detects halting by entering a fixed half-space. Every mixed derivative of the force and velocity decays at rate $O((1+t)^{-1-j})$ for time order j. Three scalar potentials lift a coded rectangle, perform its instruction, and lower its image; an unbounded logarithmic clock supplies the decay. We also realize area-changing prefix instructions on an invariant torus plane, and construct a planar Hamiltonian processor admitting periodic forcing, stationary forcing after startup, and a velocity-field detector through a companion diffusion theorem.

Original PDF | Source files and supporting material

376-09 • Result family 376

Incompressible Box Transport and Finite Computation

We realize finite positive diagonal affine maps of determinant one by effective smooth incompressible flows on neighborhoods of entire closed rational solid boxes. The source and target families are each disjoint, but may overlap each other. The construction uses localized curls, evacuation to storage and obstacle detours. A balanced three-stack recorder then assigns every machine and finite input a smooth Navier-Stokes force with one compact spatial support, periodic after a loading interval, at any fixed positive computable viscosity. The fluid starts at rest, and one fixed particle enters one fixed open cube exactly when the machine halts. Further constructions give fixed torus charts, periodicity from time zero, alternative history guards and bounded or slab observers. Onto slow clocks yield separate decaying forces. Each construction includes an all-time observation proof, effective derivative bounds and a stated pressure comparison class.

Original PDF | Source files and supporting material

377-01 • Result family 377

Uniform Interior $C^{1,\alpha}$ Estimates for Infinity-Harmonic Functions

We prove a uniform interior C^{1,α_d} estimate for bounded infinity-harmonic functions in every dimension d ≥ 3. For some $\alpha_d \in (0, 1/3]$ depending only on the dimension, the gradient's supremum norm and αd-Hölder seminorm on B1/2 are bounded by a dimension-dependent constant times the oscillation on B1. The exponent is not explicit, and no endpoint or boundary regularity claim is made. In particular, infinity-harmonic functions are locally C1 on every open subset of $\mathbb{R}$d, for d ≥ 3.

Original PDF | Source files and supporting material

Withdrawals and revision history

The following three manuscripts were withdrawn because of proof gaps. Withdrawal does not establish that their mathematical statements are false. They are excluded from the 719 active-paper catalogue.

Algebraicity of Kuga-Satake Correspondences for K3 Surfaces

The withdrawal notices identify a sign error in a stabilization-trace construction. The required signed double-point cancellation is not established, leaving the asserted algebraicity result unproved, directly or through its dependency. Consult each notice for its precise role in the chain.

October 6, 2026

Withdrawal notice

Algebraicity of Weil classes on split abelian eightfolds

The withdrawal notices identify a sign error in a stabilization-trace construction. The required signed double-point cancellation is not established, leaving the asserted algebraicity result unproved, directly or through its dependency. Consult each notice for its precise role in the chain.

October 6, 2026

Withdrawal notice

The rational Hodge conjecture for products of K3 surfaces

The withdrawal notices identify a sign error in a stabilization-trace construction. The required signed double-point cancellation is not established, leaving the asserted algebraicity result unproved, directly or through its dependency. Consult each notice for its precise role in the chain.

October 6, 2026

Withdrawal notice

Sources and licensing

Manuscripts, English abstracts and source files are drawn from openai/math under the Apache License 2.0. Consult the repository LICENSE and any file-specific notices. Chinese paraphrases, reading guidance and layout are derivative editorial work, not an official OpenAI endorsement or certification.

Typography uses Noto Sans TC (SIL Open Font License 1.1), DejaVu Sans (its accompanying font license) and Latin Modern Math (GUST Font License). Font subsets are embedded, and mathematics is typeset with vector fonts.

openai/math LICENSE

Noto Sans TC font and OFL

Source commit / 1581d4cbdd69a0faf34b2089614fe2a814316a6a